

Fragile Prudence: Loan-to-Value Limits and Macroeconomic Stability*

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Abstract

Loan-to-value (LTV) caps are a widely used borrower-based macroprudential tool, premised on the idea that safer household balance sheets make the economy more resilient to downturns. We evaluate this premise in a general-equilibrium model with heterogeneous households, long-term defaultable mortgages, endogenous housing adjustment, and constrained banks. We find that lowering the maximum loan-to-value ratio from 100 to 20 percent eliminates almost all mortgage debt and downturn foreclosures, yet reduces the house-price trough after a productivity shock only marginally. The reason is *fragile prudence*: larger down payments protect incumbent balance sheets but prevent renters and owners from buying or upgrading when income, liquid wealth, and home equity weaken together. Mortgage limits stabilize more when shocks raise bank funding costs because such limits reduce banks' exposure to long-duration mortgages; this bank-balance-sheet benefit diminishes as banks become better capitalized. More generally, even the modest stabilization after productivity shocks can reverse when banks' marginal funding cost is externally priced. Borrower-based limits therefore provide effective microprudential insurance but are unreliable as macroprudential stabilizers.

Keywords: Macroprudential policy, financial stability, housing market, mortgages, banks.

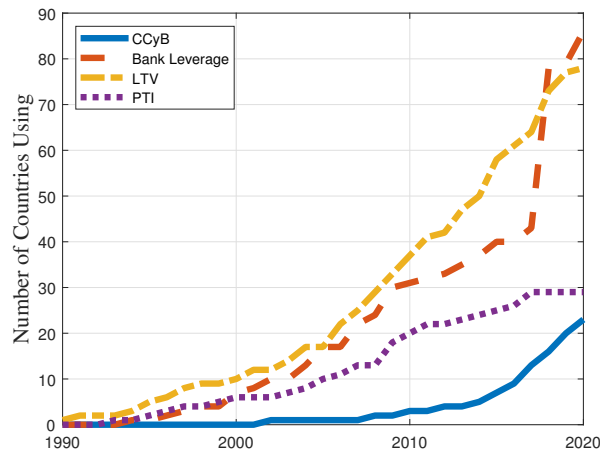
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1 Introduction

Loan-to-value (LTV) limits are the most widely used borrower-based macroprudential instrument. Their appeal rests on a simple inference. Household leverage amplified the Great Recession (Mian et al., 2013; Mian and Sufi, 2014); requiring buyers to enter with more equity should therefore make future downturns less severe. More equity reduces negative equity and default risk when house prices fall, limits lender losses, and should weaken adverse feedback between credit, house prices, consumption, and bank lending. This logic underlies policy guidance from the IMF, the BIS, and the ECB (International Monetary Fund, 2014; Committee on the Global Financial System, 2023; European Central Bank, 2022) and national implementation such as the Bank of England’s mortgage-market recommendations (Bank of England Financial Policy Committee, 2022), and it has made borrower-based rules standard practice around the world (Figure 1).

FIGURE 1 – Use of Macroprudential Tools



Notes: The graph plots the number of countries that use each macroprudential policy tool. CCyB: countercyclical capital buffer for banks. LTV: loan-to-value restriction on mortgage borrowing. PTI: payment-to-income ratio. Some sources also use debt-service-to-income ratios. Source: IMF Integrated Macroprudential Policy (iMaPP) Database, 1990-2020 vintage; see Alam et al. (2025) for database construction.

The inference from safer balance sheets to a more stable economy is incomplete, however, because an origination cap changes more than the losses borne by existing borrowers. It also changes who can buy a house, move, or trade up when prices fall. A tighter LTV cap induces households to accumulate more wealth before purchase, but it also requires buyers and movers to finance a larger share of each transaction from internal funds. In a downturn, when income, non-housing financial wealth, and home equity weaken together, renter entry and owner upgrading become harder. The pool of marginal buyers therefore contracts precisely when lower prices would otherwise attract new demand.

We call this mechanism *fragile prudence*. The policy delivers prudence ex ante by lowering

leverage and increasing borrower and lender buffers. It is fragile ex post because housing-market adjustment becomes more dependent on non-housing financial wealth in bad times. Tighter caps consequently generate two opposing forces: fewer foreclosures and forced sales support house prices, but fewer entrants and upgraders reduce the demand available to absorb housing supply. The central question is therefore not whether LTV caps make individual balance sheets safer—they do—but whether those microprudential gains survive the endogenous response of housing demand and financial intermediation.

To answer this question, we build a quantitative general-equilibrium model in which the policy can operate through the key household, housing-market, and intermediary margins. Overlapping generations of heterogeneous households face uninsurable labor-income risk over the life cycle and choose consumption, saving, renting, buying, housing size, refinancing, selling, and default; mortgages are long term, fixed rate, and defaultable, and origination is subject to the LTV cap. A rental sector with sluggish supply adjustment and a construction sector using labor and land permits let tenure, rents, and the housing stock respond to regulation. Banks intermediate deposits into mortgages and working-capital loans to firms subject to a balance-sheet constraint, so mortgage losses and funding conditions transmit to credit spreads, labor demand, wages, and output; a government runs a pension system and a mortgage-interest subsidy. This structure allows an LTV cap to operate simultaneously through household leverage, marginal housing demand, bank asset composition, and the cost of firm credit.

We calibrate the model to U.S. housing, mortgage, household, and bank-balance-sheet moments and compare economies that operate permanently under different maximum LTV ratios at origination. This permanent-regime comparison is distinct from estimating the short-run response to a policy tightening: an origination rule gradually changes the stock of borrowers, mortgages, and bank assets, and its stabilization properties need not equal the transitional effect of announcing a tighter cap. We study two shocks that emphasize the two sides of the mechanism: (i) a persistent productivity decline reduces household income and housing demand; (ii) a bank funding-cost shock reprices bank liabilities against long-term fixed-rate mortgage assets.

Our first result is that tighter caps achieve their intended balance-sheet objectives but deliver little stabilization after a productivity-driven downturn. In steady state, lowering the maximum LTV from 100 to 20 percent reduces mortgage debt from about 53 percent of GDP to nearly zero, eliminates most mortgage intermediation, and raises household net financial assets, while homeownership falls from about 64 to 58 percent as wealth-constrained households enter later.¹ We use this deliberately extreme comparison to reveal the maximum quantitative force of the mechanism. Despite the near elimination of mortgage leverage, the trough house-price decline

¹A 20-percent cap is far tighter than any observed policy. We deliberately report this wide range: the extreme regimes make the mechanisms easiest to see, and they show that even such drastic limits are unlikely to deliver robust stabilization. The intermediate regimes that we study cover the more moderate limits used in practice.

after the productivity shock improves only from 6.9 to 6.1 percent, the consumption trough from 5.4 to 5.0 percent, and the wage trough from 7.8 to 7.3 percent. Large changes in leverage and default therefore translate into small changes in the depth of the aggregate downturn.

The asset-reliance channel explains why. Tighter caps eliminate foreclosures in the benchmark downturn, but they also reduce the demand response to lower prices. Crisis-induced renter entry falls from 1.3 percent of baseline renters under the loose regime to 0.25 percent under the tightest regime, and crisis-induced upgrading by incumbent owners falls by roughly two thirds. The buyers who still enter under tighter caps finance the transaction by drawing down more non-housing wealth and cutting non-housing consumption by more. The policy thus protects incumbent borrowers while shifting adjustment costs toward the smaller group of marginal buyers and movers.²

A set of counterfactuals shows how the household margins aggregate. When all LTV regimes face the loose economy's wage path and fixed interest rates, the loose-tight difference in the house-price trough is 0.45 percentage points. Restoring non-bank general-equilibrium prices raises the gap to 0.67 percentage points, and restoring bank balance sheets raises it to the full-model value of 0.80. These numbers are path-dependent diagnostics rather than invariant channel shares, but they make the hierarchy clear. Direct household channels provide only modest net stabilization, because lower leverage and fewer defaults are offset by weaker entry, less upgrading, and greater reliance on internal funds. Banks contribute only 0.13 percentage points to the productivity-shock house-price gap, but they account for about half of the consumption gap and essentially all of the cross-regime wage difference through working-capital spreads.

Our second result is that LTV caps stabilize more when the shock hits bank funding costs. Long-term fixed-rate mortgages expose banks to a duration mismatch: mortgage cash flows are fixed while deposit funding reprices. Under the 100 percent LTV regime, the funding-cost shock reduces bank net worth by 35.3 percent on impact, raises the lending rate to about 10.6 percent, and lowers house prices by 3.81 percent. Under the 20 percent regime, bank net worth is almost unchanged, the lending rate peaks at 8.4 percent, and house prices fall by only 1.41 percent. Shutting down the link from bank net worth to lending conditions removes a big part of this cross-regime difference in the short run, and even reverses the effects in the medium run. The stabilization in this experiment therefore comes from the mortgage exposure removed from bank balance sheets, not mechanically from the safer household balance sheets that the cap produces.

²Household data provide descriptive discipline for this margin (Appendix E). In the PSID, buyers who remain stockholders through the purchase window experience weaker stock growth than matched renter-stayers, and many buyers exit direct stockholding altogether. In the SCF, recent mortgaged buyers have lower stock and retirement participation and lower balances conditional on participation than comparable long-time buyers. These patterns indicate that internal funds matter around home purchase; they do not identify the causal effect of an LTV change, which the structural model supplies.

Because much of this stabilization operates through bank balance sheets, the effectiveness of LTV regulation should depend on the capitalization of the banking system, and it does. Varying an exogenous bank leverage limit shows state dependence: the incremental stabilization provided by LTV caps is large when permitted leverage and mortgage exposure are high, and small when capital requirements are already tight; under the funding-cost shock the caps' insurance value concentrates entirely in thin-capital states. Explicit capital requirements, imposed on the benchmark calibration with all prices endogenous, stabilize the same bank-facing shocks and partly substitute for borrower-based limits on the stabilization margin.

Since the cost of a tight cap is the housing demand it suppresses in the bust, one might expect countercyclical implementation, relaxing the cap when the downturn arrives, to restore the tool's effectiveness. It does not: the release reopens entry and supports prices on impact, but the induced mortgage demand raises the bank lending rate, crowds out firm credit, weakens wages, and increases household debt service, so the trough improvement is small while consumption becomes less stable. When prices other than the house price are held fixed, the release is about three times as potent, which pins the lost potency on general-equilibrium crowding out (Appendix D.6).

These conclusions are robust to the sign of the shock and to the design of the instrument. Boom shocks are near-mirrors of busts: collateral gains loosen the cap's effective bite exactly when restraint matters most, while depleted wealth makes it most restrictive in busts, so the bite is procyclical in the unwelcome direction. A crisis-scale productivity shock reinforces the distinction between microprudential insurance and persistent stabilization: an 80 percent cap almost eliminates a 14.9 percent foreclosure wave and protects bank net worth, yet improves the trough declines in house prices and consumption by only 0.3 and 0.2 percentage points. Replacing LTV caps with payment-to-income limits produces nearly identical results, while adjustable-rate mortgages change which balance sheet bears funding-rate risk—bank duration or borrower cash flow—without changing the ranking across regimes. To the extent that the funding-cost shock captures the bank-funding leg of a monetary tightening, this last set of results also implies that borrower-based regulation can mute that leg of monetary transmission, a coordination consideration for monetary policy.

Our benchmark domestic-funding closure provides an endogenous lending-rate cushion that favors tighter caps in the productivity experiment. Bank funding costs fall with domestic activity: a downturn lowers deposit rates, cheaper funding passes through to lending rates, and the pass-through cushions the housing demand that survives the cap. This easing is what keeps the productivity-shock ranking in the cap's favor. In economies whose banks price marginal funding abroad, in wholesale or foreign-currency markets, the easing disappears: the demand-suppression and asset-reliance channels dominate, the ranking reverses, and tighter caps deepen the house-price and consumption declines. Even the small stabilization the caps deliver is

therefore not guaranteed; it depends on a funding structure that loosens when the economy weakens.

The paper’s contribution is to separate the microprudential and macroeconomic roles of borrower-based regulation. LTV caps are effective microprudential insurance: they lower leverage, reduce defaults and foreclosures, and protect lenders from mortgage losses. They are not general-purpose macroeconomic stabilizers. For shocks that reduce household income, the direct household channel is largely self-offsetting, because the same rule that makes borrowers safer also removes marginal buyers. For shocks that impair intermediary balance sheets, the caps stabilize by shrinking bank exposure. The relevant policy assignment therefore depends on the source of the shock, the structure of bank funding, and the resilience of intermediary balance sheets. We do not provide a welfare ranking, which would require weighing delayed homeownership and its distributional incidence against lower default and reduced intermediary fragility.

Related literature. Our closest connection is to quantitative work on borrower-based macroprudential policy. [Gete and Reher \(2016\)](#) study the trade-off between homeownership access and mortgage-default externalities; [Ferrero et al. \(2024\)](#) derive welfare-based countercyclical LTV rules in a New Keynesian housing model; [Grodecka \(2020\)](#) studies how LTV and payment-to-income constraints interact. Complementary work by [Elenev et al. \(2021\)](#) analyzes intermediary capital regulation and its general-equilibrium trade-offs. We differ in both object and mechanism: our exercise is positive rather than normative, compares economies operating permanently under alternative origination regimes, and makes renter entry, owner upgrading, long-term mortgage risk, and bank funding jointly endogenous, which reveals two effects of the same rule: suppressing marginal housing demand while reducing intermediary exposure. In models with pecuniary and aggregate-demand externalities, decentralized borrowing can be inefficiently high because borrowers do not internalize its effects on collateral prices and aggregate demand in constrained states ([Bianchi, 2011](#); [Farhi and Werning, 2016](#); [Korinek and Simsek, 2016](#)), although [Benigno et al. \(2013\)](#) show that normal times can also feature underborrowing, and [Jeanne and Korinek \(2020\)](#) study the optimal mix of ex-ante regulation and ex-post intervention. Those models typically study short-term collateralized debt or optimal corrective taxes, whereas we evaluate a specific origination instrument in a market with long-term contracts and endogenous movement along the housing ladder.

A second connection is to work on household leverage, marginal buyers, and the role of internal funds in housing transactions. Evidence from the Great Recession links housing-net-worth losses and household leverage to large consumption declines ([Mian et al., 2013](#)) and to local employment losses through household-demand channels ([Mian and Sufi, 2014](#)); [Berger et al. \(2018\)](#) provide a quantitative framework for the aggregate consumption response, and models of constrained deleveraging show how such cuts generate aggregate-demand amplification

(Eggertsson and Krugman, 2012; Guerrieri and Lorenzoni, 2017). Housing-ladder models emphasize that down payments, liquid wealth, and home equity determine which households become marginal buyers (Stein, 1995; Ortalo-Magné and Rady, 2006; Landvoigt et al., 2015), and related evidence links internal funds to housing demand: Boar et al. (2022) quantify homeowners' demand for home-equity liquidity, while Fuster and Zafar (2021) show that down-payment requirements and non-housing wealth shape willingness to pay for housing. Most directly, van Bakkum et al. (2024) show that Dutch LTV limits reduce household leverage and default risk while draining liquid buffers, and Aastveit et al. (2024) document greater consumption vulnerability after job loss among affected Norwegian borrowers. We place this internal-funds margin at the center of aggregate stabilization and show that it can offset the benefit of safer borrower balance sheets; our PSID and SCF evidence documents the purchase-financing margin and is deliberately not interpreted as causal evidence on LTV policy.

Third, we build on quantitative general-equilibrium models of housing, mortgages, and financial intermediation. Existing work debates how credit conditions affect house prices, leverage, ownership, and foreclosure (Favilukis et al., 2017; Kaplan et al., 2020; Justiniano et al., 2019; Greenwald and Guren, 2025; Arslan et al., 2025). Other contributions develop long-term defaultable mortgages (Jeske et al., 2013; Chatterjee and Eyigungor, 2015); long-term prepayable mortgages with LTV and payment-to-income constraints (Greenwald, 2016); equilibrium tenure choice, housing illiquidity, and foreclosure dynamics (Sommer et al., 2013; Hedlund, 2016; Arslan et al., 2015a; Garriga and Hedlund, 2020; Guren and McQuade, 2020); mortgage design (Guren et al., 2021a; Guler, 2015; Arslan and Guler, 2025); bank-driven housing cycles (Landvoigt, 2016; Arslan et al., 2025); and the distributional consequences of house-price fluctuations (Kiyotaki et al., 2024). Our contribution is to combine these margins in a policy experiment that traces how borrower regulation simultaneously changes household adjustment and bank portfolios; this combination is what makes the macroeconomic effect shock-dependent.

Finally, a large empirical literature estimates the effects of macroprudential tightening. Cross-country studies generally find that borrower-based tightening slows household credit, while effects on house prices and output are more heterogeneous and identification-sensitive (Kuttner and Shim, 2016; Cerutti et al., 2017; Akinici and Olmstead-Rumsey, 2018; Richter et al., 2019; Alam et al., 2025). Loan-level and credit-registry studies show that borrower and bank regulations bind at the loan level and reallocate credit across borrowers, lenders, and locations (DeFusco et al., 2020; Jiménez et al., 2017; Acharya et al., 2022; Peydró et al., 2024). Our object is different. These studies identify the transitional or partial-equilibrium effects of policy changes; we ask how cyclical dynamics differ once an economy has adjusted to a permanent origination regime. The two objects need not have the same sign, because an LTV rule gradually changes the distribution of buyers, the legacy mortgage stock, bank portfolios, and the set of households able to enter when prices fall.

The funding-cost experiment connects two empirical literatures that have developed largely separately: work on how mortgage-market structure shapes the transmission of interest-rate shocks to housing and consumption (Calza et al., 2013; Garriga et al., 2017; Arslan and Guler, 2025), and work on the interest-rate risk that banks carry on their balance sheets (English et al., 2018; Gomez et al., 2021; Drechsler et al., 2021; Jiang et al., 2024; Atkeson et al., 2019). Our contribution is to make the exposure at the center of this evidence a policy outcome: the LTV regime determines the size and duration of the mortgage book that banks carry, and contract design determines whether the remaining rate risk sits on bank balance sheets or in borrower cash flows.³

2 Quantitative Model

This section develops a quantitative general-equilibrium framework designed to assess how borrower-based macroprudential tools, implemented as fixed origination rules, shape housing-market outcomes and macroeconomic stability. The environment integrates six core sectors, plus a residual service sector: heterogeneous households who make housing tenure and mortgage decisions over the life cycle, a banking sector that intermediates credit subject to an enforcement constraint, rental companies that own part of the housing stock, construction companies that build new housing and sell it to households and rental companies, final-goods-producing firms that require working-capital financing, and a government that operates a pay-as-you-go pension system, runs a mortgage subsidy program, and sells land permits to construction companies.

Because we focus on transitional dynamics rather than stochastic fluctuations, the model abstracts from aggregate uncertainty. Unexpected shocks generate aggregate fluctuations; outside the shock periods the economy evolves under perfect foresight. Each model period is two years. This biennial timing is a computational compromise required by the high-dimensional household problem; it smooths within-year refinancing and foreclosure timing but preserves the life-cycle, leverage, housing-adjustment, and bank-balance-sheet margins that are central to the policy experiments. Given the extensive heterogeneity across households (in age, income, wealth, housing, and debt), all equilibrium prices, value functions, and policy functions depend implicitly on the aggregate state and the cross-sectional distribution. To ease notation we suppress these dependencies throughout. The text presents the objects needed to follow the policy experiments; Appendix A states all decision problems and the equilibrium definition.

2.1 Households

The household sector is the core of the model, capturing realistic housing tenure choices and mortgage decisions that are essential for analyzing borrower-based constraints.⁴

³Appendix B reviews this evidence alongside the model’s transmission magnitudes.

⁴The household block draws on elements from Arslan et al. (2015a) and Guler (2015), extended here to incorporate flexible housing and rental sizes together with refinancing options.

Demographics and Income Households enter at age 25, work until the mandatory retirement age J_r (age 65), and live until a terminal age J (age 85). During working life, after-tax labor income comprises a deterministic age profile $f(j)$ and a stochastic component z_j following an AR(1) process:

$$y(j, z_j) = \begin{cases} w(1 - \tau^s - \tau^m) \exp(f(j) + z_j), & \text{if } j \leq J_r \\ w(1 - \tau^m) y_R(z_{J_r}), & \text{if } j > J_r \end{cases} \quad (1)$$

$$z_j = \rho z_{j-1} + \varepsilon_j, \quad \varepsilon_j \sim i.i.d. \quad N(0, \sigma_\varepsilon^2),$$

where w denotes the wage per efficiency unit, τ^s is the social-security tax rate, τ^m is the labor-income tax that finances the mortgage subsidy, and $y_R(z_{J_r})$ is a replacement function calibrated to approximate the U.S. social-security system following [Guvenen and Smith \(2014\)](#). Labor supply is inelastic, although equilibrium wages depend on aggregate labor utilization as detailed below.

Preferences Households derive utility from non-housing consumption c_j and housing services s_j . The period utility aggregator is

$$u(c, s) = \frac{(c^{1-\epsilon} + \gamma_s s^{1-\epsilon})^{\frac{1-\sigma}{1-\epsilon}}}{1-\sigma}, \quad \epsilon \neq 1,$$

with the Cobb-Douglas limit used when $\epsilon = 1$. Lifetime expected utility is

$$E_0 \left[\sum_{j=1}^J \beta^{j-1} u(c_j, s_j) \right],$$

where β is the discount factor, σ controls curvature, ϵ controls substitutability between non-housing consumption and housing services, and γ_s is the housing-service weight.

Housing Market Participation All households begin as *active renters*, able to rent housing at price p_r per unit of service. At any age, an active renter may purchase a home and transition to homeownership. Homeownership involves several transaction and financing costs: a fixed origination cost φ_f and a proportional origination cost φ_m when taking out a mortgage, and a selling cost φ_s (as a share of the sale price) upon resale.

Homeowners may remain in their current home, refinance, sell (and either rent or buy a different home), or default. Default is costly. It takes the form of temporary exclusion from the housing market and higher foreclosure transaction costs $\varphi_e > \varphi_s$. Defaulters become *inactive renters*, regaining access to ownership with probability χ each period. Consequently, households occupy one of three tenure states: homeowner, active renter, or inactive renter.

Mortgage Contracts Mortgages are long term and carry a fixed interest rate. To preserve tractability, we assume each mortgage matures at the household’s terminal age, so that the borrower’s current age determines the remaining maturity. This assumption preserves the long-duration exposure of fixed-rate mortgages while avoiding an additional contract-age state variable. Let $n_j = J - j + 1$ denote the number of scheduled payments remaining, including the current payment. The periodic payment m satisfies the standard amortization formula evaluated at the bank lending rate r_ℓ :

$$m(d) = d \frac{r_\ell(1 + r_\ell)^{J-j}}{(1 + r_\ell)^{J-j+1} - 1}, \quad (2)$$

where d is the outstanding principal. The formula follows the convention that the household pays before interest accrues within the period; outstanding debt next period is therefore $(d - m)(1 + r_\ell)$.

Because households are heterogeneous, in principle each borrower would face an individual mortgage rate. To avoid tracking an additional state variable, we build on the mortgage-price formulation in [Hatchondo et al. \(2015\)](#) and follow [Kaplan et al. \(2020\)](#) in computing amortization at a common reference mortgage rate r_ℓ , while individual default risk enters the mortgage price q_m at origination. Equivalently, all borrowers pay upfront “points” that reduce the contract rate to r_ℓ .

Borrowing Constraints at Origination New buyers and refinancers face two origination constraints that are central to the analysis. The loan-to-value constraint caps debt at a fraction of house value:

$$d \leq p_h h (1 - \varrho), \quad (3)$$

where ϱ is the minimum down-payment ratio. The payment-to-income constraint limits the periodic mortgage payment relative to current income:⁵

$$m(d) \leq \varsigma y(j, z). \quad (4)$$

These origination rules tie borrowing capacity directly to endogenous prices and income, making them potentially procyclical even when they reduce steady-state leverage.

Household Choices Each period an active renter chooses between renting, at price p_r per unit of housing services, and buying, in which case she also chooses house size, mortgage debt subject to the origination constraints, and savings. A homeowner chooses among staying, refinancing, selling, and defaulting. Sellers re-enter the rent-or-buy decision with their housing equity;

⁵While the model accommodates both LTV and PTI constraints, the main analysis focuses on LTV regulation. Appendix D.4 repeats the central experiments with PTI limits and finds nearly identical stabilization patterns, so the mechanisms emphasized here attach to origination constraints in general rather than to the collateral ratio in particular.

defaulters surrender the house at the foreclosure discount, receive any residual equity, and become inactive renters. Because the foreclosure discount exceeds the selling cost, a household with positive net equity strictly prefers selling, so negative equity is necessary but not sufficient for default. The recursive formulations of all decision problems appear in Appendix A.

2.2 Production and the Working-Capital Channel

A representative competitive firm produces the final good with capital and effective labor $N_g u$, where N_g is the firm's labor demand and u is the utilization rate of labor, and must finance a fraction μ of the wage bill in advance through bank loans (Arslan et al., 2025). The wage per efficiency unit rises with utilization,

$$w(\bar{w}, u) = \bar{w} + \vartheta \frac{u^{1+\psi}}{1+\psi},$$

so the effective cost of labor is $(1 + \mu r_\ell) w(\bar{w}, u)$. A higher bank lending rate raises this effective cost, lowers utilization, and depresses output and wages. This working-capital channel links credit conditions to household income and is central to the transmission results. The firm's problem and first-order conditions are in Appendix A.

2.3 Housing Supply and the Rental Market

Construction companies build new housing from labor, N_h , and government land permits, L_p , and sell it to households and to rental companies. Rental companies own the rental stock H_r and incur a quadratic cost, governed by $\eta_r \geq 0$, when adjusting it. The friction makes rental supply sluggish, so tenure shifts move rents rather than being absorbed frictionlessly by supply adjustment. No-arbitrage between shares in rental companies and physical capital delivers the rental-price equation

$$p_r = \kappa + p_h + \eta_r p_h (H_r - H_r^{-1}) - \frac{(1 - \delta_h) p'_h + \eta_r p'_h (H'_r - H_r)}{1 + r'},$$

where κ is the per-unit maintenance cost, H_r^{-1} is the rental stock in the previous period, and H'_r is the rental stock in the next period. The adjustment terms vanish in steady state, so the equation reduces to the standard user cost and η_r matters only for dynamics. The problems of both firms are in Appendix A.

2.4 Banks

Banks intermediate household deposits into two assets: long-term fixed-rate mortgages and working-capital loans to firms. An enforcement friction limits bank leverage: depositors lend only up to the point at which the bank's continuation value deters diverting a fraction of assets,

which yields the balance-sheet constraint

$$(1 - \eta)(1 + r'_\ell)L' \geq (1 + r')B'$$

where L denotes the bank's assets, B denotes the bank's liabilities, and η is an endogenous collateral limit that depends on the future lending and deposit rates. In a stationary allocation, with a binding constraint, the spread between the lending and deposit rates becomes

$$r_\ell - r = \frac{1 - \beta_L(1 + r)}{\hat{\lambda} \beta_L},$$

which links the lending spread to the banker's discount factor β_L and the bank leverage ratio $\hat{\lambda}$. Lenders price mortgages fairly at origination: each loan's default risk, given the borrower's full state, enters its origination price. The banker's problem, the mortgage-valuation recursion, and the pricing condition are in Appendix A.

2.5 Government, Market Clearing, and Equilibrium

The government operates a balanced pay-as-you-go pension system and a mortgage-interest subsidy financed by land-permit sales and a labor-income tax, with any surplus rebated to households. A labor-intensive service sector supplies transaction and maintenance services. In equilibrium, households, final-goods firms, construction and rental companies, and banks optimize; the labor, asset, housing, rental, land-permit, and loan markets clear; both government programs balance; and the cross-sectional distribution is stationary, or follows its perfect-foresight transition after an unexpected shock. The formal definition is in Appendix A.

3 Calibration

This section describes how we parameterize the model to match salient features of the U.S. economy, focusing on housing and mortgage markets, financial intermediation, and household life-cycle dynamics. Each model period corresponds to two years. Annual rate, spread, and persistence targets become model-period objects before we solve the model. Households enter at age 25, retire at 65, and live until 85.

Table I summarizes parameter values, distinguishing between those set externally, from standard sources, and those calibrated internally to match model-generated moments to data targets.

Table II reports the empirical moments used to discipline the internally calibrated parameters and the corresponding model moments. In Table I, parameters marked with an asterisk are reported as annual equivalents of their biennial values. Depreciation rates are reported in percent.

TABLE I – Parameters (externally set and internally calibrated)

Panel A. Externally Set			Panel B. Internally Calibrated		
Parameter	Explanation	Value	Parameter	Explanation	Value
σ	risk aversion	2	β	discount factor	0.961*
ϵ	substitution parameter	1	$\underline{h}$	minimum house size	0.663
α	capital share	$\frac{1}{3}$	τ^d	mortgage subsidy	0.242
ψ	utilization curvature	0.5	γ_s	weight of housing services in utility	0.174
ρ	persistence of income	0.96*	μ	share of wage bill financed from banks	0.384
σ_ϵ	std of innovation to AR(1)	0.17*	β_L	bank discount factor	0.845*
φ_s	selling cost for a household	7%	ζ	divertible asset share	0.226
φ_e	selling cost for foreclosures	25%	κ	rental maintenance cost	0.020
φ_f	fixed cost of mortgage origination	1%	δ_k	capital depreciation rate	8.4%*
φ_m	variable cost of mortgage origination	0.75%	δ_h	house depreciation rate	2.5%*
χ	prob. of becoming an active renter	0.14*	θ_h	land share in house production	0.667
ϱ	down payment requirement	0	$\bar{L}$	land permit supply	0.069
ς	PTI requirement	0.45			
η_r	rental adjustment cost	3			

Notes: Variables marked with an asterisk (*) appear as the annual equivalents of their biennial values.

TABLE II – Calibration Moments and Data Targets

Statistic	Data	Model
Capital-output ratio	2.5	2.5
Homeownership rate (aggregate)	64%	64%
Mortgage debt-to-GDP ratio	53%	53%
Housing wealth-to-GDP ratio	1.625	1.625
Share of housing services in GDP	12.5%	12.5%
Ratio of mortgage loans to total loans in bank assets	0.54	0.54
Bank lending spread	0.015	0.015
Bank leverage ratio	10	10
House-price-to-rent ratio	13	13
Non-residential investment / final-good output ratio	20%	20%
Residential investment / GDP	4%	4%
Residential construction employment share	2.5%	2.5%

Notes: We report all flow variables as annual equivalents.

Preferences and Income We use the period utility aggregator defined in the model section. We set $\sigma = 2$, a standard choice in the literature, and $\epsilon = 1$, implying an elasticity of substitution between consumption and housing services of one. The computations use the Cobb-Douglas limit of the CES aggregator for this case, a common specification in quantitative housing models surveyed by [Piazzesi and Schneider \(2016\)](#). The weight γ_s ensures that imputed housing services constitute 12.5 percent of aggregate output. We calibrate the discount factor, β , to match a capital

to annual final-good output ($\frac{K}{Y_g}$) ratio of 2.5.⁶

The stochastic income component has annual persistence $\rho = 0.96$ and innovation standard deviation $\sigma_\varepsilon = 0.17$. Persistence is close to the 0.95 that [Storesletten et al. \(2004\)](#) estimate, and 0.17 is an acyclical midpoint between their expansion and recession estimates of 0.12 and 0.21. Post-retirement income follows the U.S. social-security replacement schedule as parameterized in [Guvenen and Smith \(2014\)](#).

Production and Working-Capital Finance The capital share is $\alpha = \frac{1}{3}$. We normalize TFP $Z = 1$ and aggregate labor $N = 1$. The ratio of non-residential investment to final-good output (I_k/Y_g) pins down the capital depreciation rate δ_k .

The utilization-curvature parameter is $\psi = 0.5$, implying an elasticity of 2 on the utilization margin. The coefficient ϑ in the wage function then follows from the firm’s first-order conditions at the steady-state allocation to target a normalized utilization rate of $u = 1$. The coefficient ϑ is a scale normalization with no independent economic content; only the curvature ψ matters for the dynamics of utilization and wages.

Housing Market Inactive renters become active with annual probability $\pi = 0.14$, reflecting the seven-year horizon over which default flags remain on U.S. credit reports. We set transaction costs to 7 percent for household sales (φ_s) and 25 percent for foreclosure sales (φ_e), with the household cost following [Gruber and Martin \(2003\)](#) and the foreclosure cost close to the 27 percent foreclosure discount estimated by [Campbell et al. \(2011\)](#). Mortgage origination involves a fixed cost—a lump sum equal to 1 percent of the average loan size—and a variable cost of 0.75 percent of the loan amount, calibration choices guided by the mortgage-fee ranges reported by the [Federal Reserve Board \(2008\)](#).

In the baseline calibration, we impose no regulatory down-payment floor ($\varrho = 0$); nonetheless, households endogenously choose positive down payments to obtain favorable mortgage terms. We set the maximum mortgage payment-to-income ratio at $\varsigma = 0.45$, which never binds in equilibrium. The mortgage subsidy τ^d targets the mortgage-debt-to-GDP ratio of 53 percent. The minimum owner-occupied unit $\underline{h}$ targets a 64 percent homeownership rate in steady state.

The housing depreciation rate δ_h targets a residential-investment-to-aggregate-output ratio of 4%. The share of labor in the house production function, $1 - \theta_h$, targets a residential-construction employment share of 2.5%. Because the targeted moments identify housing wealth rather than the separate scale of the housing stock and the house-price level, we cannot separately pin down house supply and house prices. We normalize $H = 1$, calibrate κ to match a house-price-to-annual-rent ratio of 13, which lies within the empirical range of 8 to 15.5 and is close to the

⁶The aggregate output in our model is $Y = Y_g + p_r H + p_h I_h + TC + c_B$. For calibration purposes, it is easier to target the $\frac{K}{Y_g}$ ratio than the wealth-to-output ratio. Given this target for $\frac{K}{Y_g}$, the model-implied wealth-to-output ratio is around 4.

mean of about 12 reported by [Sommer et al. \(2013\)](#), and calibrate the supply of land permits $\bar{L}$ to match a housing-wealth-to-aggregate-output ratio of $p_h H/Y = 13 \times 0.125 = 1.625$.

Following [Arslan et al. \(2025\)](#), we calibrate the adjustment-cost parameter η_r in the real estate companies' problem to match the evidence in [Greenwald and Guren \(2025\)](#) reported as the ratio of the percentage change in the price-rent ratio to the percentage change in the homeownership rate following a credit supply shock. This gives us $\eta_r = 3$.

Financial Sector Following [Shin \(2009\)](#), the “banking sector” encompasses deposit-taking institutions, issuers of asset-backed securities, and GSE-related entities. We match balance-sheet aggregates to the 1985–1994 U.S. average using Federal Reserve Z.1 data. We use this window because it predates the rapid growth of securitization, so the aggregates reflect a banking system closer to the model's deposit-funded intermediaries. Residential mortgages constitute 54 percent of bank assets; the remainder finances firms' working-capital needs, pinning down μ .

In steady state the spread between the bank lending rate and the deposit rate satisfies

$$r_\ell - r = \frac{1 - \beta_L(1 + r)}{\hat{\lambda} \beta_L},$$

where $\hat{\lambda}$ is the endogenous leverage ratio. We target $r_\ell - r = 1.5$ percent annually, the average annual gap between the 30-year mortgage rate and the 10-year Treasury rate. The leverage ratio $\hat{\lambda}$ equals 10, following [Gertler and Kiyotaki \(2015\)](#). These targets, together with the haircut recursion $\eta = \xi^{1-\beta_L}((1+r)/(1+r_\ell)-(1-\eta))^{\beta_L}$, identify the banker's discount factor β_L and diversion parameter ξ .

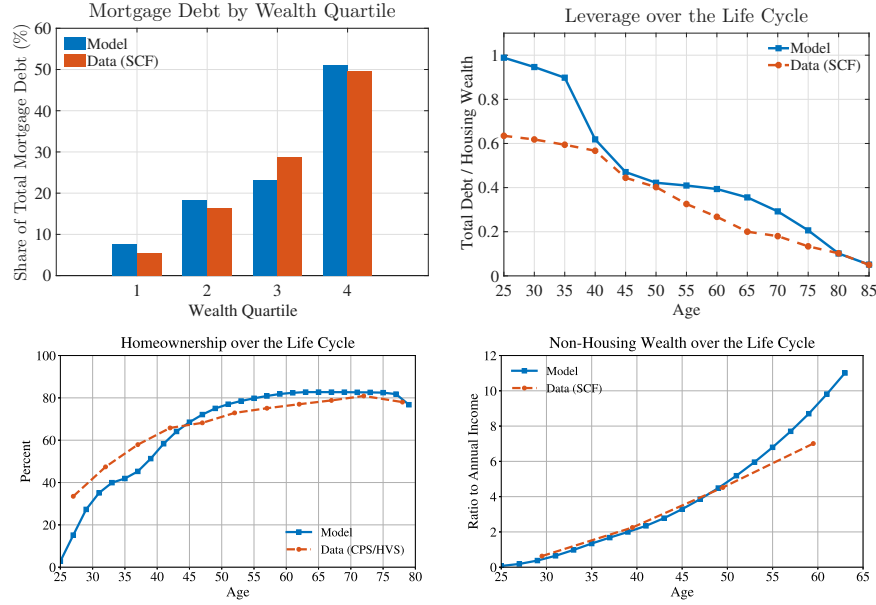
4 Results: External Validation of the Model

This section provides external validation of the quantitative model by documenting the match between the model's untargeted conditional and unconditional moments and their counterparts in the data.

External Validation of the Steady State We first briefly assess the model's ability to replicate salient features of the data that the calibration does not target. Figure 2 presents four such comparisons. The top left panel compares the distribution of mortgage debt across wealth quartiles against the 2016 Survey of Consumer Finances (SCF). In both the model and the data, the top wealth quartile holds roughly half of all outstanding mortgage debt, while the bottom quartile holds less than 10 percent. The model slightly overstates the bottom-quartile share (7.6 versus 5.4 percent in the data) but captures the monotonic concentration of leverage among wealthier households.

The top right panel plots aggregate leverage, defined as the ratio of total mortgage debt to total housing wealth within each age bin, over the life cycle. The model reproduces the steep decline

FIGURE 2 – Model Performance: Untargeted Life-Cycle and Cross-Sectional Moments



Notes: Model moments computed from the benchmark steady state (100 percent LTV). Top left: share of total mortgage debt held by each net-worth quartile; data from the 2016 SCF. Top right: ratio of aggregate mortgage debt to aggregate housing wealth within each five-year age bin; data from the 2016 SCF. Bottom left: homeownership rate by age; the model is the benchmark steady state and the data are CPS/HVS 2024 annual averages by five-year age group, plotted at bin midpoints; the figure omits the terminal model period, in which households sell their houses. Bottom right: non-housing wealth over the life cycle, the ratio of aggregate net worth excluding home equity to aggregate annual income within each age bin (model and 2016 SCF); SCF values at age-group midpoints.

in leverage from near unity for the youngest cohort to roughly 0.3 by retirement age. The data exhibit a similar shape, though model leverage is somewhat higher than the SCF for households under 40, consistent with the model's assumption of zero initial wealth at entry. Both series converge after age 40.

The bottom left panel compares the life-cycle homeownership profile with the Census CPS/HVS age-group rates: the model rises more steeply than the data before age 45 and tracks the data closely thereafter, with 74 percent owning at ages 45 to 54 versus 71 in the data, 24 versus 37 under age 35, and 83 versus 79 to 81 between ages 65 and 74. The early gap reflects entry with zero wealth; the small late gap reflects the absence of the family and health shocks that move older households out of ownership. The bottom right panel compares non-housing wealth with the SCF, measured as the ratio of aggregate net worth excluding home equity to aggregate income within each age bin. The model tracks the accumulation profile closely through the early fifties, 4.6 versus 4.5 at ages 45 to 54, and moves above the data at ages 55 to 64, 8.7 versus 7.0, because the single liquid asset absorbs all retirement saving.

Table III adds untargeted moments for the purchase margin at the center of the mechanism. The median house-value-to-annual-income ratio at purchase is 2.0 in the model, 2.3 for PSID recent movers, and 2.4 for SCF recent purchasers. Financial assets rise ahead of purchase, by

0.41 log points over the preceding four years in the model and 0.28 in the PSID event study, and drop discretely at closing; the model's 0.70-log-point drawdown sits next to the 0.77-log-point stock drawdown of matched PSID buyers (Table E.1). In any two-year period, 4.7 percent of households purchase a home in the model and 6.0 percent in the weighted SCF, and prior renters account for 64 percent of those purchases in the model and 65 percent among recent buyers in the 2023 American Housing Survey. Selling flows are of the same order: owner-to-owner moves—each a sale followed by another purchase—run at 1.7 percent of households per two years in the model and 2.9 percent in the PSID. Together with the ownership profile in the figure, these moments show that the model reproduces when households buy, how much house they buy relative to income, and the asset accumulation and drawdown that finance the purchase, the margins on which the transition experiments turn.

TABLE III – Untargeted Purchase-Margin Moments: Model and Data

Moment	Model	Data
House value / annual income at purchase (median)	2.0	2.3
Pre-purchase asset growth, four years (log points)	0.41	0.28
Asset drawdown at purchase (log points)	-0.70	-0.77
Home purchases in two years (percent of households)	4.7	6.0
Purchases by prior renters (percent of all purchases)	64.0	65.0
Owner-to-owner moves in two years (percent of households)	1.7	2.9
Renter-to-owner transitions in two years (percent of households)	3.0	5.5

Notes: Model moments come from the benchmark (100 percent LTV) steady-state simulation; none is targeted. The three flow rows and the composition row are computed from the same steady-state panel on a single base, all households aged 27 and older, so they satisfy the accounting identity: renter-to-owner purchases (3.0) plus owner-to-owner moves (1.7) equal total purchases (4.7), and prior renters account for 64 percent. Data sources: value to income from PSID recent movers (2.3; 2.4 among SCF recent purchasers); the four-year run-up from the PSID liquid-asset event study; the drawdown from the matched continuing-stockholder estimate in Table E.1 (a buyer-minus-matched-renter gap; the model column is the buyer's own within-household change, so the comparison is qualitative); the purchase rate from SCF recent purchasers as a share of all households aged 25 and older (the buyer-rate series in the transition figures uses working-age households); purchase composition from recent owner-buyers in the 2023 American Housing Survey; owner-to-owner moves from PSID households whose reference person ages consistently across waves; renter-to-owner transitions from direct PSID tenure changes, 1999–2021, which include entries into ownership without a purchase (marriage and inheritance), so the data value exceeds the product of the purchase rate and the prior-renter share. Exits from ownership are rarer in the model than in the data because the model has no family-composition or job-mobility shocks; purchases and sales need not be equal because construction and rental companies also trade housing.

External Validation of Rate Transmission The model's dynamic responses to interest-rate shocks also line up with identified estimates. A one-period 100 basis point funding shock moves the lending rate by about 40 basis points, a conservative pass-through relative to market-wide estimates (Gropp et al., 2007; Harimohan et al., 2016). Per 100 basis points of the induced mortgage-rate movement, house prices and non-housing consumption respond by 1.9 and 0.8 percent, within the ranges from direct mortgage-rate variation and identified monetary shocks (Akgündüz et al., 2023; Gorea et al., 2024; Jordà et al., 2015; Cloyne et al., 2020; Holm

et al., 2021; Di Maggio et al., 2017; Flodén et al., 2021). Firm working-capital credit responds by 0.93 percent per 100 basis points of the lending-rate movement, about half of the estimates in Altavilla et al. (2022), and the wage moves by 1.9 percent, in line with the direction and order of the employment response to credit-condition shocks in Gertler and Gilchrist (2018). Feeding the funding-shock house-price path through the household block in partial equilibrium yields about 5 cents of non-housing consumption per dollar of housing wealth, within the identified range (Guren et al., 2021b; Aladangady, 2017; Mian et al., 2013; Foulis et al., 2026). Appendix B develops each comparison.

Taken together, these untargeted comparisons, including the purchase margin that drives the asset-reliance channel and the transmission of identified interest-rate shocks, suggest that the model provides a reasonable laboratory for studying the interaction between borrower-based regulation and housing-market dynamics.

5 Results: Steady-State Implications of LTV Regulations

This section reports the model’s predictions for steady-state allocations under alternative LTV regimes.

Steady-State Analysis of LTV Regulations Figure 3 plots the steady-state values of key macroeconomic and financial aggregates as the LTV cap tightens from 100 percent to 20 percent. The main trade-off is between lower debt and higher capital accumulation on the one hand and reduced housing-market access on the other.

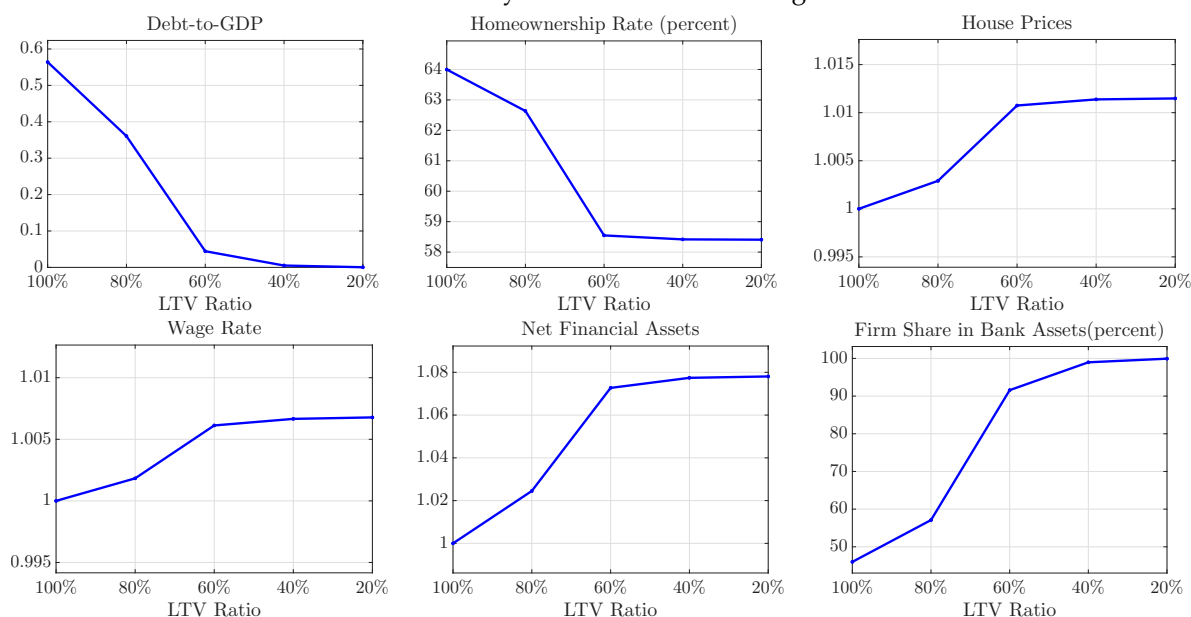
In steady state, tighter LTV limits reduce leverage and shrink financial-sector balance sheets (Figure 3). As the maximum LTV ratio falls from 100 percent to 20 percent, debt to GDP declines from about 53 percent to essentially zero, and total bank assets contract by more than half as mortgage intermediation recedes. Homeownership also falls, from about 64 percent to roughly 58 percent, because stricter down-payment requirements delay or exclude wealth-constrained households, especially younger households.

These effects extend beyond deleveraging. Larger required down payments lead households to accumulate more savings before entering homeownership. In general equilibrium, this stronger saving motive raises the aggregate capital stock by about 1 percent and equilibrium wages by a similar amount.

House prices also rise modestly as the LTV cap tightens. Although this may seem counter-intuitive given the near-elimination of mortgage debt, offsetting general-equilibrium forces dominate: higher wages and lower bank lending rates improve affordability and put upward pressure on house prices in steady state.

Gross household assets decline as housing leverage falls, but net assets rise because liabilities contract much faster. Bank portfolios also shift: as mortgage lending declines, banks reallocate

FIGURE 3 – Steady-State Effects of LTV Regulations



Notes: Steady-state values across maximum LTV limits from 100 to 20 percent. Top row: mortgage-debt-to-GDP ratio, homeownership rate, and house price. Bottom row: wage, net household financial assets, and the share of firm loans in bank assets. The aggregate capital stock (not shown) rises by about 1 percent as the cap tightens, mirroring the wage panel; the text discusses gross household assets and total bank assets.

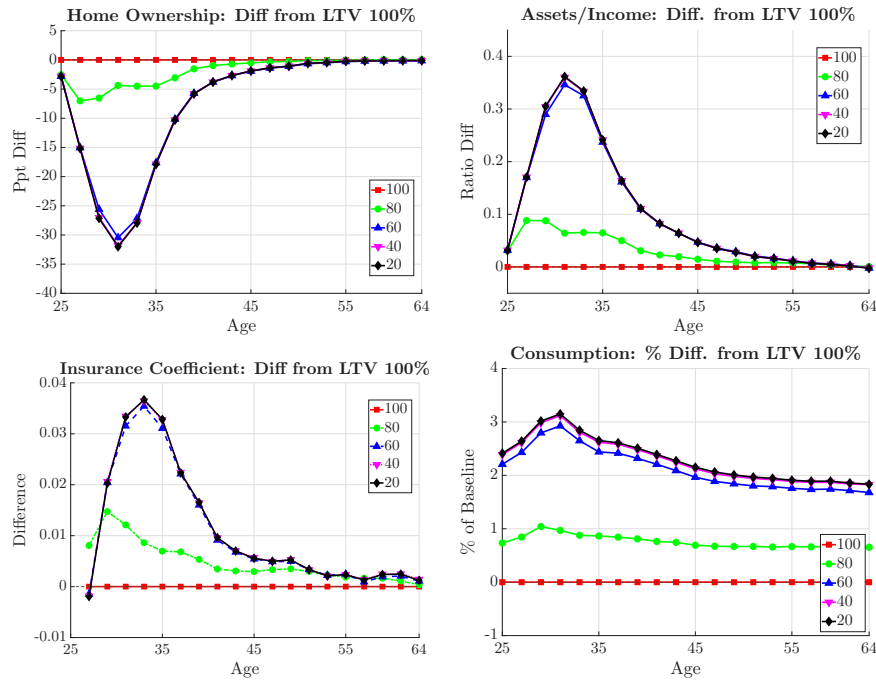
from housing credit toward loans to firms, with the firm share of bank assets rising from roughly 45 percent under loose LTV limits to nearly 100 percent under the tightest cap.

Life-Cycle Dynamics Aggregate steady-state outcomes mask substantial life-cycle heterogeneity. Figure 4 shows that tighter LTV limits do more than reduce access to mortgage credit; they shift early adulthood from leveraged home purchase toward asset accumulation.

The most immediate effect is on the extensive margin of homeownership. Under the 100 percent LTV benchmark, young households use leverage to buy housing early, and homeownership exceeds 40 percent by age 30. Under tighter limits, such as a maximum LTV of 20 percent, the down-payment constraint binds for many more young households. Homeownership at age 30 therefore falls by more than 30 percentage points relative to the 100 percent LTV limit, and entry into owner-occupation slips by roughly a decade. The effect is not permanent: as households accumulate the required equity, ownership rates converge and the gap largely closes by age 50.

This delay in housing access comes with a large portfolio shift. When young households cannot finance housing with high leverage, they must accumulate internal funds before entering owner-occupation. Under tight LTV limits, net financial assets by age 30 reach nearly 40 percent of annual income, compared with negligible balances in the baseline economy. This is the life-cycle counterpart of the asset-reliance channel developed below: tighter LTV limits require more pre-purchase financial wealth, but at the cost of delayed homeownership.

FIGURE 4 – Life-Cycle Dynamics under LTV Regulations



Notes: Life-cycle profiles across maximum LTV limits from 100 to 20 percent. Top row: homeownership rate and net financial assets by age. Bottom row: consumption-insurance coefficients following [Blundell et al. \(2008\)](#) and average non-housing consumption by age. The figure omits mortgage debt and housing services by age; the text reports their patterns.

The insurance-coefficient profiles point in the same direction. Constructed following the [Blundell et al. \(2008\)](#) approach, the coefficient measures how much income-shock variation households absorb rather than pass through to consumption. Relative to the no-cap benchmark, tighter limits raise insurance mainly for young households, with the effect peaking in the early 30s and then declining over the life cycle. Because this profile closely mirrors net financial assets, the increase in early-life self-insurance appears to operate primarily through greater asset accumulation.

Tighter LTV limits also raise non-housing consumption over the life cycle. In the tightest regimes, non-housing consumption is about 2.5 to 3.5 percent higher across much of the life cycle. This reflects two forces: higher saving increases aggregate capital and wages in general equilibrium, while tighter borrowing limits reduce mortgage obligations among households that remain renters longer. Young households therefore devote fewer resources to debt service and more to consumption and financial saving. The trade-off is lower housing consumption early in life: in the tight-LTV economy, young households consume roughly 10 percent less housing services while they remain in smaller rental units before transitioning into homeownership.

6 Results: Productivity-Shock Transitions across LTV Regimes

We begin with a negative productivity shock because it is a real-side shock with a direct household-income effect: it moves labor utilization, wages, the return on capital, and firms' credit demand together. The experiment asks whether safer borrower balance sheets under tighter LTV limits stabilize housing when income falls. We study the response to an unexpected and persistent 5 percent decline in productivity under alternative LTV regimes. Productivity follows $\log Z_t = \rho_Z \log Z_{t-1} + \varepsilon_t^Z$ around its steady-state level, with $\rho_Z = 0.6$ at the model's biennial frequency, about 0.77 at an annual rate. The experiment is a single unanticipated innovation that lowers productivity by 5 percent on impact; the process then decays back to the steady state with a half-life of about 2.7 years. The crisis experiment scales the same innovation to a 15 percent impact decline with unchanged persistence.

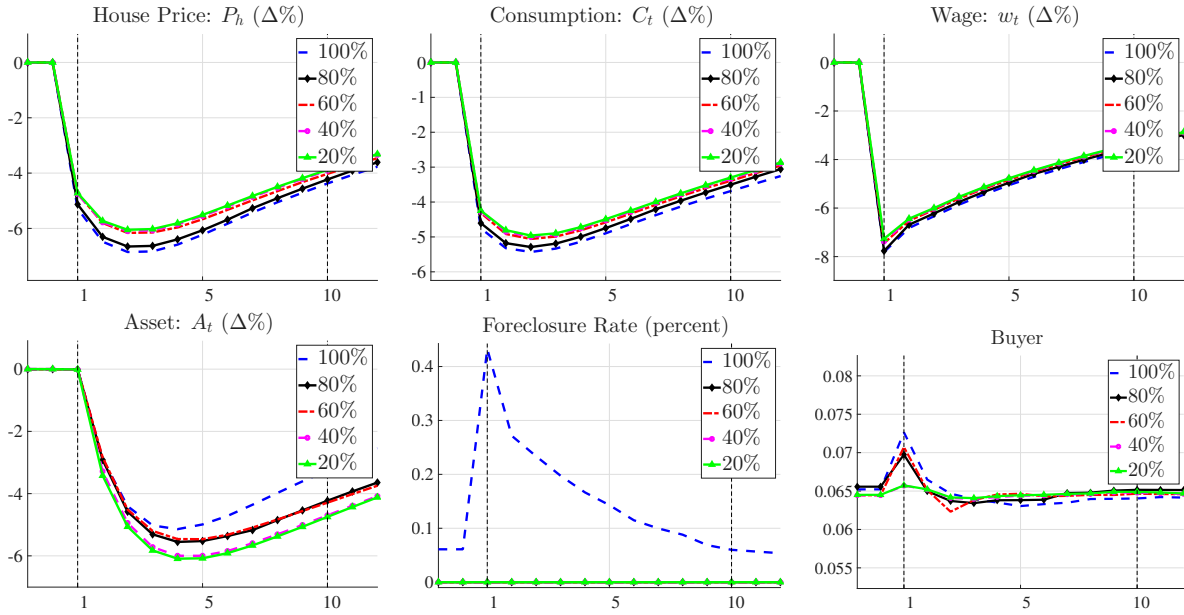
On impact, lower productivity reduces firms' desired labor input and output, lowering wages and household income. Because firms finance part of the wage bill through bank credit, the production decline also reduces working-capital demand and changes equilibrium lending rates. Lower household income weakens housing demand, puts downward pressure on house prices, and reduces consumption through both labor-income and housing-wealth channels. The fall in house prices then activates the two opposing consequences of the LTV regime. Under loose regulation, leveraged incumbents are exposed to negative equity and default, but low down-payment requirements let marginal renters and movers finance purchases of cheaper housing with debt. Under tight regulation, incumbents are safer, but purchases and moves require internal funds precisely when income, financial wealth, and sale equity weaken together.

Under domestic bank funding, an interest-rate response partly offsets this fragility. Weaker capital and working-capital demand lower the equilibrium lending rate after the shock. Because tight-LTV banks hold few mortgages and a large share of firm loans, the contraction in firm-credit demand moves their lending rate by more, and the lower rate supports labor utilization, wages, and buyer affordability. The cross-regime comparison below is therefore the net of three forces: balance-sheet insurance, weaker marginal housing demand, and this endogenous lending-rate cushion.

Figure 5 shows that tighter LTV regulation eliminates foreclosures and strengthens private-sector balance sheets, but provides only limited aggregate stabilization. House prices fall by 6.9 percent in the loose economy and by 6.1 percent in the tight economy. Wages decline by 7.8 percent versus 7.3 percent, and consumption falls by 5.4 percent versus 5.0 percent. The two economies also converge at similar speeds, so tighter regulation mainly shifts the level of the response rather than accelerating recovery.

Although the tight-LTV economy enters the recession with higher net worth and lower leverage, total household assets decline more during the downturn, by 6.1 percent rather than 5.1 percent.

FIGURE 5 – Dynamics Following the Benchmark Productivity Downturn



Notes: Transition dynamics following the unexpected, persistent 5 percent productivity decline described in the text, across maximum LTV limits from 100 to 20 percent. Top row: house price, consumption, and wage, as percentage deviations from steady state. Bottom row: total household assets (percentage deviation), the foreclosure rate, and the buyer rate (levels). The text reports the capital-stock and homeownership responses.

Households with larger initial buffers draw them down more aggressively. As a result, the capital stock contracts by similar amounts across regimes, 6.6 percent in the loose economy and 6.2 percent in the tight economy, leaving only modest differences in equilibrium wages. Section 7.1 examines this asset-reliance mechanism in detail.

The housing market displays an asymmetry between supply and demand. On the supply side, tighter regulation delivers its intended effect: foreclosures rise under loose regulation but are absent under tight regulation. Refinancing also varies across regimes. At the onset of the bust, the refinancing rate increases to 1.2 percent in the 60 percent economy, followed by the 100 percent economy at 0.9 percent and the 80 percent economy at 0.8 percent, and it is negligible in the tightest regimes. This non-monotonic pattern reflects three forces. Tighter regulation reduces the stock of outstanding mortgages and thus the pool of potential refinancers. Moderate LTV limits leave households with enough equity to qualify after house prices fall. Under very loose regulation, by contrast, some borrowers enter with too little equity and become ineligible after the downturn; under very tight regulation, the mortgage stock is too small for refinancing to matter quantitatively.

On the demand side, the response reverses. Under loose regulation, falling house prices induce entry, with the buyer rate rising from 6.5 to 7.3 percent of working-age households, or by about 12 percent. Under tight regulation, the buyer rate increases by only about 2 percent. Homeownership mirrors this difference: ownership rises by about 1.2 percentage points in

the loose economy, compared with less than 0.1 percentage points under tight regulation. The household decomposition below shows that the missing buyers are the households whose entry would otherwise provide stabilizing demand during the downturn.

Taken together, these results indicate that tighter LTV limits alter default outcomes and the composition of balance-sheet adjustment, but do little to stabilize aggregate quantities after a persistent productivity shock. They protect incumbent owners through fewer foreclosures and, at intermediate LTV levels, greater scope for refinancing. But these supply-side gains are largely offset by suppressed buyer entry, which otherwise would absorb excess supply and put a floor under house prices. None of this is specific to the collateral instrument: replacing the LTV cap with payment-to-income limits delivers nearly identical steady states and dynamics (Appendix D.4).

Financial Transmission and Bank Portfolio Composition LTV regulation also changes financial-sector transmission. Figure 6 reports changes in total bank lending and loans to firms in the top row, and household debt and the bank lending rate in the bottom row. The central mechanism is bank-asset composition. Under tight LTV limits, household mortgage borrowing is lower, so banks hold fewer residential mortgages and more working-capital loans to firms. This portfolio shift shapes the financial response to the downturn.

When productivity falls, firms reduce labor utilization and working-capital demand. Because banks in the tight-LTV economy are more exposed to corporate credit, this decline in firm borrowing generates a larger contraction in total lending. Bank loans fall by 7.4 percent in the tight economy, compared with 2 percent in the loose economy. The contraction reflects weaker credit demand rather than default-driven deterioration in bank balance sheets.

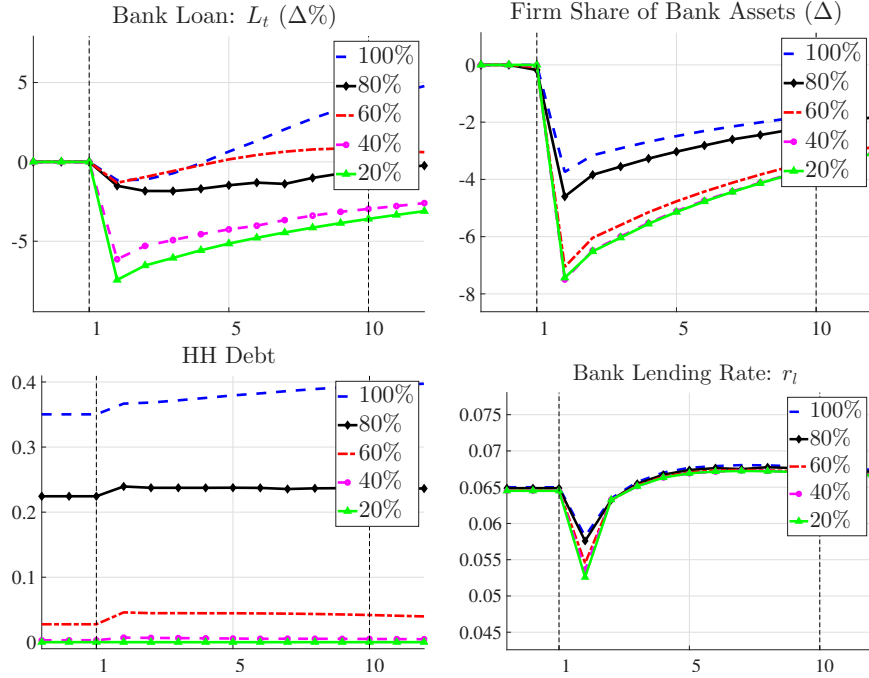
Interest rates move consistently with this demand-driven adjustment. In the tight economy, weaker loan demand pushes the bank lending rate down by about 120 basis points, from 6.5 percent to 5.3 percent. In the loose economy, the lending rate falls by only 65 basis points. Because mortgages account for a larger share of bank assets under loose LTV regulation, the working-capital channel is less important, and stronger household entry at lower house prices partly offsets the fall in firm borrowing.

Household debt dynamics reinforce this divergence. In the loose economy, household debt rises during the crisis as falling prices open buying opportunities. In the tight economy, debt remains negligible, insulating household balance sheets from leverage dynamics but also limiting consumption smoothing through mortgage markets.

Overall, tighter LTV regulation changes financial transmission primarily through bank-portfolio composition rather than greater aggregate financial resilience. Banks in the tight economy avoid mortgage-related distress, but they are more exposed to fluctuations in firm credit demand.

Financial quantities therefore respond by more strongly, even though the underlying mechanism is lower loan demand rather than a contraction in credit supply.

FIGURE 6 – Financial Transmission and Bank Portfolio Dynamics



Notes: Transition dynamics following the benchmark productivity shock, across maximum LTV limits from 100 to 20 percent. Top row: total bank loans and loans to firms, as percentage deviations from steady state. Bottom row: household debt and the bank lending rate, in levels. The lending rate is annualized.

7 Why Do Tighter LTV Caps Stabilize So Little?

The transition results pose the puzzle this section explains: reducing the maximum LTV from 100 to 20 percent eliminates most household leverage and all downturn foreclosures, yet narrows the house-price trough by less than one percentage point. We locate the explanation in the interaction between household adjustment margins and general-equilibrium feedback, and we proceed in steps. Throughout, fragile prudence is the umbrella household mechanism, with an extensive margin (fewer households can enter or upgrade) and an intensive margin (the households that still transact commit more internal funds and cut other spending). We first document the mechanism at the household level: regulated buyers hold larger buffers, but the buyers who still adjust in a downturn must spend those buffers, so purchases become more asset dependent. We then decompose the market-level margins through which this operates—entry by renters, upgrading by owners, and distressed exit—and aggregate them: with wages and interest rates held fixed, the household margins nearly cancel, and much of the remaining cross-regime difference runs through bank portfolios and the working-capital wage channel. We close by examining the role of the bank funding closure and by asking whether a crisis-scale

shock overturns the verdict.

7.1 Fragile Prudence: The Asset-Reliance Channel

The mechanism at the center of the puzzle is the *asset-reliance channel*: when regulation tightens, households must finance purchases and moves with more internal funds. Buyers and movers therefore enter the downturn with larger buffers but draw them down more aggressively when they adjust. This is *fragile prudence*: regulation makes households safer in normal times while making housing adjustment more dependent on liquid-asset depletion in bad times. The channel is symmetric: in booms, rising prices revalue the collateral that finances down payments, loosening the constraint precisely when the boom builds (Appendix D.1). This subsection isolates the intensive margin; the next documents the extensive margin and the opposing balance-sheet-insurance margins.

The channel is visible at purchase. As Section 5 showed, tighter caps delay ownership and induce pre-purchase saving, so steady-state buyers have more equity. But they also finance a larger share of housing with liquid wealth rather than mortgage credit. The same regulation that raises *ex ante* buffers therefore makes downturn purchases and moves more cash-intensive.

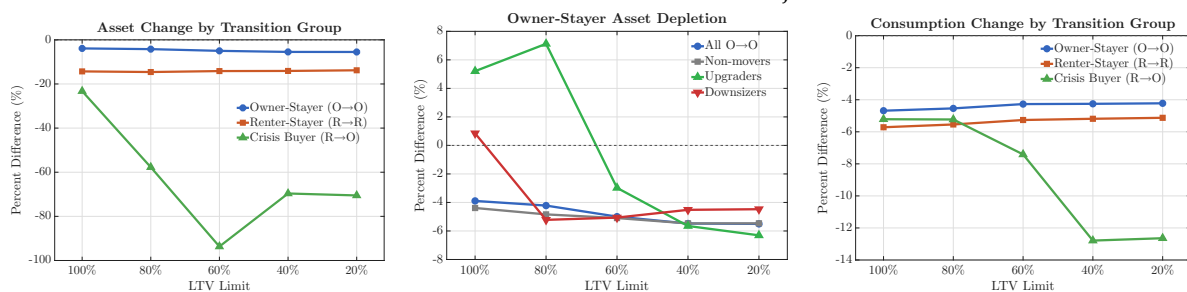
To measure this margin, we compare crisis-period changes in liquid assets and consumption to a no-crisis counterfactual for owner-stayers, renter-stayers, and crisis-induced buyers. The samples hold household identity fixed across LTV regimes: owner-stayers remain owners in both paths in all five regimes; renter-stayers and crisis buyers come from households that are steady-state renters in every regime.

Figure 7 shows the offset. Tighter caps modestly improve consumption outcomes for owner-stayers and renter-stayers, but force crisis-induced buyers to liquidate more wealth and compress consumption by more. Lower leverage helps incumbents; tighter down-payment constraints make the small group of crisis purchasers bear larger adjustment costs.

The first panel reports liquid-asset changes relative to the no-crisis path. Owner-stayer drawdowns rise only slightly, from about 4 percent under 100 percent LTV to about 6 percent under the tightest cap, while renter-stayer drawdowns remain around 14–15 percent. Crisis buyers are different: their liquid-asset decline deepens from about 23 percent under loose LTV to 94 percent at the 60 percent cap, before partially recovering to roughly 70 percent in the tightest regimes. This intensive-margin depletion occurs as crisis-buyer entry falls by about 80 percent.

The third panel gives the consumption counterpart. Owner-stayer consumption losses shrink from 4.7 to 4.3 percent as regulation tightens, and renter-stayer losses shrink from 5.7 to 5.2 percent. Crisis buyers again move the other way: their loss rises from 5.3 to about 12.7 percent. The households that clear the tighter down-payment constraint smooth housing adjustment by cutting non-housing consumption.

FIGURE 7 – Household Balance Sheet Adjustments



Notes: Balance-sheet adjustments at the household level under a persistent 5 percent negative productivity shock, expressed as the percentage difference between the crisis path and a no-crisis counterfactual, plotted against the LTV cap (x-axis ranges from 100 percent on the left to 20 percent on the right). Left panel: change in liquid (non-housing) assets for three transition groups: owner-stayers (owners in both crisis and no-crisis paths), renter-stayers (renters in both paths), and crisis-induced buyers (renters in the no-crisis path who buy during the crisis). Middle panel: liquid-asset changes for owner-stayer subgroups: non-movers, upgraders, and downsizers. Right panel: change in non-housing consumption for the three transition groups. More negative values indicate larger depletion under tighter regulation, the asset-reliance channel.

The second panel decomposes owner-stayers by mover status. Under loose caps, upgraders increase liquid assets by about 5–7 percent because they can exploit lower prices with mortgage finance. Once the cap tightens from 80 to 60 percent, the pattern reverses: upgrader liquid-asset changes fall from about +7 to -3 percent, and depletion grows as the cap tightens further. At that point, upgrading becomes increasingly self-financed.

Non-movers reduce liquid assets by about 5 percent even under full leverage, with slightly larger drawdowns under tighter caps. Downsizers move from a small gain under 100 percent LTV to drawdowns of about 4–5 percent in tighter regimes, because the replacement-home down payment absorbs funds that downsizing would otherwise release.

The asset and consumption responses diverge for movers (not shown separately). Upgrader consumption remains roughly flat across regimes, at 4.4–4.8 percent below the no-crisis path, because larger asset drawdowns preserve consumption. Non-movers and aggregate owner-stayers show modest consumption gains as regulation tightens.

These micro patterns explain the small aggregate consumption effect. Tighter LTV regulation improves the aggregate consumption response by only 0.4 percentage points, from -5.4 to -5.0 percent, despite reducing debt-to-GDP from about 53 percent to nearly zero. Incumbent owners and renter-stayers gain about 0.4 and 0.5 percentage points, respectively, but crisis buyers' consumption loss more than doubles, from 5.3 to about 12.7 percent. Regulation matters, but its gains for incumbents are partly offset by losses for marginal entrants.

The empirical appendix asks whether the purchase-financing margin appears in household data, without claiming to estimate an LTV-policy effect. In the PSID, among matched pairs of stockholders, buyers who remain stockholders through the purchase window fall roughly 54 percent behind matched renter-stayers in stock growth; this log estimate excludes zero holdings

and is scale invariant. Separately, four in ten initial stockholder buyers exit direct stockholding altogether. The appendix reports the supporting design, inference, and robustness.

The SCF supplies further evidence. Using the Federal Reserve’s replicate-weight procedure, recent mortgaged buyers are 2.57 percentage points less likely to hold stocks and 2.64 points less likely to hold retirement assets than comparable long-time buyers. Among positive holders, their stock and retirement balances are about 24 and 26 percent lower, while their deposit difference is small and insignificant, consistent with deposits’ role as the transaction vehicle around closing. The 2023 AHS shows that substantial down payments are widespread well beyond these survey subsamples. Together, these facts point to an internal-funds margin at purchase; the structural model supplies its response to an LTV cap.

The downturn also lowers the return on the buffers themselves. The equilibrium return r , which is both the return on productive capital and the return on household financial wealth, falls with the shock, from about 10.3 to 7.4 percent per two-year model period (roughly 5.0 to 3.6 percent at annual rates). The fall is common to both regimes, so it shapes their relative performance only through the balance sheets and financing requirements that the LTV cap induces. For a given predetermined asset stock, the lower realized return reduces cash on hand, $a(1+r)$, and with it down-payment capacity; at the same time, the lower prospective return reduces the payoff to carrying liquid wealth forward, which pushes the households that can still buy or move to spend down their buffers. The first is an income effect and the second a substitution effect, and they need not move housing demand in the same direction; lower r also raises the discounted value of future housing services, supporting house prices. Their common implication across regimes is selection: wealth-poor households stay out of the market, while the smaller group that still transacts relies more on internal funds and, under tighter caps, holds more of the assets whose return has fallen.

Overall, tighter caps produce financially stronger buyers in steady state but make downturn adjustment more cash-intensive. Crisis buyers nearly exhaust liquid assets and cut consumption; upgraders preserve consumption by running down wealth. Better *ex ante* balance sheets thus coexist with greater *ex post* liquid-asset depletion, leaving aggregate capital and wage dynamics only modestly more stable.

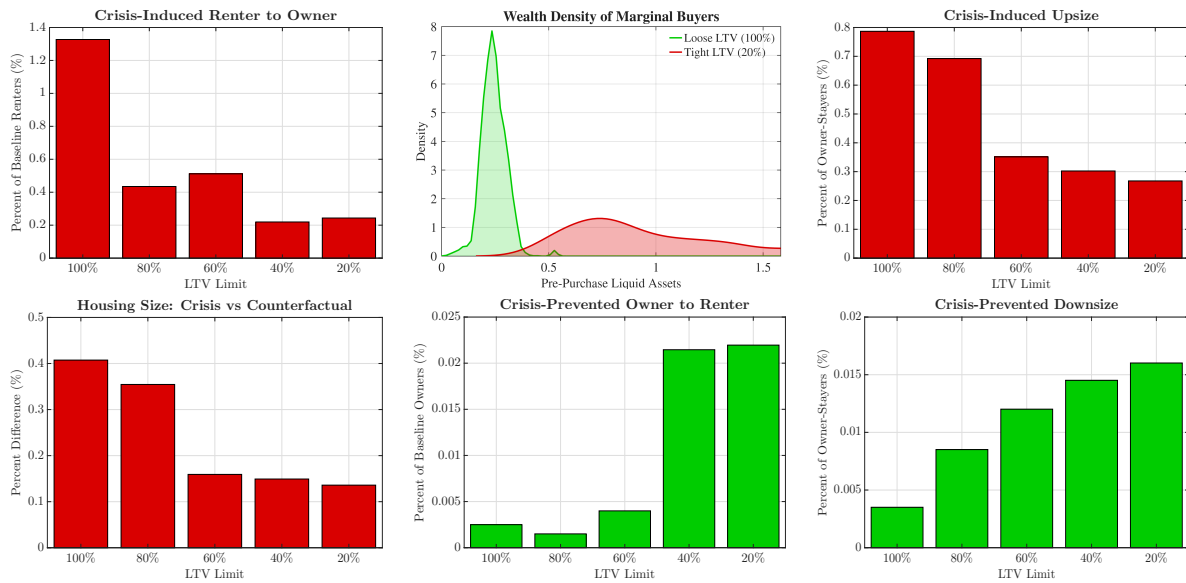
7.2 Marginal-Buyer and Balance-Sheet-Insurance Channels

The previous subsection traced the intensive asset-reliance margin among households that still transact. This subsection turns to the extensive margins. We first examine the marginal-buyer channels, renter entry and owner upgrading, through which falling prices normally raise housing demand. We then examine the opposing balance-sheet-insurance margins, fewer distressed exits and less downsizing, that support incumbent ownership and reduce the housing

supplied under stress. The following counterfactuals ask how these offsetting household margins aggregate once general-equilibrium prices and bank intermediation are restored.

Renter Entry Tighter LTV regulation stabilizes post-shock house prices less than the usual balance-sheet rationale suggests, because that rationale omits the demand response. Safer borrowers and fewer forced sales reduce distressed supply, but tighter caps also reduce crisis-induced renter entry and owner upgrading. We measure these margins by comparing realized one-period-ahead transitions in the crisis economy with a composition-controlled no-crisis counterfactual. The exercise separates the reduction in distress-driven supply from the weakening of buyer demand.⁷

FIGURE 8 – Household Margins after the Shock: Entry, Upgrading, and Retention



Notes: Composition-controlled household margins one period after the shock, by maximum LTV limit (100 to 20 percent), each measured against the regime's own no-crisis counterfactual. Panel (1): crisis-induced renter-to-owner entry, as a percent of baseline renters. Panel (2): kernel density of pre-purchase liquid assets of crisis-induced entrants. Panel (3): crisis-induced upsizing, as a percent of owner-stayers. Panel (4): average housing-size change of owner-stayers, crisis versus counterfactual, in percent. Panel (5): crisis-prevented owner-to-renter exits, as a percent of baseline owners. Panel (6): crisis-prevented downsizing, as a percent of owner-stayers.

Figure 8 collects the main household margins in one exhibit; its first panel reports crisis-induced entry: renters who would have remained renters absent the downturn but buy when prices fall. This margin is 1.3 percent of the baseline renter population under 100 percent LTV and only 0.25 percent under the 20 percent cap. Lower prices thus generate less new demand when renters must finance larger down payments. Crisis-prevented entry also declines, from about 0.4 to 0.15 percent (not shown), because tight-LTV economies have fewer marginal would-be buyers. Quantitatively, the dominant effect is the decline in opportunistic entry.

⁷These margins echo classic housing-market models in which down-payment constraints affect transaction volume and price dynamics, including [Stein \(1995\)](#) and [Ortalo-Magné and Rady \(2006\)](#), where shocks propagate through trade-up dynamics across housing-market segments.

Figure 8 shows in its second panel which renters enter. Under loose caps, crisis-induced buyers are young and have little liquid wealth; mortgage finance lets them substitute future income for current savings. Under tight caps, this group almost disappears. The wealth distribution of entrants shifts right, so only older and richer renters can buy. Their numbers are too small to generate the demand needed to stabilize prices.

Tighter caps therefore weaken a stabilizing force that operates through the entry margin. Under loose regulation, falling prices attract marginal renters into ownership; under tight regulation, those renters remain out because they cannot finance the required down payment.

Owner Upgrading The same demand mechanism operates among incumbent owners. Even when households remain owners in both the crisis and no-crisis paths, tighter caps can prevent them from upgrading when lower prices make larger homes more affordable.

Figure 8 turns to continuous owners in its third and fourth panels. The third panel reports crisis-induced upgrading: owners who buy a larger home in the downturn but would not have upgraded in the no-crisis path. This margin falls from about 0.8 percent of owner-stayers under loose LTV to about 0.27 percent under the tightest cap. The mechanism mirrors the entry channel: upgrading requires a larger purchase and mortgage, but tight caps force households to finance more of the transaction with internal funds, often when the crisis has already reduced current-home equity.

The fourth panel shows the implication for housing consumption. Continuous owners increase housing size by about 0.4 percent in the loose-LTV economy but only about 0.13 percent in the tight-LTV economy. Lower prices therefore translate into realized housing-demand gains mainly when mortgage finance can support the move.

Entry and upgrading show the same pattern: tighter caps reduce leverage and foreclosure risk, but also mute the demand response to falling prices. Housing reallocation becomes more dependent on internal funds, so marginal renters and incumbent owners are less able to absorb housing when prices decline.

Balance-Sheet Insurance: What Does Tight Regulation Deliver? Any assessment must weigh the demand-side costs of tighter caps against the supply-side benefits. Larger equity stakes reduce negative equity, foreclosures, forced sales, and other forms of distressed supply. Figure 8 focuses on two composition-controlled retention margins, in addition to the foreclosure margin documented above.

Figure 8 shows in its final two panels that these supply-side gains are real but small. Foreclosures rise under loose regulation and are essentially absent under tight regulation. The fifth panel adds crisis-prevented owner-to-renter transitions: owners who would have rented in the no-crisis path but remain owners during the crisis. This margin rises from roughly 0.002 percent of

baseline owners under loose regulation to 0.022 percent under the 20 percent cap, small relative to the entry margins in Figure 8. The sixth panel shows crisis-prevented downsizing, which rises from about 0.0035 to 0.016 percent of owner-stayers. This supports prices by keeping larger homes off the market and sustaining incumbent housing demand.

Why does an origination cap bind more tightly when house prices fall, given that the cap scales with the price and the required down payment on any given house therefore declines? Because the financing side deteriorates by more: income, financial wealth, and, for movers, the sale equity that funds the next down payment all fall with the same downturn that lowers prices. The evidence above shows the net effect: crisis-induced entry falls by about 80 percent under tight caps, the entrants who remain come from further right in the wealth distribution, and the buyers who still transact draw down more assets. What binds is the ratio of required cash at closing to available internal funds, and in the bust the numerator falls by less than the denominator.

The household margins therefore offset. Tighter caps reduce crisis-induced entry and upgrading, but also reduce foreclosures, raise owner retention, and limit downsizing. Section 7.3 quantifies the net aggregate effect of these offsetting margins.

7.3 Do Household-Level Channels Stabilize the Economy?

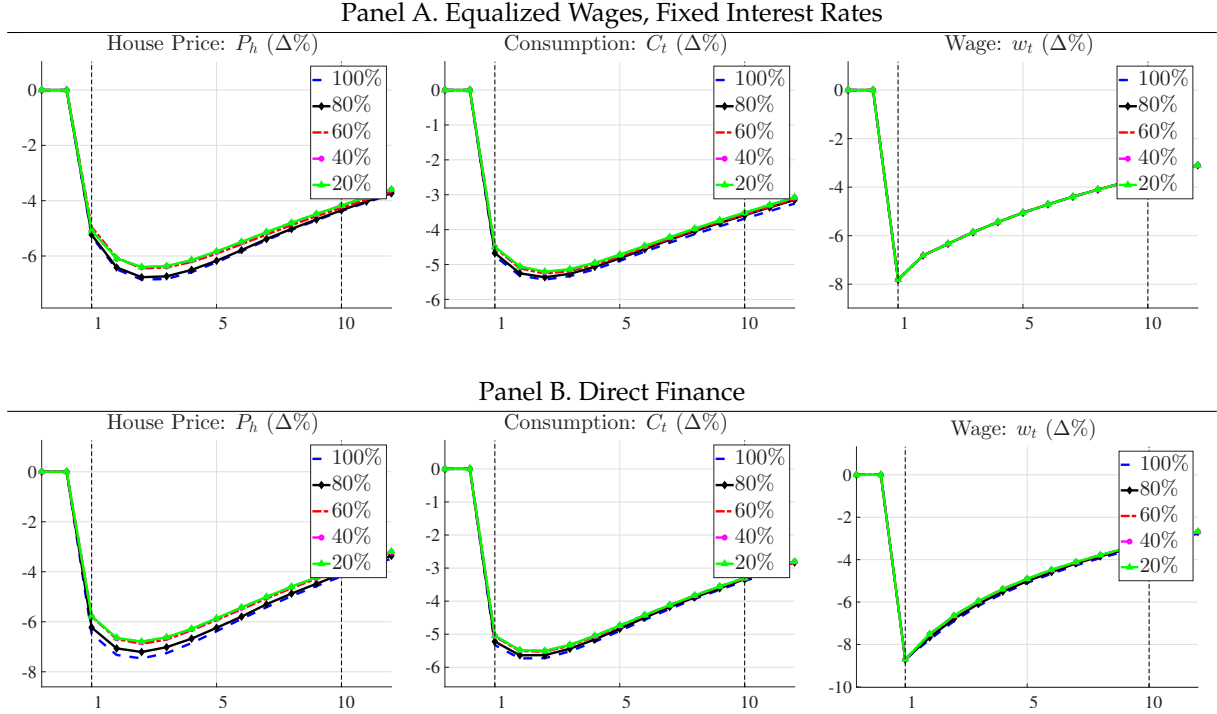
The preceding decomposition shows offsetting household-level forces. To isolate household channels, we impose the loose-LTV economy's wage path on every regime and hold all interest rates at their steady-state values. House prices and household choices remain endogenous. Differences across regimes then reflect only the household mechanisms documented above, not feedback through capital, wages, or endogenous bank intermediation.

Figure 9 reports the equalized-wage experiment in Panel A. By construction, wage dynamics are identical across regimes. House-price and consumption dynamics also become similar: trough house prices fall 6.9 percent in the loose economy and 6.4 percent in the tightest economy, a 0.45 percentage-point gap (computed from unrounded paths) versus 0.80 percentage points in the full-GE benchmark. Equalizing wages therefore removes about half of the cross-regime house-price difference.

Because all household mechanisms remain active, the result implies that they largely cancel. The tight-LTV economy still has stronger balance sheets and virtually no foreclosures; the loose-LTV economy still has more crisis-induced entry and housing-market flexibility. With income held fixed, these forces generate little house-price separation.

Consumption tells the same story. The trough decline is 5.4 percent in the loose economy and 5.2 percent in the tightest economy, a 0.23 percentage-point gap versus 0.47 percentage points in full GE. Lower debt service in the tight economy is partly offset by greater reliance on liquid assets during the downturn.

FIGURE 9 – Counterfactual Transitions: Equalized Wages and Direct Finance



Notes: Impulse responses to the benchmark negative productivity shock in two counterfactual economies, across maximum LTV limits from 100 to 20 percent. Panel A: all interest rates held at steady-state levels and all regimes receiving the wage path of the loose (100 percent LTV) economy. Panel B: direct-finance counterfactual, with mortgages funded through zero-profit originators at the market rate and firm working capital supplied without an endogenous intermediation spread; endogenous wages and interest rates restored. Panels show house prices, consumption, and wages as percentage deviations from steady state. The narrowing of the loose-tight house-price wedge relative to the full model underlies the decomposition in the text.

Thus the direct household channels provide limited net stabilization once we fix wages and rates. The additional stabilization in the full model must come mainly from indirect general-equilibrium channels, especially wages and financial returns.

7.4 The Role of Bank Intermediation

After equalizing wages and rates, the house-price gap falls from 0.80 to 0.45 percentage points. The remaining full-GE difference can come from production-side effects through capital and factor prices, or from financial effects through endogenous bank intermediation.

To isolate the contribution of endogenous bank intermediation, we use a direct-finance counterfactual. We remove the bank capital constraint, fund mortgages through zero-profit originators at the market rate, and supply firm working capital without an endogenous intermediation spread. This exercise removes the entire bank-intermediation block, both the balance-sheet constraint and endogenous spread setting, rather than only a pure bank-net-worth channel. The difference from the full model therefore measures the contribution of bank intermediation as a whole.

Figure 9 reports the direct-finance transition in Panel B. House prices fall by 7.5 percent in the loose economy and 6.8 percent in the tight economy, a 0.67 percentage-point gap. The direct-finance counterfactual therefore reduces the full-GE gap by about 0.13 percentage points, roughly one-sixth, while leaving most of the house-price difference intact.

Capital dynamics are similar: the cross-regime gap remains about 0.3 percentage points. Consumption stabilization falls more, from 0.47 to 0.23 percentage points. The key change is wages: the trough decline is about 8.7 percent under both LTV regimes, so the wage gap falls from 0.54 percentage points in full GE to essentially zero. Removing endogenous bank intermediation eliminates the wage advantage of tight regulation while preserving much of the house-price and capital gap.

The wage result identifies the working-capital channel. In the full model, firms finance a fraction μ of the wage bill at the bank lending rate r_ℓ , which includes an endogenous spread. When productivity falls, firm credit demand contracts. Because tight-LTV banks hold relatively more working-capital loans and fewer mortgages, the lending rate falls more in the tight economy, about 120 basis points versus 65 in the loose economy. This larger decline in the lending rate lowers the effective labor cost, $(1 + \mu r_\ell)w$, and supports wages. Under direct finance, working-capital costs move with the common market rate, so wage dynamics converge.

Table IV summarizes this ordered decomposition. The equalized-wage experiment measures direct household channels: lower leverage and default, but also weaker entry, reduced upgrading, and greater internal-fund reliance. The direct-finance experiment adds non-bank GE channels from wages and rates while excluding endogenous bank intermediation. The residual between the full model and the direct-finance economy is the bank-intermediation block.

Along this ordering, the household-focused step leaves a 0.45 percentage-point house-price gap and a 0.23 percentage-point consumption gap over the full 100-to-20 percent LTV comparison. Restoring non-bank general-equilibrium adjustment adds 0.22 percentage points to the house-price gap. Restoring endogenous bank intermediation adds a further 0.13 percentage points to the house-price gap, 0.24 to the consumption gap, and the full 0.54-point wage gap. These increments describe this sequence of counterfactuals rather than an order-invariant attribution. The stability of the tight regime reflects the joint response of household margins, capital accumulation, factor prices, and bank portfolio composition.

Two implications follow. First, the household-balance-sheet narrative is incomplete: lower leverage and fewer defaults are real microprudential gains, but they do not deliver robust house-price stabilization once entry and upgrading are endogenous. Second, endogenous bank intermediation matters for real transmission through the lending rate and the working-capital wage channel, even though its direct contribution to the house-price gap is modest. Overall, borrower-based regulation provides microprudential insurance, but against productivity shocks

TABLE IV – An Ordered Decomposition of the Stabilization Benefit of Tighter LTV Limits

	<i>House Price</i> (trough, pp)	<i>Consumption</i> (trough, pp)	<i>Wage</i> (trough, pp)
(a) Household-level channels (same wages, fixed rates)	0.45	0.23	0.00
(b) Non-bank GE channels (direct finance – same wages)	+0.22	+0.00	+0.00
(c) Bank-intermediation block (full GE – direct finance)	+0.13	+0.24	+0.54
Full GE gap (100% – 20%)	0.80	0.47	0.54
Direct-finance gap (100% – 20%)	0.67	0.23	0.00

Notes: Each entry is the difference in the absolute magnitude of the trough decline between the loose (100 percent LTV) and tight (20 percent LTV) economies, $|x_{100}| - |x_{20}|$; a positive value means the tighter regime has a smaller decline. The sequence is path dependent: row (a) first imposes the loose economy's wage path and fixed rates, row (b) restores non-bank equilibrium prices, and row (c) restores endogenous bank intermediation. The entries are therefore ordered diagnostics, not a Shapley allocation, and need not be invariant to a different ordering. Row (a) is the cross-regime gap when all economies face the same wage path and fixed interest rates (Section 7.3). Row (b) is the increment from allowing wages and interest rates to adjust endogenously in the direct-finance economy (this section minus Section 7.3). Row (c) is the increment from restoring endogenous bank intermediation (full GE minus the direct-finance economy). Positive values indicate that a channel widens the stabilization advantage of tight LTV; negative values indicate it narrows it.

its aggregate stabilization is limited and channel-dependent. Section 8.1 returns to this dependence on bank capitalization and its implication for the assignment of macroprudential instruments.

7.5 Externally Priced Marginal Bank Funding

The benchmark productivity-shock results use domestic marginal bank funding. Let r_t^B denote banks' marginal liability cost and let $s_t \equiv r_t^\ell - r_t^B$ denote the intermediation spread, so $r_t^\ell = r_t^B + s_t$. In the domestic benchmark, $r_t^B = r_t + \varepsilon_t$ with $\varepsilon_t = 0$, where r_t is the return earned on household savings. After the productivity shock, weaker demand for capital and firm credit lowers r_t^B , helping the lending rate remain low. This endogenous rate cushion supports affordability and wages and partly offsets fragile prudence.

Appendix C considers a counterfactual in which $r_t^B = \bar{r}^B$ at its steady-state level; equivalently, $\varepsilon_t = \bar{r}^B - r_t$. Wages, house prices, rents, the return on capital, the intermediation spread, and the lending rate remain endogenous. The exercise is therefore a general-equilibrium closure with externally priced marginal bank funding, not a complete small-open-economy model: households and firms do not borrow or save directly at the external rate.⁸

Removing the domestic funding-rate cushion reverses the ranking. At the trough, house prices fall 6.5 percent under tight LTV and 5.2 percent under loose LTV; non-housing consumption

⁸As the market clearing condition for the asset market changes, we recalibrate the model to match the same moments,

falls 5.4 and 5.0 percent, respectively. Tight caps still prevent foreclosures, but the lending-rate decline is brief and small. The tight-LTV economy therefore retains the demand-suppression and asset-reliance channels while losing the sustained rate cushion; the loose-LTV economy retains leveraged entry by marginal buyers. The result strengthens rather than drives the conclusion: fixed LTV caps are not robust aggregate stabilizers and may deepen house-price and consumption declines when banks' marginal funding cost is insensitive to domestic activity.

7.6 Larger Shocks

Sections 7.1 and 7.2 showed that, under the benchmark 5 percent productivity shock, tighter LTV regulation delivers only modest aggregate stabilization: the trough house-price gap between loose and tight economies is 0.80 percentage points. A key reason is that foreclosures remain rare even in the loose economy. The peak foreclosure rate is only 0.43 percent, so the supply-side benefits of tighter regulation never become quantitatively large. A natural question is whether this changes in a crisis-scale downturn, when defaults and distressed sales become widespread.

To address this question, we consider a 15 percent productivity shock, three times as large as the benchmark, and compare the loose economy with a 100 percent LTV limit to a tighter economy with an 80 percent cap. We focus on the 80 percent regime because it corresponds to a commonly observed policy threshold. Table V reports trough responses for house prices, consumption, wages, and bank net worth, together with peak foreclosure rates, under both the benchmark and crisis scenarios.

The larger shock activates a much stronger foreclosure channel. Under the benchmark recession, foreclosures in the loose economy peak at 0.43 percent, and the 80 percent LTV cap eliminates them. Under the crisis-scale shock, foreclosures in the loose economy peak at 14.9 percent, compared with 0.03 percent in the tighter regime. The gains for intermediary solvency are also large: bank net worth falls by 28.5 percent at its trough in the loose economy and by 9.5 percent in the tighter one. On these margins, tighter LTV regulation provides substantial microprudential protection, and that protection grows with shock severity.

The persistent aggregate stabilization effect nevertheless remains small. Tightening LTV from 100 to 80 percent improves the house-price trough by only 0.3 percentage points, from 19.0 to 18.7 percent, and the consumption trough by 0.2 points, from 15.4 to 15.2 percent. The wage trough improves by 0.7 points. Thus, a 14.9-point reduction in peak foreclosures and a 19.0-point improvement in the bank-net-worth trough translate into little additional stabilization of house prices or consumption at their troughs.

Part of the explanation is the demand-suppression mechanism emphasized earlier. Two further pieces complete it: the timing of the foreclosure channel, which the default counterfactual below isolates, and the return-on-assets channel from the benchmark analysis, which the larger shock deepens.

TABLE V – Trough and Peak Responses under Benchmark and Crisis-Scale Shocks

	<i>Loose (no cap)</i>	<i>Tight (80% LTV)</i>	<i>Difference</i>
Panel A: Benchmark Recession (5 percent productivity shock)			
House Price (trough, %)	−6.9	−6.7	0.2
Consumption (trough, %)	−5.4	−5.3	0.1
Wage (trough, %)	−7.8	−7.8	0.05
Foreclosure Rate (peak, %)	0.43	≈ 0	0.43
Bank Net Worth (trough, %)	−2.8	−2.4	0.4
Panel B: Crisis Scenario (15 percent productivity shock)			
House Price (trough, %)	−19.0	−18.7	0.3
Consumption (trough, %)	−15.4	−15.2	0.2
Wage (trough, %)	−23.4	−22.6	0.7
Foreclosure Rate (peak, %)	14.9	0.03	14.9
Bank Net Worth (trough, %)	−28.5	−9.5	19.0

Notes: Each row reports the most extreme deviation from steady state over the transition path. Difference entries are gaps in the absolute magnitude of the decline between the loose and tight regimes; a positive value means the tighter regime has a smaller decline. We compute differences from unrounded transition paths, so they may not exactly equal differences of the rounded entries in the table. House prices, consumption, wages, and bank net worth are trough (minimum) percentage deviations from steady state; the foreclosure rate is the peak (maximum) level. The benchmark recession uses the same 5 percent productivity shock as in Section 6. The crisis scenario applies a 15 percent productivity shock. The loose economy imposes no binding LTV limit (100 percent); the tight economy imposes an 80 percent LTV cap.

The homeownership dynamics illustrate the interaction between supply and demand. In the loose economy, the crisis generates a two-stage adjustment. First, foreclosures displace a large mass of owners into renting, pushing the homeownership rate well below its steady-state level. Second, lower house prices induce new leveraged buyers to enter, and this demand response eventually more than offsets the initial displacement. Under the 80 percent cap, the foreclosure wave disappears, but so does much of the entry response. Because down-payment constraints remain binding and returns on accumulated savings are low, the market does not generate enough new demand to absorb the price decline.

The crisis case therefore sharpens rather than overturns the main message. Tighter LTV regulation is effective microprudential insurance: it prevents defaults, protects incumbent owners, and preserves intermediary net worth. Its aggregate effects remain limited because it also suppresses the demand margins through which housing markets adjust.

Isolating the Default Channel To separate foreclosure prevention from other mechanisms, we re-run the 15 percent productivity shock in a counterfactual with the mortgage default channel closed.⁹ Because foreclosures are already essentially absent under the 80 percent LTV cap, most non-bank responses in the tight economy change little; the experiment isolates default in the

⁹Households cannot default, so no foreclosures occur and banks incur no default-related losses.

loose economy. It also shows that default affects aggregate dynamics mainly on impact, a pattern that trough-only comparisons miss.

TABLE VI – Crisis-Scale Shock with and without the Default Channel

	<i>Default channel active</i>			<i>Default channel closed</i>		
	<i>No cap</i>	<i>80% LTV</i>	<i>Diff</i>	<i>No cap</i>	<i>80% LTV</i>	<i>Diff</i>
House Price, impact (%)	−16.9	−14.6	2.3	−15.1	−14.6	0.4
House Price, trough (%)	−19.0	−18.7	0.3	−19.2	−18.7	0.4
Consumption, impact (%)	−14.1	−13.5	0.6	−13.8	−13.4	0.4
Consumption, trough (%)	−15.4	−15.2	0.2	−15.6	−15.2	0.4
Wage, trough (%)	−23.4	−22.6	0.7	−22.8	−22.5	0.3
Bank Net Worth, trough (%)	−28.5	−9.5	19.0	−5.9	−8.3	−2.4

Notes: Trough and impact percentage deviations from steady state under the 15 percent productivity shock, for the loose (100 percent LTV) and tight (80 percent LTV) economies. The left panel reports the full model with the default channel active; the right panel reports a counterfactual in which households cannot default on their mortgages. The “Diff” columns report the gap in the magnitude of the decline between the loose and the tight regimes; a positive value indicates that the tight economy stabilizes more. We compute differences from unrounded transition paths, so they may not exactly equal differences of the rounded entries in the table. The foreclosure rate under the default-closed experiment is zero in both regimes by construction.

Table VI reveals two features that a trough-only comparison would miss. First, default prevention is most visible on impact. With default active, the house-price impact gap between the loose and tight regimes is 2.3 percentage points; closing default reduces it to 0.4 points. Foreclosure prevention therefore accounts for about 1.9 points of the impact gap. The corresponding contribution to the consumption impact gap is about 0.2 points, and to the wage trough gap about 0.4 points. At the house-price and consumption troughs, however, the active-default gaps are only 0.3 and 0.2 points, no larger than the 0.4-point gaps when we shut down default. Default prevention is therefore an important impact stabilizer but does not produce a larger persistent trough advantage in these outcomes.

Second, the bank-net-worth ranking reverses when we shut down default. With default active, the 80 percent cap reduces the trough loss in bank net worth from 28.5 to 9.5 percent, an improvement of 19.0 percentage points. Without mortgage default, net worth falls by 5.9 percent in the loose economy and by 8.3 percent in the tight economy, so the tight regime performs 2.4 points worse. The crisis-scale bank-resilience advantage of tighter LTV regulation therefore comes from avoiding mortgage-default losses; absent those losses, portfolio composition and working-capital stress favor the loose regime at the trough.

The crisis-scale shock magnifies the channels discussed in the benchmark analysis: demand suppression, asset reliance, foreclosures, and bank net-worth losses all scale up. While the larger shock disproportionately increases foreclosures, the foreclosure wave’s aggregate effect is transitory: it moves house prices on impact, but the loose-LTV economy replaces foreclosed owners quickly, because low down-payment requirements let new leveraged buyers enter as

prices fall, while the tight-LTV economy avoids the wave yet suppresses that entry. This is why the trough gap stays small. The return-on-assets channel from the benchmark analysis also deepens: the equilibrium return falls on impact to 2.1 percent per two-year model period (about 1.0 percent annualized), against 7.4 percent (3.6 percent annualized) under the benchmark shock, cutting cash on hand and the payoff to holding buffers precisely when down payments must come from internal funds.

Together, the experiments separate microprudential success from persistent aggregate stabilization. The tighter cap nearly eliminates a crisis foreclosure wave and protects bank net worth, yet improves the house-price and consumption troughs by only 0.3 and 0.2 percentage points. The default channel explains much of the impact response, while the asset-reliance and entry margins offset its longer-run support for housing demand.

8 When LTV Regulation Stabilizes: Bank Funding-Cost Shocks

The previous sections studied a real-side productivity shock with a direct household-income effect. Fixed LTV limits stabilize that productivity-driven downturn only modestly: they lower leverage and strengthen household balance sheets, but also suppress the housing-demand margins that support prices when income and fundamentals weaken. This section turns to the complementary shock, one that stresses the banking sector rather than household income directly. When funding costs rise against banks' fixed-rate mortgage assets, tight LTV caps look more stabilizing because they insulate banks from duration risk and therefore support bank net worth and house prices. But the same insulation attenuates the pass-through of rate changes to credit spreads and asset prices. Stabilization under bank funding-cost shocks is therefore a banking-transmission result, not an unqualified policy success. Appendix D collects the underlying robustness exercises; Section 8.1 draws out the main implication below.

We use a bank funding-cost shock to isolate a banking-sector source of instability. Let r_t^B denote banks' marginal liability cost and write $r_t^B = r_t + \varepsilon_t$, where r_t is the return earned on household savings. Let $s_t \equiv r_t^\ell - r_t^B$, so $r_t^\ell = r_t^B + s_t$. The benchmark sets $\varepsilon_t = 0$. The experiment raises the wedge by 1.5 percentage points on impact, moving r_t^B from 5 to 6.5 percent, after which ε_t returns to zero with annual persistence 0.75. The wedge is not paid to households: it is an iceberg liability-side cost borne by banks, leaving the household portfolio problem and asset-market clearing unchanged. Agents have perfect foresight after the initial realization. Unlike the productivity shock, this experiment does not directly lower the marginal product of capital or household labor income; its first-round effects operate through bank funding, mortgage valuation, and credit supply.

Two features drive the transmission of bank funding-cost shocks. First, mortgages are long-term fixed-rate contracts, so payments by existing borrowers do not adjust when market rates rise.

Second, banks hold these fixed-rate mortgages while funding themselves with shorter-duration deposits whose cost reprices immediately. A rate increase therefore opens a duration mismatch: fixed-rate mortgage values fall while funding costs rise one-for-one, reducing bank net worth.

Lower bank net worth tightens the enforcement constraint and contracts lending. As credit supply tightens, the equilibrium lending rate r_ℓ rises. In perfect foresight equilibrium, mortgages must earn the same return as other bank assets, so the increase in r_ℓ lowers mortgage prices. Because mortgages serve as bank collateral, lower mortgage valuations reduce lending capacity, which pushes lending rates up further. The result is a feedback loop between lending rates, mortgage prices, and bank net worth.

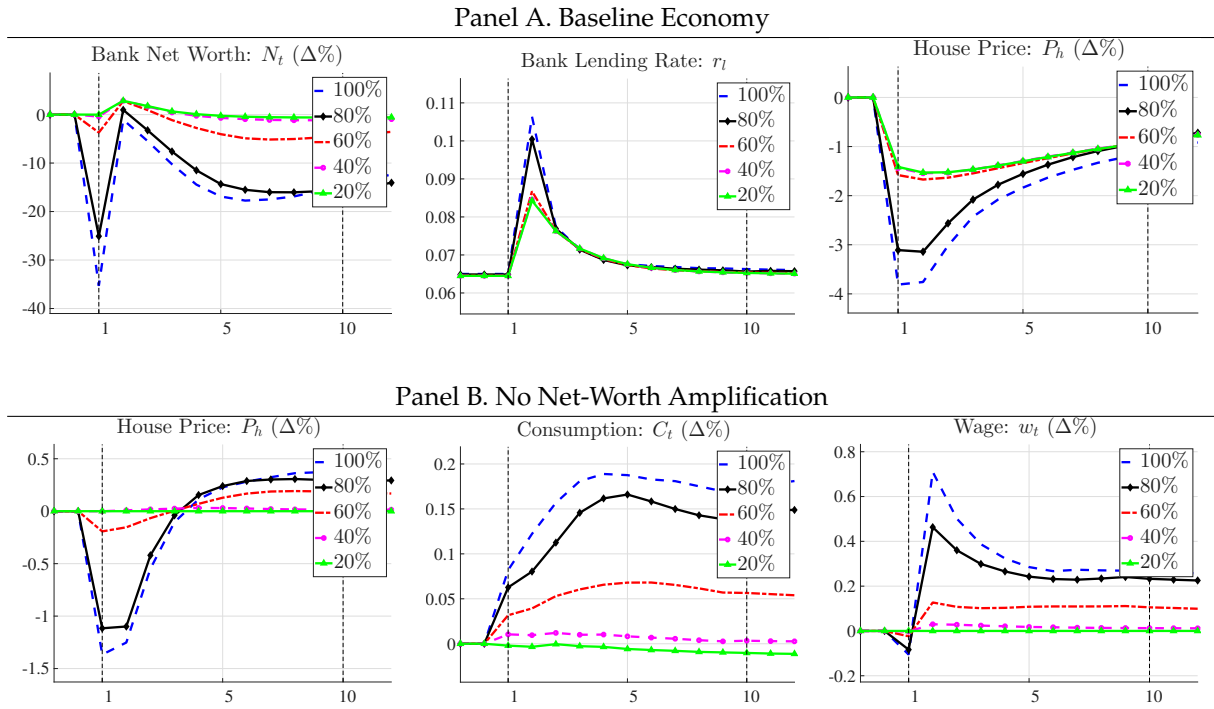
These financial effects reach both households and firms. Higher lending rates reduce housing affordability and discourage entry into homeownership; falling house prices also erode the equity of highly leveraged households, raise default risk, and feed additional losses back to banks. On the production side, higher working-capital borrowing costs increase the effective cost of labor, $(1 + \mu r_\ell)w$, reducing labor demand and wages. Lower wages then depress household income, consumption, and housing demand.

Tighter LTV regulation matters more in this environment because it directly reduces banks' exposure to duration risk. Under loose regulation, banks hold a large stock of fixed-rate mortgages, so the duration mismatch and the amplification mechanism are at their strongest. Under tight regulation, mortgage lending is smaller and bank assets concentrate in shorter-duration working-capital loans that reprice with market conditions. Bank net worth is therefore more insulated from the bank funding-cost shock, and the amplification loop weakens.

This mechanism differs from the productivity-shock case because the source of stress is different. When productivity falls, the shock directly lowers household income and generates a purchase response to lower house prices that tight LTV limits suppress by constraining marginal buyers. With a bank funding-cost shock, the first-round stress is on banks: funding costs rise without the same deterioration in productive fundamentals or household labor income. The tight-LTV economy also has more financial wealth and less mortgage debt, a more resilient portfolio in this environment. The asset-reliance and marginal-buyer channels that were central under productivity shocks are therefore less important here.

Figure 10 shows in Panel A that the bank funding-cost shock generates a pronounced bank-driven amplification in the loose economy. Under a 100 percent LTV limit, bank net worth falls by 35.3 percent on impact. Bank loans fall by 12.0 percent initially and reach a trough of about 16.5 percent after 6 periods. The lending rate rises to about 10.6 percent, amplifying the decline in housing demand and economic activity. On impact, consumption falls by 2.0 percent, wages by 5.2 percent, and house prices by 3.81 percent.

FIGURE 10 – Bank Funding-Cost Shock: Baseline and No Net-Worth Amplification



Notes: Transition dynamics following a persistent increase in banks' marginal funding rate (steady-state value 5 percent; wedge of 1.5 percentage points on impact, annual persistence 0.75), across maximum LTV limits from 100 to 20 percent. Panel A, baseline economy: bank net worth (percentage deviation from steady state), the bank lending rate (level), and the house price (percentage deviation). Panel B, no-net-worth-amplification counterfactual (funding-rate changes pass through one-for-one to the lending rate): house price, consumption, and wage (percentage deviations). The text reports bank loans and the remaining baseline responses.

Tighter LTV limits attenuate these responses. Under the 20 percent LTV regime, house prices fall only 1.41 percent on impact, compared with 3.81 percent in the loose economy. Consumption falls by 1.13 percent, roughly 44 percent less than under the 100 percent LTV regime. The banking response is nearly shut down: bank net worth declines by only 0.02 percent on impact, bank loans fall by 2.61 percent, and the lending rate peaks at 8.4 percent rather than 10.6 percent.

The mechanism follows from the steady-state portfolio composition induced by tighter regulation. Smaller long-duration mortgage books generate smaller valuation losses, preserve bank net worth, and prevent the feedback loop between asset prices, lending capacity, and borrowing costs from taking hold. The same portfolio reallocation that shapes financial transmission under productivity shocks becomes a stronger source of resilience here because it reduces banks' exposure to mortgage-duration risk. The external-validation discussion in Appendix B benchmarks these responses against identified interest-rate estimates.

Isolating Bank Net-Worth Amplification To isolate intermediary balance-sheet effects, we shut down the link from bank net worth and lending capacity to the lending rate. Changes in banks' marginal funding cost then pass through one-for-one to the bank lending rate. Transition

dynamics in this counterfactual therefore reflect mainly household-side responses to higher rates, rather than amplification through bank balance sheets.

Figure 10 reports the results in Panel B. The contrast with the baseline funding-cost-shock economy is sharp. When the banking channel is active, house prices in the loose-LTV economy fall by 3.81 percent on impact. When net-worth amplification is shut down, the same economy experiences a decline of only 1.37 percent on impact. Thus, roughly two-thirds of the initial house-price decline in the baseline arises from intermediary balance-sheet amplification rather than from the direct effect of higher discount rates alone.

Consumption dynamics also change once we remove net-worth amplification. In the loose-LTV economy without balance-sheet amplification, non-housing consumption does not fall on impact; it rises slightly and remains above steady state. Higher mortgage rates delay home purchases and shift expenditure away from housing services toward non-housing consumption. Weaker housing demand also supports saving and capital accumulation, raising wages, while lower house prices reduce rents and ease expenditure pressure on renters. Because the return on capital does not fall as it does under a productivity shock, households with financial wealth have no reason to liquidate assets to exploit lower house prices. The household-side amplification channels that matter under productivity shocks are therefore less active in this environment.

The sign of transmission thus depends on whether the shock operates through bank net worth or through household portfolio substitution between housing and non-housing uses of funds. A related lesson appears in a different setting in [Hirano and Stiglitz \(2025\)](#), where credit and interest-rate changes reallocate savings between land and productive capital, so the same financial impulse can expand or contract the economy depending on which asset absorbs the funds.

Overall, this counterfactual confirms that stabilization under the funding-cost shock operates mainly through bank balance sheets. By reducing banks' exposure to long-duration fixed-rate mortgages, tighter regulation limits transmission to credit conditions, wages, and house prices. Because we hold household returns and the capital return fixed on impact, the experiment isolates a bank funding-cost channel rather than a complete monetary-policy shock.

Robustness Two exercises show that the bank-portfolio interpretation is not specific to contractionary shocks or to fixed-rate mortgage design. First, reversing the sign of the funding-cost shock yields the same qualitative mechanism in booms: tighter LTV limits damp credit-driven expansions, with the peak house-price increase falling from 5.5 percent under the loosest regime to about 1.5 percent under the tightest (Appendix D.1). Second, replacing fixed-rate with adjustable-rate mortgages changes the locus of transmission but not the cross-regime ranking. With adjustable rates, higher funding costs are passed through to existing borrowers, so adjustment operates through borrower cash flows and default rather than bank duration losses; the

contraction under loose regulation is correspondingly deeper, while tighter LTV limits continue to attenuate the response (Appendix D.5). These exercises indicate that the stabilization result reflects the reduction in exposure induced by tighter origination limits, while contract design determines whether the remaining interest-rate risk is borne primarily by banks or borrowers.

8.1 Are LTV Caps Needed for Stabilization When Banks Are Well Capitalized?

The funding-cost experiment locates the stabilization value of LTV caps in bank balance sheets, the margin that capital regulation targets directly, and the two instruments work on it differently. An LTV cap does not add capital to banks; it shrinks their exposure: a smaller book of long-duration fixed-rate mortgages, and smaller default losses when borrowers fail. Capital regulation instead adds loss-absorbing equity, although in general equilibrium tighter requirements also raise lending spreads and shrink mortgage credit, so both instruments can reduce the underlying exposure. Because both instruments target the same bank vulnerability, the incremental stabilization value of a borrower-based cap should fall as bank balance sheets become more resilient.

Two appendix experiments confirm this prediction and quantify it. Appendix D.2 fixes bank leverage at an exogenous limit—an explicit capital requirement—and repeats the LTV comparison at each limit. At a limit of 15, the cross-regime house-price gap rises from 0.80 to 1.31 percentage points under the productivity shock and reaches 8.5 points under the funding-cost shock, where the loose economy loses nearly all bank net worth; the productivity-shock gap is 1.17 percentage points at a limit of 10 and 0.90 at 9.5 and the funding-cost shock barely transmits in any LTV regime. Appendix D.3 varies the capital requirement itself, holding the calibration fixed and leaving the LTV cap at its loose benchmark: tightening the requirement reduces the productivity-shock house-price trough from 7.6 percent at the loosest limit to 5.3 percent at the tightest, roughly three times the stabilization delivered by the full LTV range, and lowers the funding-cost trough from 8.8 percent at a limit of 15 to 0.5 percent at the benchmark limit of 10, with mortgage credit contracting endogenously as spreads rise. Comparing regulation-fixed and market-determined leverage at the same steady-state leverage ratio isolates a further channel: busts are roughly half as deep when the regulatory ceiling binds as when the enforcement constraint lets leverage contract with net worth, so capital regulation stabilizes partly by severing the leverage-net-worth feedback.

On the stabilization margin, the two instruments therefore act on a common vulnerability: both reduce the economy's exposure to losses on bank mortgage portfolios, although they do so through different margins. This overlap implies that the incremental stabilization benefit of tighter LTV limits is largest when bank balance sheets are relatively fragile and diminishes as capital regulation already provides greater loss-absorbing capacity. This result is best interpreted as an instrument-assignment insight rather than a ranking of policies. First, the experiments do

not hold mortgage-market structure fixed: tighter capital requirements raise lending spreads and reduce mortgage intermediation, so part of their stabilizing effect operates by shrinking the underlying exposure itself. Second, capital regulation has its own general-equilibrium costs and distributional consequences (Elenev et al., 2021). We therefore do not infer that capital requirements dominate borrower-based limits; rather, the results show that the stabilization value of LTV regulation depends on how much intermediary vulnerability remains after bank-side regulation is taken into account.

9 Would Countercyclical LTV Policy Be More Effective?

The previous sections showed that fixed LTV limits weaken housing-market adjustment in downturns by preventing wealth-constrained households from converting lower prices into purchases. A natural policy response is to relax borrowing limits temporarily during the downturn and re-tighten them once conditions normalize. This section asks whether such a countercyclical LTV policy can preserve microprudential benefits while restoring the demand margins that support house prices in recessions.

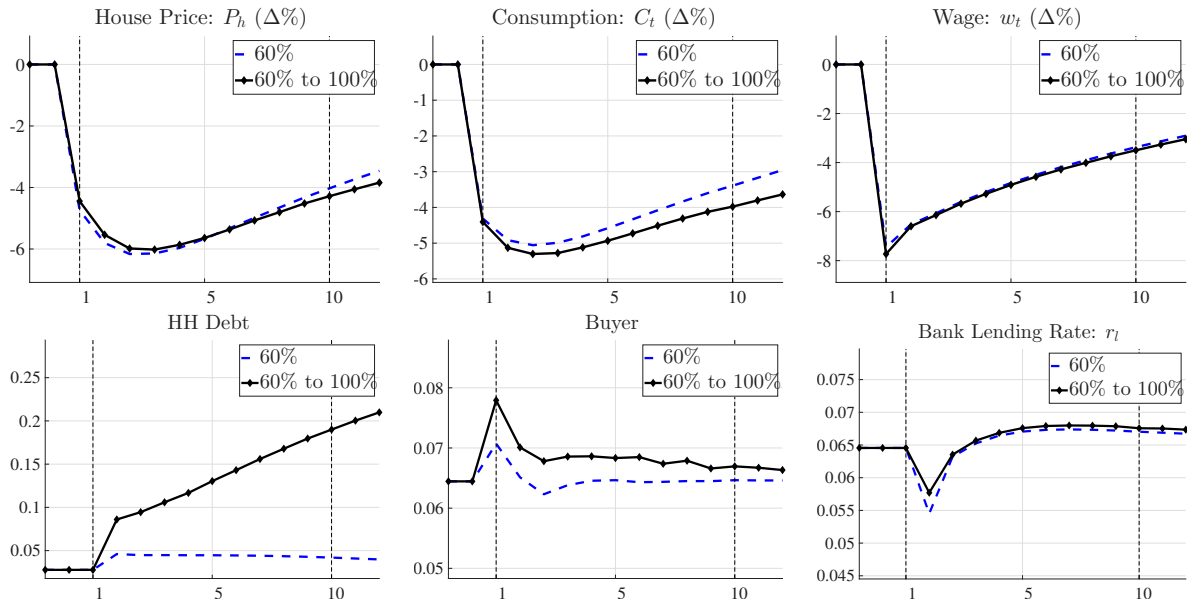
We consider an economy with a 60 percent LTV limit in steady state. When the benchmark productivity shock occurs, the macroprudential authority raises the cap immediately to 100 percent for 20 periods (40 years), after which it returns to 60 percent. This experiment targets the mechanism identified in Sections 7.1–7.2: by suspending the down-payment constraint during the downturn, the policy reopens entry for households that otherwise could not buy despite lower house prices. The 20-period window makes the exercise favorable to countercyclical policy by showing the extreme effect of reopening high-LTV borrowing during the bust. We therefore read the experiment as an upper bound on the mechanism rather than as a realistic implementation horizon.¹⁰

Figure 11 compares this countercyclical rule with a fixed 60 percent LTV cap in full general equilibrium. The policy operates on the intended margin. The buyer rate rises at the start of the release window, and the homeownership rate increases gradually relative to the fixed-regime path. Household debt rises from its steady-state level and continues to increase over the displayed horizon, while household assets are higher than under the fixed cap. The relaxation therefore allows households to enter homeownership with less reliance on asset drawdowns, but it does so by replacing internal financing with mortgage debt.

The house-price effect stays small. On impact, the house-price decline is 4.45 percent under the countercyclical rule and 4.73 percent under the fixed 60 percent cap, an improvement of 0.29 percentage points (differences here use unrounded paths). At the trough, the declines are

¹⁰We have also computed transitions with shorter release windows; the results are qualitatively unchanged. We report the long window because it makes the mechanisms most visible.

FIGURE 11 – Countercyclical LTV Policy: Buffer Release during the Bust (General Equilibrium)



Notes: Transition dynamics following the benchmark 5 percent negative productivity shock under two regimes: a fixed 60 percent LTV cap (dashed) and a countercyclical regime in which the cap rises immediately from 60 to 100 percent, remains at 100 percent for 20 model periods, and then returns to 60 percent (solid). Top row: house price, consumption, and wage, as percentage deviations from steady state. Bottom row: household debt, the buyer rate, and the bank lending rate, in levels. The lending rate is annualized. The text discusses household assets and the homeownership rate.

6.02 and 6.16 percent, so the improvement narrows to 0.14 points. After the first few periods, the house-price paths are close, and the countercyclical path eventually falls slightly below the fixed-regime path. The release therefore provides short-run price support without producing a meaningfully shallower housing downturn in general equilibrium.

The countercyclical rule does not improve consumption at the trough. Consumption falls by 5.30 percent under the release and by 5.06 percent under the fixed cap, a deterioration of 0.25 percentage points. Relaxing the down-payment constraint shifts financing toward mortgage debt, and the resulting debt-service obligations absorb resources during a period of weak income.

General-equilibrium price responses mute the effects of countercyclical policy. The bank lending rate is higher under the countercyclical rule after the initial shock, and the wage trough worsens from 7.45 to 7.73 percent, a difference of 0.28 percentage points. The increase in mortgage demand raises the lending rate and crowds out firm loans, reducing labor demand and wages and offsetting the expansionary effect of easier entry.

A partial-equilibrium diagnostic isolates the housing-demand effect of the release (Appendix D.6). We hold wages, the return on capital, and the bank lending rate at their fixed-regime paths and allow only the house price to clear endogenously. Under this restriction, the release has a

substantially larger effect: the impact decline in house prices is 3.79 percent rather than 4.73 percent, an improvement of about 0.9 percentage points—roughly three times the corresponding general-equilibrium gain—and the increase in homeownership is nearly twice as large. The comparison identifies the general-equilibrium offset. In the full model, the additional mortgage demand generated by the release raises the bank lending rate, increases firms’ working-capital costs, and depresses labor demand and wages. At the same time, the resulting increase in household leverage raises debt-service burdens and weakens consumption later in the transition (Figure D.7).

10 Reconciling the Model’s Results with Earlier Evidence

An extensive literature estimates the effects of macroprudential policies. A meta-analysis by Araújo et al. (2024) synthesizes more than 6,000 estimates from 58 studies and finds that borrower-based tools tend to restrain household credit, while estimated effects on house prices and real activity are more heterogeneous (Table VII). However, effects vary widely across studies, and many estimates are statistically insignificant.

TABLE VII – Empirical Results on the Effectiveness of Macroprudential Policies

	Borrower-based mortgage limits				Bank-based measures			
	No. of estimates	Mean	Min	Max	No. of estimates	Mean	Min	Max
Macroprudential Policies in Place (0,1)								
House Prices	38	-0.17	-0.52	0.34	21	0.01	-0.53	0.36
Household Credit	33	-0.12	-1.21	0.51	22	-0.03	-0.45	0.93
Macroprudential Policies Change (-1, 0,1)								
Economic Activity	50	-0.02	-0.20	0.10	98	-0.04	-0.33	0.23
House Prices	182	-0.01	-1.44	2.69	57	-0.04	-0.23	0.22
Household Credit	193	-0.18	-1.58	1.02	63	-0.08	-1.59	0.86

Notes: Ranges of estimated effects from the meta-analysis of Araújo et al. (2024), which synthesizes more than 6,000 estimates from 58 studies. Entries are the number of estimates and the mean, minimum, and maximum standardized effects of macroprudential policy on each outcome. The left panel covers borrower-based limits on mortgages; the right panel covers bank-based measures. The policies-in-place rows use indicators for a tool being active; the policy-change rows use tightening and loosening actions. Estimates are regression coefficients on policy indicators, coded +1 for tightening and -1 for loosening in the policy-change rows and 0/1 in the policies-in-place rows, normalized by the standard deviation of the outcome variable: an entry of -0.17 means the outcome, typically a growth rate, is 0.17 standard deviations lower. Means are unweighted averages across estimates.

Our results do not map cleanly into a stabilizing-versus-destabilizing verdict. Under the benchmark domestic-funding calibration, tighter LTV limits deliver clear microprudential benefits (lower leverage, fewer foreclosures, safer bank balance sheets) and modest macro stabilization in productivity-driven downturns. But the aggregate effect is small relative to the microprudential gains (Sections 7.1–7.2), is fragile across environments, and can reverse when banks’ marginal funding cost is externally determined (Section 7.5 and Appendix C). Even under crisis-scale shocks, where foreclosures are first-order, eliminating mass defaults produces

only a small persistent improvement in the house-price and consumption troughs (Section 7.6). These findings are compatible with the existing evidence, but the comparison requires care.

First, most empirical work estimates how a change in macroprudential policy affects changes in outcome variables, whereas we compare cyclical dynamics across economies operating permanently under different regimes. The two objects differ mechanically: because an LTV limit applies at origination, it binds new borrowers while leaving the legacy mortgage stock unchanged, so for many years outstanding loans reflect pre-regulation conditions, and estimates based on policy changes or policy-existence indicators mix transitional dynamics with legacy effects. When we do study a transition, in the countercyclical experiment (Section 9), we recover the standard short-run response: loosening LTV during a bust increases entry, initially supports house prices, and expands credit, though the benefits are small and short-lived. Across steady states, by contrast, mortgage credit falls while house prices can rise modestly, because higher saving, higher wages, and lower lending rates change the user cost of housing in the long-run regime comparison.

Second, our model combines households, banks, and firms, so sectoral balance sheets interact. LTV regulation targets households but spills over to banks by reducing losses and improving solvency, and to firms through the reallocation between housing leverage and productive capital. These cross-sector interactions are central to our results. The equalized-wage experiment in Section 7.3 quantifies this: when all general-equilibrium price channels are shut down, the house-price stabilization gap is 0.45 percentage points, around half of the 0.80 percentage-point gap in the full general-equilibrium model. The remainder comes from wages and interest rates, not from the mortgage-market mechanisms that micro studies aim to detect. Empirical work typically observes one segment of the economy at a time; it can capture microprudential benefits (lower leverage, fewer defaults) but may miss the general-equilibrium channels that determine macro stabilization.

Third, empirical estimates face well-known identification challenges. Regulators may act when vulnerabilities are rising, biasing estimates toward finding no effect or even adverse effects; actions taken near the end of a crisis, when political feasibility is greatest, may overstate stabilization. Many cross-country studies address endogeneity with lagged policy indicators, which requires strong identifying assumptions (see [Cerutti et al. \(2017\)](#) and [Akinci and Olmstead-Rumsey \(2018\)](#), among others), while micro-data designs using cross-sectional heterogeneity and matched borrower-lender data identify partial-equilibrium effects in specific markets ([Ayyagari et al., 2018](#); [Arslan et al., 2015b](#); [Jiménez et al., 2017](#); [Acharya et al., 2022](#)). Household-level studies point the same way: [Peydró et al. \(2024\)](#) document how UK mortgage limits reshape the allocation and distribution of credit; [van Bakkum et al. \(2024\)](#) find lower leverage and default alongside depleted liquid buffers under Dutch LTV limits; and [Aastveit et al. \(2024\)](#) show that constrained Norwegian borrowers hold less liquidity and cut consumption more

after job loss. Our analysis suggests that general-equilibrium effects, especially the offsetting demand-suppression channel, may be as important as the direct microprudential benefits these designs detect.

Finally, the result that tight LTV limits can become destabilizing when banks' marginal funding cost is externally determined helps rationalize heterogeneity in empirical estimates. International wholesale funding weakens the domestic-rate adjustment that cushions demand suppression in the domestic-funding benchmark. We therefore treat the externally priced closure as a disciplined limiting case rather than an empirically estimated threshold: available cross-country data conflate endogenous policy adoption, slow-moving regimes, and changing shock composition too coarsely to identify the interaction.

11 Conclusions

Loan-to-value limits reliably make borrower and lender balance sheets safer. Whether these microprudential gains translate into aggregate stabilization is more contingent: it depends on the source of the shock, the balance sheet on which the relevant exposure resides, and the structure of financial intermediation. In our model, tighter origination rules stabilize more when they reduce the exposure on which the shock loads—bank duration risk under fixed-rate mortgages or borrower payment risk under adjustable-rate mortgages—and much less when the downturn simultaneously weakens the internal funds required for housing transactions.

Against productivity-driven downturns, even a cap tight enough to eliminate most household leverage and downturn foreclosures changes the depth of the bust only modestly. The reason is fragile prudence. Larger required down payments tie entry and upgrading more closely to internal funds; when income, non-housing financial wealth, and home equity weaken together, the marginal buyers who would otherwise respond to lower prices withdraw from the market. The effective bite of the constraint is therefore procyclical in an undesirable direction: collateral gains loosen it during booms, while depleted internal funds tighten it during busts. Our counterfactual decomposition shows that the resulting stabilization reflects offsetting household margins together with equilibrium price and intermediary responses; in particular, the cross-regime wage gap in our decomposition operates through bank intermediation and the working-capital channel. Even under a crisis-scale shock, large reductions in foreclosures and bank-net-worth losses translate into little persistent stabilization of house prices or consumption.

The stabilization benefit is larger when the shock originates in bank funding costs. Tighter LTV limits leave banks with smaller fixed-rate mortgage books and therefore less duration exposure, weakening the feedback from mortgage valuations to bank net worth, lending capacity, and lending rates. This result locates much of the stabilization value of borrower-based limits on intermediary rather than household balance sheets and naturally raises the comparison with

bank-capital regulation. The two instruments act on a common vulnerability through different margins: LTV limits shrink the underlying mortgage exposure, whereas capital requirements increase loss-absorbing capacity and, in equilibrium, also affect lending spreads and mortgage intermediation. Consistent with this overlap, the incremental stabilization benefit of tighter LTV limits is largest when bank balance sheets are relatively fragile and diminishes as bank-side resilience increases.

Two additional exercises underscore the absence of a general stabilization result. First, under the externally priced marginal-funding closure, the productivity-shock ranking reverses: tighter caps continue to prevent foreclosures but deepen the declines in house prices and consumption. The exercise represents a limiting case in which banks' marginal funding cost is insulated from domestic saving and investment conditions. This case is particularly relevant for small open economies, but the underlying mechanism is more general and can also arise in larger economies when banks price marginal funding in wholesale or international markets. The modest stabilization in the domestic-funding benchmark therefore depends partly on an endogenous decline in bank funding and lending rates; as that funding-rate response weakens, the asset-reliance and demand-suppression channels become more important.

Second, countercyclical relaxation of the LTV cap restores the entry margin and provides short-run support to house prices, but does not generate a stronger aggregate recovery. Additional mortgage demand raises lending rates, crowds out working-capital credit, weakens wages, and increases household debt-service burdens. The limitation is therefore not resolved simply by making the origination rule countercyclical. These conclusions are also robust to replacing LTV limits with payment-to-income constraints and to allowing adjustable-rate mortgages; contract design changes the incidence of funding-rate risk without overturning the central cross-regime comparison.

The policy implication is therefore to distinguish microprudential insurance from macroeconomic stabilization. Borrower-based limits are effective at reducing leverage, preventing foreclosure waves, and limiting lenders' exposure to mortgage losses. But evaluating them as aggregate stabilizers requires tracking the margins that conventional balance-sheet measures omit: entry into ownership, mobility along the housing ladder, the internal funds used to finance transactions, and the interaction of mortgage regulation with intermediary balance sheets and funding conditions. The appropriate stabilization role of borrower-based regulation therefore depends on the shocks the financial system is expected to absorb and on the structure of intermediation through which those shocks propagate. Our analysis addresses this positive stabilization question, not the welfare ranking of alternative regimes. Such a ranking would additionally have to weigh delayed homeownership and its distributional incidence against lower default and intermediary fragility across prospective buyers, incumbent owners, renters, banks, and cohorts.

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APPENDIX

A Model Details

This appendix states the household decision problems; the problems of final-goods firms, real estate and construction companies, and banks; the government budgets; and the definition of the stationary equilibrium summarized in the main text.

A.1 Household Decision Problems

We solve the household problems backward from terminal age. We impose the standard terminal condition $V_{J+1}^r = V_{J+1}^e = V_{J+1}^h = 0$ and require assets and mortgage balances to satisfy the same feasibility constraints as at earlier ages at the terminal date. The mortgage-payment convention above fully amortizes any mortgage by age J , so no terminal debt remains.

Active Renters An active renter chooses between continuing to rent and purchasing a home:

$$V^r = \max\{V^{rr}, V^{rh}\}.$$

If she continues renting, she selects consumption c , rental size s , and savings a' :

$$V_j^{rr}(a, z) = \max_{c, s, a' \geq 0} \left\{ u(c, s) + \beta EV_{j+1}^r(a', z') \right\} \quad (\text{A.1})$$

subject to the budget constraint

$$c + a' + p_r s = y(j, z) + a(1 + r),$$

where r is the return on savings; labor and payroll taxes enter through the after-tax income process in equation (1).

If she purchases a home, she additionally chooses house size h and mortgage debt d . The origination price $q_m(d; a', h, z, j)$ reflects default risk and depends on household characteristics. The purchaser's problem is

$$V_j^{rh}(a, z) = \max_{c, d, h, a' \geq 0} \left\{ u(c, h) + \beta EV_{j+1}^h(a', h, d, z') \right\} \quad (\text{A.2})$$

subject to

$$c + p_h h + \varphi_f + a' = y(j, z) + a(1 + r) + d \left(q_m(d; a', h, z, j) - \varphi_m + \tau^d r_\ell \right),$$

where τ^d is the rate of the government-provided mortgage subsidy, which is proportional to the mortgage interest payment.

The purchase is subject to the origination constraints (3) and (4) stated in the main text.

Inactive Renters An inactive renter, excluded from ownership after default, solves

$$V_j^e(a, z) = \max_{c, s, a' \geq 0} \left\{ u(c, s) + \beta \left[\pi EV_{j+1}^r(a', z') + (1 - \pi) EV_{j+1}^e(a', z') \right] \right\} \quad (\text{A.3})$$

subject to the same budget constraint as an active renter. Each period she regains active status with probability π .

Homeowners Homeowners choose among four options each period:

$$V^h = \max\{V^{hh}, V^{hf}, V^{hr}, V^{he}\},$$

corresponding to staying, refinancing, selling, and defaulting.

Stayers A homeowner who remains in place selects consumption and savings while meeting the mortgage payment:

$$V_j^{hh}(a, h, d, z) = \max_{c, a' \geq 0} \left\{ u(c, h) + \beta EV_{j+1}^h(a', h, d', z') \right\} \quad (\text{A.4})$$

subject to

$$\begin{aligned} c + \delta_h p_h h + a' + m - \tau^d r_\ell d &= y(j, z) + a(1 + r), \\ d' &= (d - m)(1 + r_\ell). \end{aligned}$$

where δ_h is the per-unit maintenance cost to cover the depreciation of houses.

Refinancers Refinancing involves repaying the existing balance and originating a new mortgage, incurring the same costs as an initial purchase:

$$V_j^{hf}(a, h, d, z) = \max_{c, d', a' \geq 0} \left\{ u(c, h) + \beta EV_{j+1}^h(a', h, d', z') \right\} \quad (\text{A.5})$$

subject to

$$\begin{aligned} c + d + \delta_h p_h h + \varphi_f + a' &= y(j, z) + a(1 + r) + d' \left(q_m(d'; a', h, z, j) - \varphi_m + \tau^d r_\ell \right), \\ d' &\leq (1 - \varrho) p_h h \\ m(d') &\leq \varsigma y(j, z). \end{aligned}$$

Sellers A homeowner may sell the current property (paying the selling cost φ_s and the maintenance cost) and either rent or buy a different home. The value equals that of an active renter with augmented wealth:

$$V_j^{hr}(a, h, d, z) = V_j^r(a + p_h h(1 - \delta_h - \varphi_s) - d, z).$$

Defaulters A homeowner with negative equity may default, surrendering the home to the lender. The lender sells the property at a discount $(1 - \varphi_e) p_h h$ and returns any residual equity to the household. The defaulter then rents and transitions to active status with probability χ :

$$V_j^{he}(a, h, d, z) = \max_{c, s, a' \geq 0} \left\{ u(c, s) + \beta E \left[\chi V_{j+1}^r(a', z') + (1 - \chi) V_{j+1}^e(a', z') \right] \right\} \quad (\text{A.6})$$

subject to

$$c + a' + p_r s = a(1 + r) + y(j, z) + \max\{(1 - \varphi_e) p_h h - d, 0\}.$$

Because $\varphi_e > \varphi_s$, a household with positive net equity strictly prefers selling over defaulting. Hence negative equity is necessary, but not sufficient, for default.

A.2 Final-Goods Firms

A representative, perfectly competitive firm combines capital K and effective labor $N_g u$, where N_g is the quantity of household labor hired and u is the utilization rate of each unit of labor, to produce output:

$$Y_g = \mathbb{Z} K^\alpha (N_g u)^{1-\alpha},$$

where $\mathbb{Z}$ is total factor productivity.¹¹ The wage per efficiency unit depends on utilization:

$$w(\bar{w}, u) = \bar{w} + \vartheta \frac{u^{1+\psi}}{1+\psi},$$

with ϑ and ψ constants and $\bar{w}$ an equilibrium object. The firm must finance a fraction μ of the wage bill in advance through bank loans, which links credit conditions directly to production. The firm's problem is

$$\max_{K, N_g, u} \mathbb{Z} K^\alpha (N_g u)^{1-\alpha} - (r + \delta_k) K - (1 + \mu r_\ell) w(\bar{w}, u) N_g, \quad (\text{A.7})$$

where r is the return on capital and δ_k is the depreciation rate of capital. The first-order conditions are

$$\begin{aligned} \alpha \mathbb{Z} \left(\frac{K}{N_g u} \right)^{\alpha-1} &= r + \delta_k, \\ (1 - \alpha) \mathbb{Z} u \left(\frac{K}{N_g u} \right)^\alpha &= (1 + \mu r_\ell) \left(\bar{w} + \vartheta \frac{u^{1+\psi}}{1+\psi} \right), \\ (1 - \alpha) \mathbb{Z} \left(\frac{K}{N_g u} \right)^\alpha &= (1 + \mu r_\ell) \vartheta u^\psi. \end{aligned}$$

When the bank lending rate r_ℓ rises, the firm reduces labor utilization, depressing output and wages. This is a key transmission channel linking financial conditions to the real economy.

The sign of the channel follows directly from the firm's first-order conditions. Combining the capital and utilization conditions gives

$$\frac{K}{N_g u} = \left(\frac{\alpha \mathbb{Z}}{r + \delta_k} \right)^{1/(1-\alpha)}$$

and

$$u = \left[\frac{(1 - \alpha) \mathbb{Z}}{(1 + \mu r_\ell) \vartheta} \left(\frac{\alpha \mathbb{Z}}{r + \delta_k} \right)^{\alpha/(1-\alpha)} \right]^{1/\psi}.$$

A higher bank lending rate therefore raises the effective cost of the wage bill and lowers labor utilization holding productivity and the capital return fixed. The second and third first-order conditions also imply

$$\bar{w} = \frac{\psi}{1+\psi} \vartheta u^{1+\psi},$$

so the fall in utilization maps into the wage shifter used in household income.

A.3 Real Estate Companies

A real estate firm enters the period with $(1 - \delta_h) H_r$ units of rental housing, where δ_h denotes the depreciation rate of housing. The firm then chooses its end-of-period rental housing stock, H'_r . During the period, it earns net rental income $(p_r - \kappa) H'_r$, where p_r is the rental price per unit and κ is the per-unit maintenance cost. Adjusting the rental housing stock entails a quadratic cost governed by $\eta_r \geq 0$. The real estate firm chooses H'_r to maximize its market value:

$$(1 + r) V^{rc}(H_r) = \max_{H'_r} p_h (1 - \delta_h) H_r - p_h H'_r + (p_r - \kappa) H'_r - \frac{\eta_r}{2} p_h (H'_r - H_r)^2 + V^{rc}(H'_r), \quad (\text{A.8})$$

¹¹We follow [Arslan et al. \(2025\)](#) in the introduction of labor utilization. They show that this formulation is equivalent to assuming flexible labor supply together with GHH preferences on the household side.

Because capital and shares in the real estate firm are riskless both in the steady state and along the transition path, except in the period containing the unanticipated shock, they must deliver the same return. Using this no-arbitrage condition together with the firm's first-order condition yields the rental price as a function of house prices and the firm's rental housing stock:

$$p_r = \kappa + p_h + \eta_r p_h (H_r - H_r^{-1}) - \frac{(1 - \delta_h) p'_h + \eta_r p'_h (H'_r - H_r)}{1 + r'} \quad (\text{A.9})$$

where p'_h is the house price in the next period and H_r^{-1} denotes the firm's rental housing stock in the previous period. Equation (A.9) characterizes the supply side of the rental housing market: the rental price compensates the firm for maintenance costs, the opportunity cost of housing capital, and the costs associated with adjusting its rental stock, net of the continuation value of housing. Households' housing choices determine rental housing demand.

A.4 Construction Companies

Construction companies hire household labor and purchase land permits from the government to produce houses and sell them to the households and rental companies. They solve the following problem:

$$\max \{ p_h I_h - w N_h - q_p L_p \} \quad (\text{A.10})$$

where $I_h = L_p^{\theta_h} N_h^{1-\theta_h}$, N_h is the quantity of labor, L_p is the quantity of land permits, and q_p is the price of land permits. Labor is perfectly mobile, and construction companies pay the same wage as the final-goods firms. The construction firm's first-order conditions pin down labor demand and the land-permit price:

$$w = (1 - \theta_h) p_h \left(\frac{L_p}{N_h} \right)^{\theta_h}, \quad q_p = \theta_h p_h \left(\frac{N_h}{L_p} \right)^{1-\theta_h}.$$

These equations show how higher house prices raise both construction labor demand and the value of scarce land permits. Aggregate housing supply follows the usual law of motion:

$$H' = (1 - \delta_h) H + I_h \quad (\text{A.11})$$

A.5 Banking Sector

Banks intermediate credit between depositors and borrowers (households and firms). Let aggregate bank lending be $L = L_k + \int_{\theta} p_{\ell}(\theta) \ell(\theta)$, where L_k denotes loans to firms, $\ell(\theta)$ the stock of mortgages to households with characteristics $\theta \equiv (a, d, h, j, z)$, and $p_{\ell}(\theta)$ the value of a unit mortgage. The banker's problem is¹²

$$\Psi(L, B) = \max_{c^B, L'_k, \ell'(\theta), B'} \left\{ \log(c^B) + \beta_L \Psi(L', B') \right\} \quad (\text{A.12})$$

subject to

$$c^B + L'_k + \int_{\theta} p_{\ell}(\theta) \ell'(\theta) = N + B',$$

$$(1 - \eta)(1 + r'_{\ell})L' \geq (1 + r')B'$$

where c^B is banker consumption, B' borrowing from depositors, N beginning-of-period net worth, and

$$\eta = \xi^{1-\beta_L} \left(\frac{1 + r'}{1 + r'_{\ell}} - (1 - \eta') \right)^{\beta_L}$$

¹²See Arslan et al. (2025) for the details of this problem.

where ξ denotes the fraction of assets the bank can divert when defaulting on its depositors. In a stationary allocation with a binding enforcement constraint, the banker's Euler equation implies the spread expression

$$r_\ell - r = \frac{1 - \beta_L(1 + r)}{\hat{\lambda} \beta_L},$$

where $\hat{\lambda}$ is the bank leverage ratio. This expression links the lending spread to the banker's discount factor and the tightness of the intermediary balance-sheet constraint.

Net worth evolves as $N = L(1 + r_\ell) - B(1 + r)$.

In the funding-cost experiments, the marginal gross cost of bank liabilities is $1 + r^{B'}$ with $r^{B'} = r' + \varepsilon'$: the wedge replaces the deposit rate in the enforcement constraint and the net-worth law of motion, so that $N = L(1 + r_\ell) - B(1 + r^{B'})$, while households continue to receive $1 + r$ on deposits. The difference $(r^B - r)B$ is an iceberg intermediation cost rather than a payment to any agent, so household budget sets and asset-market clearing are unchanged. The externally priced closure of Appendix C replaces r^B with a fixed external funding rate in the same two equations.

The market value of a mortgage satisfies the recursion

$$p_\ell(\theta) = m_\ell(\theta) + \frac{1}{1 + r'_\ell} \int_{\theta'} p_\ell(\theta') \Pi(\theta'|\theta), \quad (\text{A.13})$$

where Π is the transition kernel over household states and $m_\ell(\theta)$ the current-period cash flow:

$$m_\ell(\theta) = \begin{cases} d & \text{if sell or refinance,} \\ p_h h(1 - \varphi_e) & \text{if default,} \\ m(d) & \text{if stay.} \end{cases} \quad (\text{A.14})$$

At origination the mortgage price $q_m(d; \theta)$ satisfies

$$q_m(d; \theta) d = p_\ell(\theta). \quad (\text{A.15})$$

Default risk lowers the origination price. With no default risk, no origination wedge, and a contract rate equal to the bank lending rate, the present value of promised payments equals face value; under the proceeds normalization in the household budget this benchmark is $q_m = 1$.

A.6 Government

The government operates two programs:

1. A balanced pay-as-you-go pension system, collecting taxes from workers and distributing benefits to retirees

$$\tau^s \sum_{j=1}^{J_R} \sum_z (\exp(f(j) + z)) \pi_j(z) = \sum_{j=J_R+1}^J \sum_z y_R(j, z) \pi_j(z), \quad (\text{A.16})$$

where $\pi_j(z)$ is the measure of age- j households with productivity z .

2. A mortgage subsidy and land-permit sale program: Mortgage subsidies, which are proportional to mortgage interest payments, draw their funding from land-permit sales and labor-income taxes on

all households, including retirees.¹³

$$\tau^d \int \tilde{m}_\ell(\theta) d\lambda(\theta) = q_p \bar{L} + \tau^m w \sum_{j=1}^{J_R} \sum_z (\exp(f(j) + z)) \pi_j(z) + \tau^m w \sum_{j=J_R+1}^J \sum_z y_R(z_{J_R}) \pi_j(z) \quad (\text{A.17})$$

where $\tilde{m}_\ell(\theta) = r_\ell d(\theta)$ is the mortgage interest payment of a household with current state θ .

A.7 Residual Service Sector

There are several transaction costs in the model: maintenance cost of rental units, house sale transaction cost, and mortgage origination costs. We assume that a labor-intensive service sector provides all these services. Denoting TC as the sum of all these costs, we assume

$$TC = wN_s$$

where N_s is the labor demand in this sector.

A.8 Stationary Equilibrium

A stationary recursive equilibrium is a set of consumption (g_c), housing services (g_s), saving (g_a), mortgage debt (g_d), and housing tenure (g_h) policy functions for the households, capital demand (K), labor demand (N_g), and labor utilization (u) for the final-goods firms, rental stock supply (H_r) for real estate companies, labor demand (N_h), housing investment (I_h), and land permit demand (L_p) for construction companies, and the associated house supply (H), residual service sector labor demand (N_s), corporate (L_k) and mortgage (ℓ) loan supply, consumption (c_B), and deposit demand (B) for the banks, tax rates for the pension system, τ^s , and mortgage subsidy system, τ^m , supply of land permits, $\bar{L}_p$, prices $\{r, w, r_\ell, p_h, p_r, q_m, q_p\}$, value functions for the households $\{V_j^i\}$ and for the lenders $\{p_\ell\}$ and a stationary distribution such that

1. Given prices $\{r, w, r_\ell, p_h, p_r, q_m\}$ and taxes $\{\tau^s, \tau^m\}$, household policy functions $\{g_c, g_s, g_a, g_d, g_h\}$ solve households' problems in (A.1)–(A.6).
2. Given prices $\{r, w\}$, $\{K, N_g, u\}$ solve final-goods firms' problem in (A.7).
3. Given prices $\{w, p_h, q_p\}$, $\{N_h, I_h, L_p\}$ solve construction companies' problem (A.10) and housing supply follows equation (A.11)
4. Given prices $\{p_h, p_r\}$, $\{H_r\}$ solves the real-estate companies' problem (A.8).
5. Given prices $\{r, r_\ell, p_\ell\}$, $\{L_k, \ell, c_B, B\}$ solve banks' problem (A.12).
6. Given household policy functions and prices, mortgage continuation values $\{p_\ell\}$ satisfy (A.13)–(A.14).
7. Given $\{p_\ell\}$, $\{q_m\}$ solves (A.15).
8. Markets clear:
 - (a) Labor market: $N_g + N_h + N_s = 1$ where $wN_s = TC$.
 - (b) Asset market: $K + V^{rc} + B = \int ad\lambda$.
 - (c) Housing market: $H = \int sd\lambda$.
 - (d) Rental market: $H_r = \int sd\lambda^r$ where λ^r is the distribution of renters and defaulters.
 - (e) Land permit market: $L_p = \bar{L}$.
 - (f) Loan market: $L = L_k + \int_\theta p_\ell(\theta)\ell(\theta)$ where $L_k = \mu w N_g$ and $\ell(\theta) = \lambda(\theta)$.

¹³When land permit sales exceed mortgage subsidies, the government rebates the balance to households as a negative labor-income tax.

9. Government runs a balanced budget for both programs:
 - (a) τ^s solves (A.16).
 - (b) Given household policy functions, τ^m solves (A.17).
10. λ is stationary and the evolution of it is consistent with household policy functions.

B External Validation of Rate Transmission

The model's dynamic responses to interest-rate shocks are in line with the available evidence. We compare the model with three complementary designs: direct variation in mortgage rates, identified monetary-policy shocks, and cross-bank variation in interest-rate exposure. Because identified innovations are transitory, we compute elasticities from a one-period 100 basis point funding shock; the persistent experiment of Section 8, whose wedge decays slowly, produces larger cumulative responses. In the loose economy the one-period shock moves the lending rate by about 40 basis points and house prices and non-housing consumption by 0.8 and 0.3 percent at the first post-shock model date; per 100 basis points of the induced mortgage-rate movement, the responses are 1.9 and 0.8 percent. The 40 basis point pass-through is conservative relative to estimates from market-wide funding movements: [Gropp et al. \(2007\)](#) estimate lending-rate responses of about 62 basis points within the quarter and 86 basis points after two quarters, and [Harimohan et al. \(2016\)](#) find near-complete pass-through of the common component of U.K. bank funding costs. One model period is two years, so the model's responses are two-year responses rather than quarterly impacts.

Direct mortgage-rate evidence places the house-price response in a plausible range. [Akgündüz et al. \(2023\)](#) exploit a temporary subsidy to mortgage rates at Turkish state-owned banks and estimate that a 100 basis point reduction in the annual mortgage rate raises individual mortgage borrowing by 3.3 percent and house prices by 1.6 percent. Their experiment is an easing, as is our one-period shock; the model's 1.9 percent house-price response per 100 basis points of the mortgage rate sits next to their estimate. [Gorea et al. \(2024\)](#), using high-frequency U.S. monetary surprises, find that a surprise raising the average 30-year mortgage rate by 25 basis points lowers list prices by 1 percent within two weeks; the initial response passes through to sale prices and persists for at least a year. At annual frequency, [Jordà et al. \(2015\)](#) estimate that a 100 basis point short-rate perturbation raises long rates by about 42 basis points on impact and lowers the real house-price-to-income ratio by 3.7 percent after two years and 4.4 percent after four years, with a peak decline of 5.0 percent after three years. The model's one-period response is on the conservative side of these persistent-innovation estimates, while the persistent experiment of the main text brackets them from above; the shock, outcome, and timing differ across designs.

For consumption, the model is plausible but not conservative relative to aggregate monetary estimates. [Cloyne et al. \(2020\)](#) estimate that a 25 basis point easing raises aggregate nondurable consumption at its peak by about 0.1 percent in the United States and 0.2 percent in the United Kingdom. The implied elasticities are 0.4 to 0.8 percent per 100 basis points. The model's elasticity with respect to the induced mortgage-rate movement, 0.8 percent per 100 basis points, is at the upper end of this range, while its response per 100 basis points of the funding-cost shock, 0.3 percent, is slightly below it. More importantly for the mechanism, [Holm et al. \(2021\)](#) use identified shocks and Norwegian administrative data to show that disposable-income changes generate sizable consumption responses across the liquid-asset distribution, while indirect effects build over time and eventually outweigh direct interest-rate-exposure effects. This supports the model's working-capital and wage channel. [Cloyne et al. \(2020\)](#) also find that mortgagors account for most of the aggregate consumption response, while [Di Maggio et al. \(2017\)](#) and [Flodén et al. \(2021\)](#) provide causal evidence for borrower cash-flow transmission under adjustable rates.

The credit-demand link also has an empirical counterpart. The one-period easing raises firm working-capital credit by about 0.4 percent while moving the lending rate by 40 basis points, a reduced-form ratio

of 0.93 percent per 100 basis points of the lending rate. Because working-capital borrowing is proportional to the wage bill in the model, this response also reflects the expansion in labor demand. [Altavilla et al. \(2022\)](#) estimate that a 100 basis point increase in firm lending rates lowers total bank-credit demand by 1.6 to 1.8 percent in log-change specifications and by 2.3 to 2.5 percent under their level estimator; the model response is conservative, about half of their log-change estimates. The comparison is a diagnostic rather than a structural elasticity: the lending rate responds endogenously to the funding shock, and working-capital credit in the model is proportional to the wage bill, while their object is total firm bank borrowing.

The labor-market link is of the right order, though the empirical and model objects differ. Per 100 basis points of the lending rate, the one-period shock moves the wage by about 1.9 percent; in the contractionary version, labor utilization falls 2.7 percent and output 1.8 percent on impact. [Gertler and Gilchrist \(2018\)](#) estimate that a 100 basis point increase in the excess bond premium lowers aggregate employment by 3.6 percent after two years. Their shock is a corporate credit premium and their outcome is employment, while households in the model supply labor inelastically and adjustment runs through utilization and the wage; we read their estimate as directional support for a quantitatively meaningful link from intermediary credit conditions to labor demand.

The consumption response to house prices is also in line with identified estimates. Feeding the funding-shock house-price path into the household block, with all other prices held at steady-state values, changes aggregate non-housing consumption by about 5 cents per dollar of predetermined housing wealth. This partial-equilibrium response reflects a combination of valuation, collateral, and expected-capital-gain effects rather than a pure marginal propensity. [Guren et al. \(2021b\)](#) estimate 2.7 to 6.5 cents across instruments, [Aladangady \(2017\)](#) finds 4.7 cents for homeowners, and [Mian et al. \(2013\)](#) report 5 to 7 cents; [Foulis et al. \(2026\)](#) obtain about 4 cents from interest-rate-induced house-price variation, the design closest to ours. The model response sits in the middle of this range. Larger shocks raise the response,¹⁴ consistent with the dependence of housing-wealth effects on balance sheets and credit conditions in [Berger et al. \(2018\)](#).

Beyond these link-by-link magnitudes, the experiment's predictions line up with two literatures that have developed largely separately: work on how mortgage-market structure shapes the transmission of interest-rate shocks to housing and consumption, and work on the interest-rate risk that banks carry on their balance sheets. [Calza et al. \(2013\)](#) find stronger house-price and residential-investment responses in countries with more flexible and developed mortgage markets, and stronger consumption responses where mortgage-equity withdrawal is common and variable-rate contracts predominate; [Garriga et al. \(2017\)](#) likewise show that mortgage-contract design changes monetary transmission; [Arslan and Guler \(2025\)](#) show that fixed-rate contracts concentrate the losses from rate shocks on bank balance sheets, while adjustable-rate contracts shift the adjustment to borrower cash flows. On the intermediary side, [English et al. \(2018\)](#) document lower bank equity values after unanticipated upward shifts in the yield curve. [Gomez et al. \(2021\)](#) show that cash-flow exposure matters for lending: following a 100 basis point increase in the federal funds rate, a bank at the 75th percentile of the income-gap distribution contracts total lending about 0.25 percentage points less over the next year than one at the 25th percentile. [Jiang et al. \(2024\)](#) estimate roughly two trillion dollars of mark-to-market losses on U.S. bank assets during the 2022–2023 tightening, while [Drechsler et al. \(2021\)](#) show that a sticky core-deposit franchise can hedge long-duration lending, and [Atkeson et al. \(2019\)](#) attribute about half of the elevated market value of U.S. bank equity before 2008 to capitalized government guarantees, with much of the remaining franchise value coming from the core-deposit business.

This evidence supports the model's emphasis on balance-sheet exposure rather than an unconditional

¹⁴A 10 percent permanent decline in house prices yields about 6 cents per dollar. For a finite decline this is an average rather than a marginal propensity; the house-price movement in the funding-shock experiment is about 1 percent, while the permanent decline is 10 percent.

rate effect. It also clarifies the calibration’s scope: the model bank resembles an intermediary with rate-sensitive marginal funding and concentrated fixed-rate mortgage exposure, and its net-worth response is stronger than representative-bank estimates.

C Productivity Shocks with Externally Priced Marginal Bank Funding

Let r_t^B denote banks’ marginal liability cost and $s_t \equiv r_t^\ell - r_t^B$ the intermediation spread, so $r_t^\ell = r_t^B + s_t$. In the domestic benchmark, $r_t^B = r_t + \varepsilon_t$ with $\varepsilon_t = 0$. After a negative productivity shock, lower capital and credit demand reduce r_t and therefore r_t^B , helping keep the lending rate low. This response supports housing affordability and softens the wage decline through the working-capital channel. The benchmark stabilization result therefore relies partly on banks’ marginal funding cost moving with domestic conditions.

External pricing is relevant whenever banks obtain marginal funds in broader capital markets rather than solely from local household saving. This includes open economies and wholesale-funded banking systems in large economies: federal funds, repo, money-market funding, brokered deposits, and commercial paper can shape the marginal cost of bank funding. When that margin decouples from domestic saving and investment, a local downturn need not generate the funding-rate decline present in the domestic benchmark.

We capture the limiting case by fixing $r_t^B = \bar{r}^B$ at its steady-state value while leaving wages, house prices, rents, the return on capital, the spread s_t , and the lending rate r_t^ℓ endogenous. Equivalently, $\varepsilon_t = \bar{r}^B - r_t$ adjusts as the domestic safe return moves. This is a general-equilibrium economy with externally priced marginal bank funding, but not a complete small-open-economy model: households and firms do not transact directly at the external rate.¹⁵

The asymmetry is deliberate. External funding reaches the real sector through bank balance sheets, while the marginal firm remains bank dependent and pays the endogenous lending rate r_t^ℓ . The exercise isolates how disconnecting banks’ marginal liability cost from domestic conditions changes the LTV comparison; it does not impose direct international borrowing by households or firms.

Cross-border capital flows are heavily intermediated by banks. [Bruno and Shin \(2015\)](#) show that the global bank leverage cycle is a central mechanism transmitting financial conditions across borders through banking-sector capital flows, and [Cetorelli and Goldberg \(2012\)](#) provide direct evidence that banking globalization transmits monetary-policy shocks through cross-border internal capital markets. The BIS Locational Banking Statistics, which capture roughly 95 percent of cross-border banking activity, exist precisely to measure these gross cross-border bank positions.¹⁶ On the funding side, the Federal Reserve’s May 2024 *Supervision and Regulation Report* reports that wholesale funding rose to 21.6 percent of total assets at year-end 2023.¹⁷ FDIC Quarterly Banking Profile releases provide the same order of magnitude: the 2023:Q2 profile reports wholesale funding equal to 22.8 percent of total assets for all FDIC-insured institutions and 22.6 percent for community banks.¹⁸ The closure imposed in this appendix

¹⁵As the market clearing condition for the asset market changes, we recalibrate the model to match the same moments. We remove the mortgage subsidy and use $\bar{r}^B$ to match the mortgage debt to GDP ratio.

¹⁶See [Avdjiev et al. \(2016\)](#) on why gross bank-intermediated positions, rather than only net capital flows, are the relevant object for assessing the international transmission of financial conditions. The BIS data portal states that the Locational Banking Statistics capture around 95 percent of cross-border banking activity.

¹⁷Board of Governors of the Federal Reserve System, *Supervision and Regulation Report*, May 2024, section “Banking System Conditions”. The report defines wholesale funding as brokered deposits under 250,000 dollars, federal funds purchased, repurchase agreements, subordinated notes and debentures, and other borrowed money.

¹⁸FDIC, Quarterly Banking Profile, second quarter of 2023. The release reports that wholesale funding as a percentage of total assets rose from 17.1 percent in the year-ago quarter to 22.8 percent in 2023:Q2 for

therefore replaces the benchmark link between domestic deposits and banks' marginal funding cost with an externally priced funding rate.

On the borrower side, the marginal firm is often bank-dependent. In the euro area, bank loans accounted for 35.4 percent of the stock of non-financial-corporate external financing in 2016:Q1, against 10.1 percent for debt securities and 29.9 percent for quoted shares; small and medium-sized enterprises remained financed mainly through bank credit.¹⁹ The U.S. is the main exception by aggregate value: Federal Reserve Z.1 data for 2025:Q4 report 7.394 trillion dollars in non-financial-corporate bonds and 1.137 trillion dollars in depository-institution loans to non-financial corporations.²⁰ These aggregate stocks overstate market access for the marginal firm because bond issuance concentrates among larger corporations. Our closure therefore targets the bank-dependent marginal borrower: external funding reaches firms through the banking sector and the working-capital margin, while abstracting from direct market borrowing by large corporations. This is a limiting case rather than a point estimate: allowing some firms to borrow directly at the international rate would attenuate the quantitative reversal below, and intermediate cases represent partial-funding sensitivities between the domestic-funding benchmark and the full external-pricing limit. We therefore do not interpret this magnitude as a threshold result for every degree of financial openness. The qualitative force of the channel depends on whether the marginal firm remains bank-dependent and whether the funding-rate decline present under domestic funding is sufficiently muted. Intermediated-credit closures of this broad family are standard in macro-finance models with bank balance-sheet constraints. Although they ask different policy questions, [Akinci and Queralto \(2022\)](#) and [Gertler et al. \(2012\)](#) share the premise that intermediary balance sheets shape shock transmission to real borrowers.

Figure C.1 shows that the externally priced funding closure reverses the domestic-funding ranking. In the benchmark, the tight-LTV economy had a modestly smaller house-price decline. Here, house prices fall by 6.5 percent in the tight economy and by 5.2 percent in the loose economy, so tighter regulation amplifies the trough decline by 1.3 percentage points, a swing of roughly 2 percentage points relative to the benchmark. Consumption also falls more in the tight economy, 5.4 percent versus 5.0 percent. Foreclosures remain close to zero, so the policy still achieves its microprudential objective even as aggregate outcomes worsen.

The reversal operates through the lending rate, which combines the marginal funding rate and the endogenous spread: $r_t^\ell = r_t^B + s_t$. In the domestic benchmark, r_t^B falls with local activity, helping keep r_t^ℓ low. This cushions the economy directly through mortgage affordability and indirectly by reducing the effective cost of labor, $(1 + \mu r_t^\ell)w_t$. Under external pricing, r_t^B is fixed, so that component of the lending-rate decline disappears. From a steady-state value of about 4.8 percent, the lending rate falls only briefly on impact, to roughly 4.2 percent in the loose economy and 4.1 percent in the tight economy, before returning rapidly toward steady state. Neither regime therefore receives a sustained reduction in r_t^ℓ , so the demand-suppression and asset-reliance channels operate with greater force.

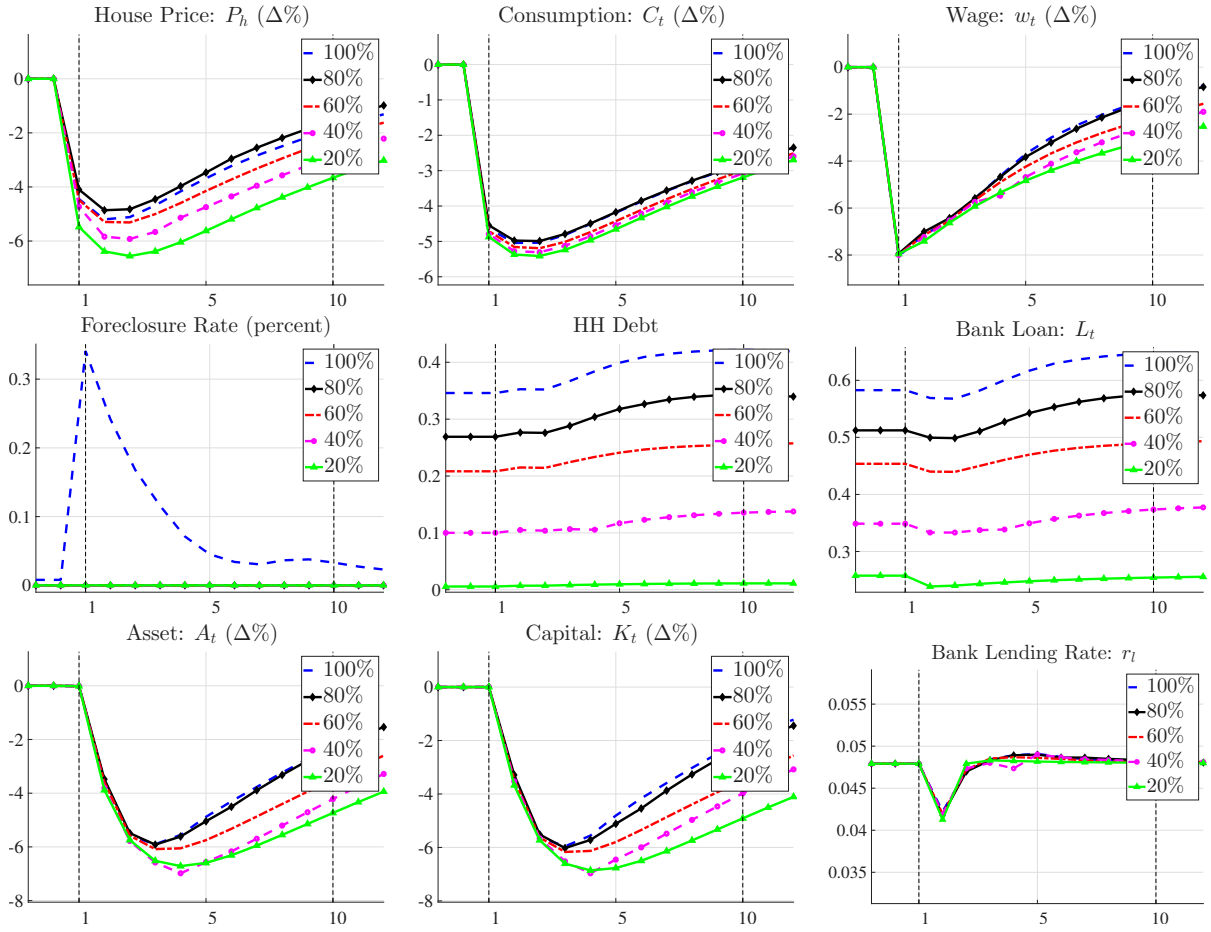
Bank credit dynamics reinforce the same point. In the loose economy, lending expands during the downturn, rising from about 0.58 to about 0.65, because banks can accommodate the mortgage demand generated by opportunistic entrants responding to lower house prices. In the tight-LTV economy, this adjustment is largely absent. The down-payment constraint prevents entry, mortgage demand contracts,

all FDIC-insured institutions. In its community-bank section, it reports 22.6 percent in 2023:Q2, up from 21.1 percent in 2023:Q1.

¹⁹ECB Economic Bulletin, Issue 5/2016, Box 5, "Trends in the external financing structure of euro area non-financial corporations". The box reports the 2016:Q1 stock shares (35.4 percent for bank loans, 10.1 percent for debt securities, and 29.9 percent for quoted shares) and notes that SMEs continue to rely mainly on bank credit.

²⁰Federal Reserve, Financial Accounts of the United States, Z.1 release, Table B.103, fourth quarter of 2025. The corresponding FRED release reports 7,393,738 million dollars for non-financial corporate bonds and 1,137,472 million dollars for depository-institution loans n.e.c.

FIGURE C.1 – LTV Regulation with Externally Priced Bank Funding



Notes: Transition dynamics following a persistent negative productivity shock under a fixed externally priced marginal bank funding rate. Lines correspond to maximum LTV limits of 100, 80, 60, 40, and 20 percent. Top row: house price, consumption, and wage (percentage deviations from steady state). Middle row: foreclosure rate, household debt, and bank loans (levels). Bottom row: total household assets, capital stock (percentage deviations from steady state), and lending rate (level).

and bank lending stagnates. The loose economy thus retains its main self-correcting mechanism, leveraged entry by marginal buyers, while the tight economy loses the decline in r_ℓ that had partly compensated for the absence of that mechanism in the domestic-funding benchmark.

The gap in capital-stock dynamics across regimes also widens. In the tight economy, the capital stock falls by 6.9 percent, compared with 6.0 percent in the loose economy. With the funding rate fixed, the return on capital does not decline as much as in the domestic-funding benchmark, yet households in the tight regime still draw down financial buffers to adjust. Total household assets therefore contract by more, the capital stock falls further, and wages weaken by more. This amplifies the difference in aggregate outcomes across regulatory regimes.

The broader implication is that the stabilizing properties of LTV caps depend on the funding environment. When banks' marginal funding cost is determined outside the domestic economy, the endogenous fall in the funding rate that supported stabilization in the domestic benchmark is absent or attenuated. Tighter LTV regulation still prevents foreclosures, but its demand-suppression effects are no longer offset by a sustained decline in lending rates. In our model, the policy can therefore amplify declines in house prices

and consumption when banks' marginal funding cost is externally priced.

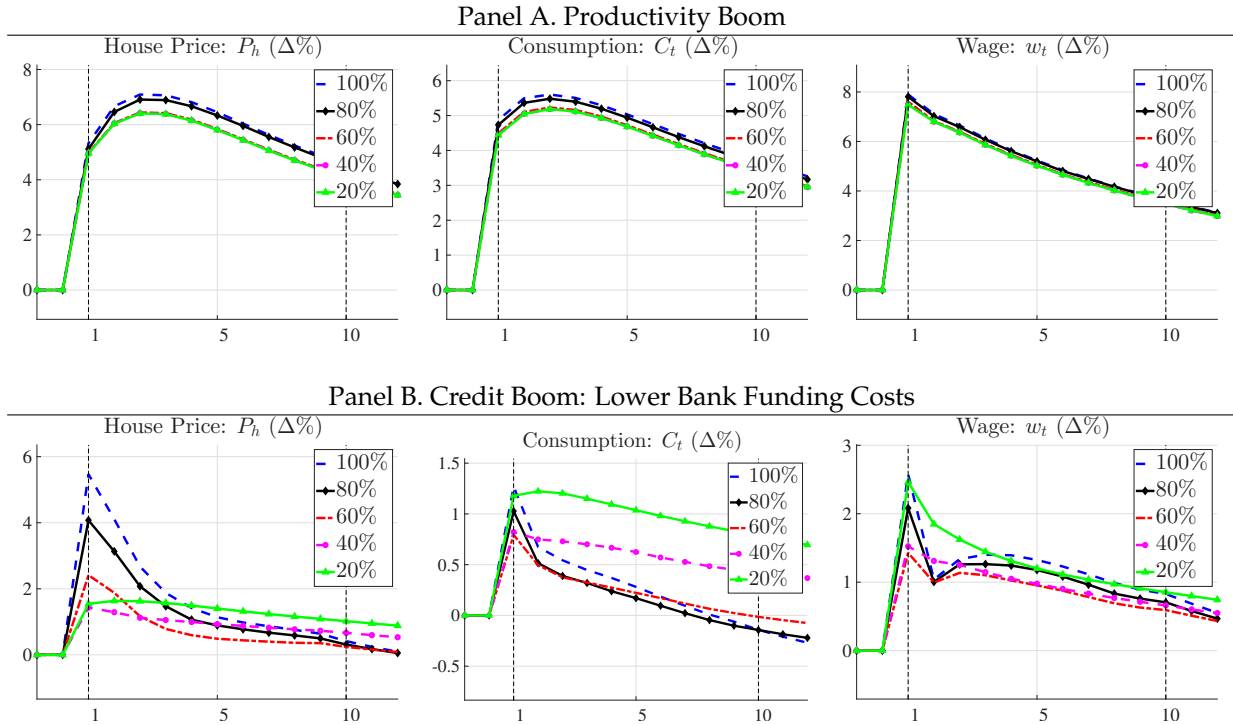
D Robustness

This appendix examines the robustness of the paper's conclusions along four dimensions: the sign of the aggregate shock, the capitalization of the banking sector, the choice of borrower-based instrument, and mortgage contract design. Two findings recur. First, the aggregate stabilization delivered by borrower-based limits continues to depend on the shock and on intermediary balance sheets rather than on household leverage itself. Second, the mechanisms of the main text are not specific to recessions, to the LTV instrument, or to fixed-rate contracts.

D.1 Boom Shocks

The main text studies recessions. Here we reverse the sign of both aggregate shocks: a persistent 5 percent productivity increase and a persistent decline in bank funding costs, the mirror images of the experiments in Sections 6 and 8. Figure D.1 reports house prices, consumption, and wages across the five LTV regimes.

FIGURE D.1 – Booms under Alternative LTV Regimes



Notes: Transition dynamics following expansionary shocks that mirror the recessions in the main text: a persistent 5 percent productivity increase (Panel A) and a persistent decline in bank funding costs (Panel B). Lines correspond to maximum LTV limits of 100, 80, 60, 40, and 20 percent. Panels show house prices, consumption, and wages as percentage deviations from steady state.

Tighter caps moderate productivity-driven booms about as weakly as they cushion busts. The peak house-price increase is 7.1 percent under the 100 percent cap and 6.4 percent under the 20 percent cap, a 0.68 percentage-point gap, slightly smaller than the 0.80 percentage-point bust gap; peak consumption and wage responses shrink by about 0.4 percentage points each. The slight asymmetry reflects the collateral channel: rising prices revalue incumbent equity, and sale proceeds finance the down payment on the next home, so the cap loses part of its bite precisely as the boom builds. Tight caps, by contrast, damp funding-cost booms: the peak house-price increase falls from 5.5 percent under the loosest regime

to about 1.5 percent under the tightest, through the same bank-portfolio mechanism that operates in the funding-cost bust. Fixed caps therefore lean against credit-driven booms through bank balance sheets, but they do not lean asymmetrically against productivity-driven booms: the tool moderates booms roughly as modestly as it cushions busts.

D.2 Bank Capital and the Effectiveness of LTV Limits

The benchmark calibration targets a bank leverage ratio of 10 and an annual steady-state lending spread of 1.5 percent. The experiment holds the bank leverage limit fixed exogenously, as an explicit capital requirement, and repeats the five-regime LTV comparison at limits of 15, 10, and 9.5 (the figures display 15 and 9.5), so the two grids show how the effectiveness of LTV regulation changes with bank capital.²¹ The choice of 9.5 comes from a sweep of leverage limits under the benchmark LTV cap: the steady-state face value of mortgage debt falls from 0.35 at the benchmark ratio of 10 to 0.21 at 9.5 and 0.02 at 9, and at 7.5 mortgage debt essentially disappears, so 9.5 is the lowest ratio that preserves a live mortgage market. Because leverage is imposed rather than market-determined, the comparison isolates the level of bank capital from the endogenous deleveraging that the enforcement constraint generates in the benchmark. We repeat both bust experiments under each calibration (Figures D.2 and D.3). Because the limit binds directly, the comparison is an experiment in bank capital rather than a joint sensitivity in leverage and spreads.

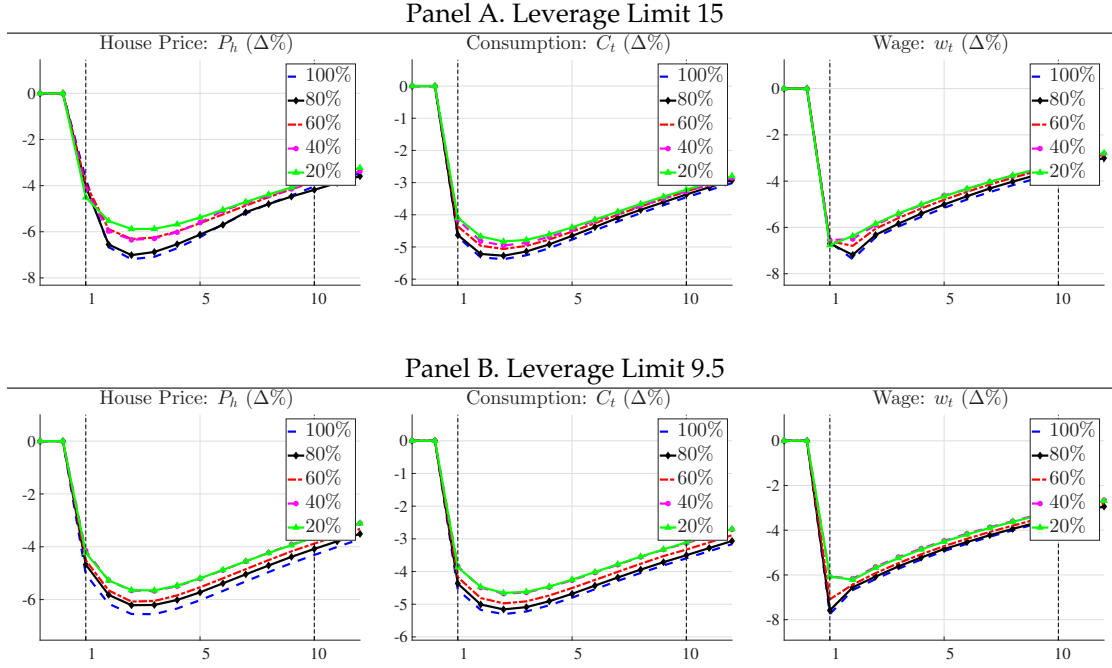
At a leverage limit of 15, the productivity-shock house-price trough is 7.19 percent under the 100 percent LTV limit and 5.88 percent under the 20 percent limit, a 1.31-percentage-point gap, compared with 0.80 in the benchmark. At a limit of 10 the gap is 1.17 percentage points and at 9.5 it narrows to 0.90 (6.55 versus 5.65), so the incremental stabilization from tighter caps rises with bank leverage. Consumption gaps are flatter, between 0.55 and 0.75 percentage points, without a clear leverage gradient, because the wage channel operates at every leverage level.

The state dependence is larger under the funding-cost shock. At a leverage limit of 15, under the 100 percent cap, bank net worth falls by up to 99 percent, the lending rate peaks near 15 percent, and house prices fall 8.83 percent at the trough, against 0.31 percent under the 20 percent cap, a gap of 8.5 percentage points. At limits of 10 and 9.5 the same shock barely transmits: house-price declines stay below half a percent at every cap and the cross-regime gap essentially vanishes. Tighter caps are therefore insurance against funding shocks precisely when bank capital is thin, and close to irrelevant when banks are well capitalized. The sweep of leverage limits under the benchmark cap tells the same story: as the limit falls from 20 to 9, the house-price trough shrinks monotonically from 7.64 to 6.11 percent and the consumption trough from 5.48 to 5.10 percent.

The alternative leverage choices also change the steady-state mortgage market. The aggregate market value of mortgages is 0.623 in the high-leverage calibration and 0.308 in the benchmark; homeownership below age 35 is 32.1 and 24.1 percent. At a ratio of 7.5 the aggregate mortgage value is approximately 0.000054 and young ownership 3.0 percent, the near-disappearance of mortgage borrowing that motivates the sweep's lower bound. The declining amplification along the sweep supports the case that the bank-mediated stabilization delivered by LTV caps weakens as bank balance sheets strengthen. The exercise therefore shows that LTV transmission depends jointly on intermediary leverage, mortgage spreads, and mortgage-market depth, but it does not establish that LTV caps and explicit capital requirements are substitutes. Establishing substitution requires a separate policy experiment that holds the benchmark structural parameters fixed, varies an explicit capital requirement, and allows lending spreads and mortgage and housing outcomes to respond endogenously while accounting for the general-equilibrium effects of capital regulation (Elenev et al., 2021). Low leverage therefore makes LTV caps less relevant partly by making the mortgage market nearly disappear: the sensitivity documents state dependence

²¹In the benchmark model, leverage, defined as $\frac{L_{t+1}}{N_t} = \hat{\lambda}_t$, is endogenous and depends on the future lending and funding rates for the bank. In this counterfactual we set $\hat{\lambda}_t = \bar{\lambda}$ for all periods.

FIGURE D.2 – Bank Capital and LTV Effectiveness: Productivity Shock



Notes: Transition dynamics after the benchmark productivity shock under alternative bank leverage. The leverage limit is imposed exogenously, as an explicit capital requirement. Panel A (top row): leverage limit 15; lines correspond to maximum LTV limits from 100 to 20 percent, and panels show house prices, consumption, and wages as percentage deviations from steady state. Panel B (bottom row): leverage limit 9.5, the lowest that preserves a mortgage market; lines and panels as in the top row.

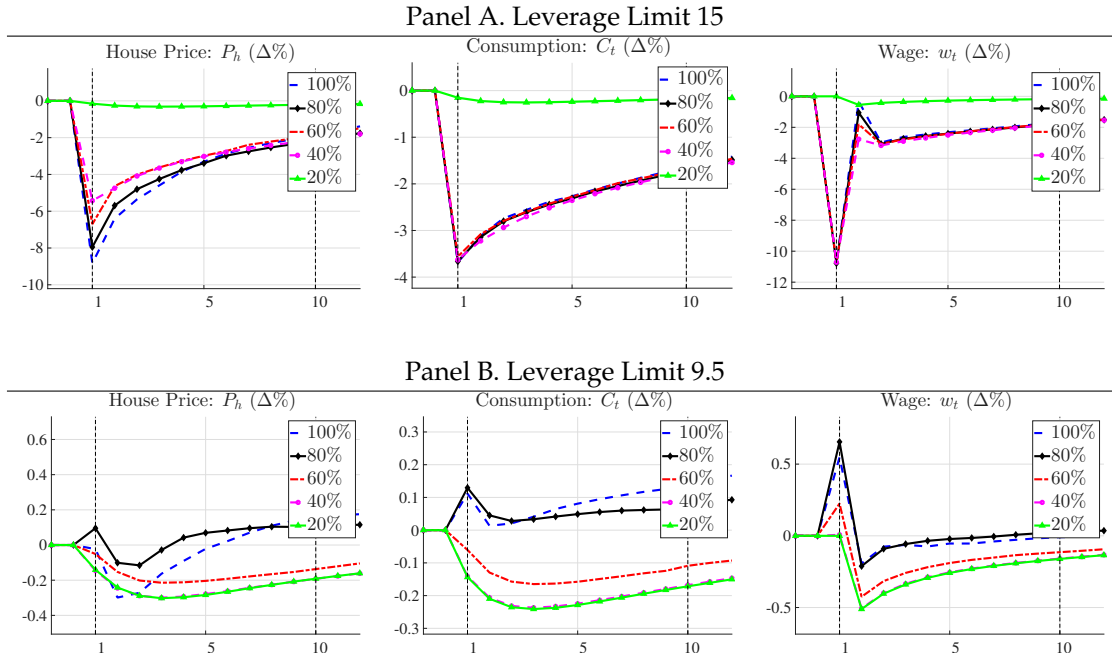
but cannot by itself establish that bank capital substitutes for borrower-based regulation. Appendix D.3 provides the complementary policy experiment.

D.3 Stabilization with Bank Capital Requirements

This subsection reports the explicit capital-policy experiment referenced in the main text. Leverage-ratio ceilings of this kind are standard prudential instruments in practice. The design complements Appendix D.2: there, the capital requirement is held fixed and the LTV cap varies; here, the LTV cap is absent and the capital requirement itself varies. In both designs the regulatory ceiling binds along the entire path, so the feedback from net worth to leverage is absent. We retain the benchmark calibration, with no borrower-based cap, and impose regulatory bank leverage caps of 20, 15, 10 (the benchmark), 9.5, 8, and 6, corresponding to capital requirements of 5 to 16.7 percent of assets. We hold all structural parameters fixed, and each policy defines a new equilibrium, so lending spreads, mortgage credit, homeownership, and house prices are outcomes of the policy rather than calibration targets (Table D.1).

Tighter requirements stabilize both shocks. The productivity-shock house-price trough shallows from 7.6 percent at the loosest cap to 5.3 percent at the tightest, and the funding-cost trough falls from 8.8 percent at a leverage cap of 15 to 0.5 percent at the benchmark cap of 10 and 0.4 percent at 9.5, the three cases for which we compute funding-cost transitions; dashes in the table mark cases we do not compute. At caps of 8 and 6 the steady-state mortgage stock is essentially zero, so a funding-cost transition there would operate only through the non-mortgage channels of the shock, working-capital lending and wages, rather than through the mortgage exposure that is this appendix's focus. The steady-state columns show the mechanism. Tighter capital raises the lending spread by roughly 140 basis points across the range, and mortgage credit responds: the mortgage stock roughly doubles relative to the benchmark when we loosen the cap to 15 or 20, and contracts to essentially zero at requirements of 12.5 percent and

FIGURE D.3 – Bank Capital and LTV Effectiveness: Funding-Cost Shock



Notes: Transition dynamics after the bank funding-cost shock under alternative bank leverage. The leverage limit is imposed exogenously, as an explicit capital requirement; top row leverage limit 15, bottom row 9.5, with lines corresponding to maximum LTV limits from 100 to 20 percent. Panels show house prices, consumption, and wages as percentage deviations from steady state.

above, with homeownership moving from 72 to 58 percent. In this model, mortgage intermediation is highly elastic to the spread at the benchmark, so capital policy operates on the same exposure margin that borrower-based caps target directly. A comparison with a market-determined leverage economy isolates a further channel: at the same steady-state leverage of 15, the productivity-shock bust is 7.19 percent deep when the regulatory ceiling binds and 14.75 percent when the enforcement constraint lets leverage contract with net worth, so a regulatory ceiling stabilizes partly by severing the feedback from bank net worth to leverage. The stabilization ranking of the two instruments is therefore an equilibrium property rather than a mechanical one: bank-facing requirements achieve it through funding costs and portfolio composition, borrower-based caps through origination constraints, and both converge on smaller mortgage exposure when set tightly.

TABLE D.1 – Capital Requirements: Steady States and Stabilization

Regulatory leverage cap	20	15	10 (benchmark)	9.5	8	6
Capital requirement (percent of assets)	5.0	6.7	10.0	10.5	12.5	16.7
Lending rate (annualized, percent)	6.0	6.1	6.5	6.6	6.8	7.4
Mortgage debt (percent of annual GDP)	111	106	53	32	≈ 0	≈ 0
Homeownership rate (percent)	72.0	70.0	64.0	62.3	58.4	58.5
House-price trough, productivity shock (percent)	-7.6	-7.2	-6.9	-6.6	-5.9	-5.3
House-price trough, funding-cost shock (percent)	--	-8.8	-0.5	-0.4	--	--

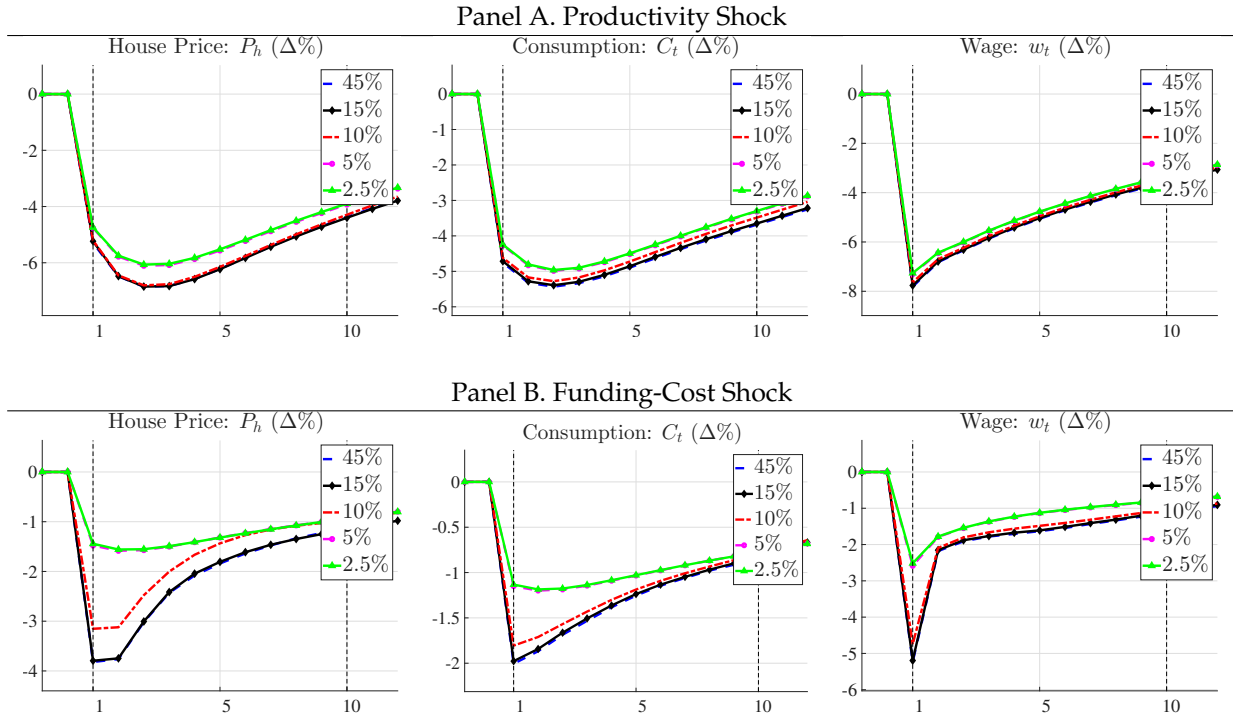
Notes: Each column is an equilibrium of the benchmark economy (100 percent LTV, no borrower-based cap) under a different regulatory cap on bank leverage, holding all structural parameters at their benchmark values. Lending rates, mortgage credit, homeownership, and house prices are equilibrium outcomes. Troughs are the largest house-price declines along the transition after the benchmark 5 percent productivity shock and the bank funding-cost shock. Funding-cost transitions are computed for caps 15, 10, and 9.5; dashes mark cases not computed.

D.4 Payment-to-Income Limits

The second most common borrower-based tool conditions the mortgage payment on income rather than the loan on collateral. We replace the LTV grid with payment-to-income limits of 45, 15, 10, 5, and 2.5 percent of current income. The 45 percent limit never binds in equilibrium and replicates the benchmark, and the constraint begins to bind between 15 and 10 percent (Figure D.4).

Tight PTI limits reproduce the LTV results almost exactly. Under the productivity shock, the house-price trough shrinks from 6.9 percent under the slack limit to 6.1 percent under the 2.5 percent limit, a 0.78 percentage-point gap, nearly identical to the 0.80 gap across LTV regimes, and the funding-cost bust shrinks from 3.8 to 1.6 percent. Fragile prudence is therefore a property of origination constraints in general, not of the LTV ratio in particular: whichever ratio rations the marginal borrower thins the same pool of downturn buyers. The equivalence is consistent with the emphasis on interacting borrower constraints in [Grodecka \(2020\)](#); our model adds the inference that aggregate dynamics depend on which constraint binds for the marginal buyer, not on its label.

FIGURE D.4 – Payment-to-Income Regulation and Macroeconomic Stabilization



Notes: Transition dynamics following the benchmark 5 percent negative productivity shock (Panel A) and the bank funding-cost shock (Panel B) when the borrower-based instrument is a payment-to-income limit. Lines correspond to maximum payment-to-income ratios of 45, 15, 10, 5, and 2.5 percent of current income; the 45 percent limit never binds and replicates the unconstrained benchmark. Panels show house prices, consumption, and wages as percentage deviations from steady state.

D.5 Adjustable-Rate Mortgages

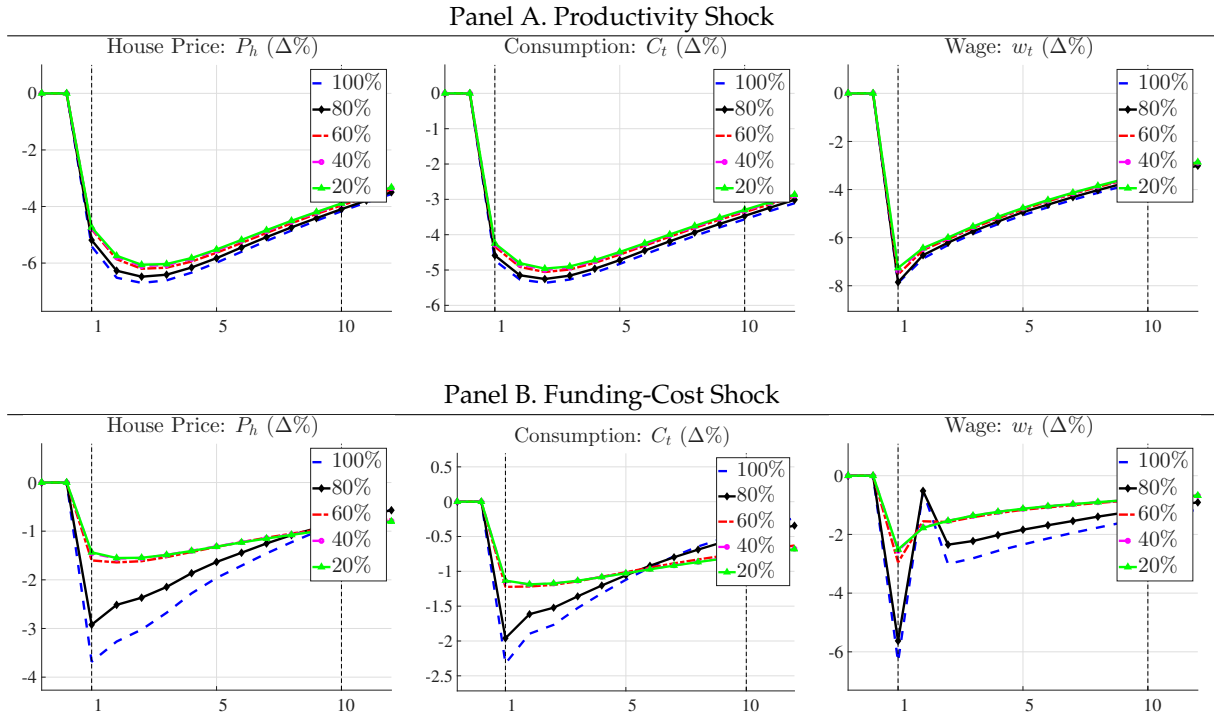
The benchmark's long-term fixed-rate contract concentrates interest-rate risk on bank balance sheets. We repeat both bust experiments in an economy with adjustable-rate mortgages, in which outstanding mortgage rates reset to the prevailing lending rate, so banks bear no duration mismatch (Figures D.5 and D.6).

Against the productivity shock, contract design is nearly irrelevant: the cross-regime house-price gap is 0.64 percentage points, against 0.80 under fixed rates, and the same holds when the instrument is a

PTI limit. Against the funding-cost shock, the contract reallocates the incidence of the shock without changing the verdict. With rate resets, higher funding costs pass through to existing borrowers' payments: foreclosures peak at 3.8 percent of owners in the loose economy, against 0.14 percent under fixed rates, and bank net worth falls by 51 percent at its trough, now through default losses rather than through the valuation of long-duration assets. Consumption and wages fall by more than under fixed rates (2.3 versus 2.0 percent, and 6.3 versus 5.2 percent), while the house-price decline is marginally smaller (3.7 versus 3.8 percent). Tight caps remain powerful, limiting the house-price decline to about 1.6 percent.

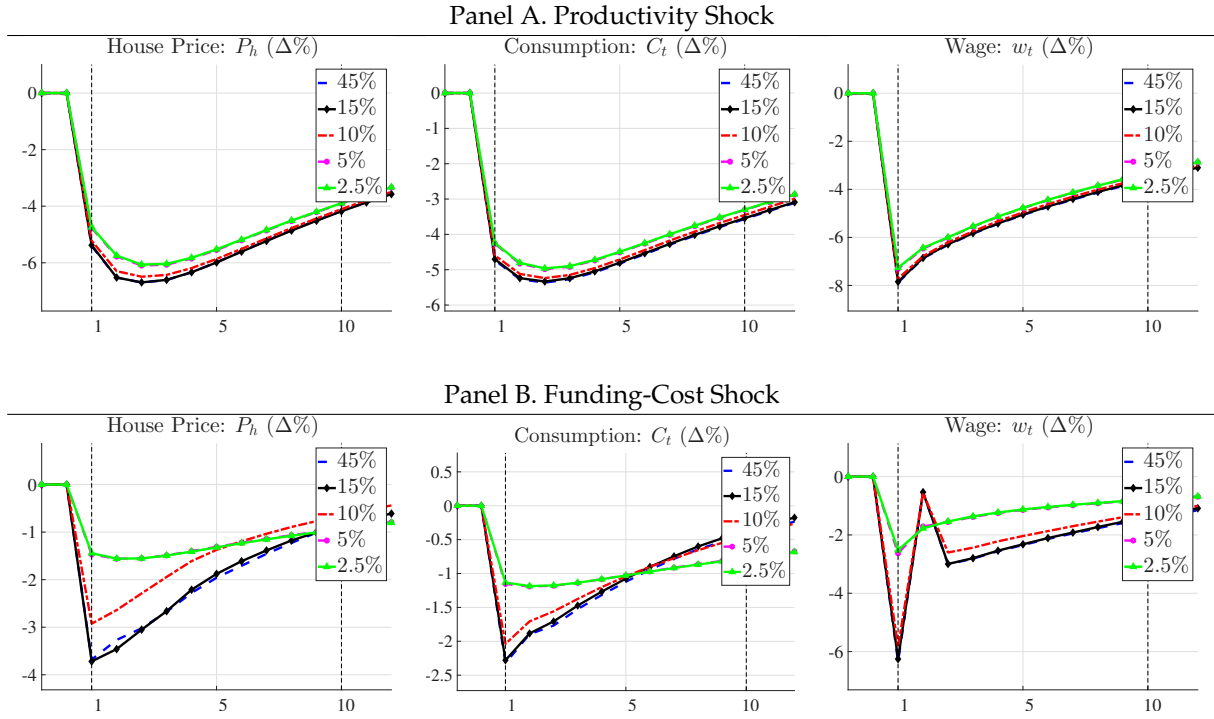
The experiment extends the funding-cost transmission result of Section 8. Under fixed rates, borrower-based regulation insulates the economy by shrinking banks' duration exposure; under adjustable rates, it insulates by shrinking the population of exposed borrowers. In both contract environments the caps mute the same funding-cost transmission, operating through whichever balance sheet, bank or household, bears the interest-rate risk. We study this trade-off between household insurance and bank stability in detail in [Arslan and Guler \(2025\)](#).

FIGURE D.5 – Adjustable-Rate Mortgages: LTV Limits



Notes: Transition dynamics in the adjustable-rate economy, in which outstanding mortgage rates reset to the prevailing lending rate, across maximum LTV limits from 100 to 20 percent. Panel A: benchmark productivity shock. Panel B: bank funding-cost shock. Panels show house prices, consumption, and wages as percentage deviations from steady state.

FIGURE D.6 – Adjustable-Rate Mortgages: Payment-to-Income Limits



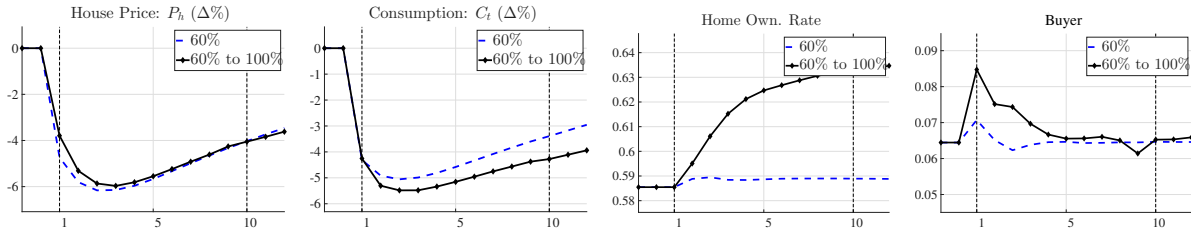
Notes: As in the previous figure, with payment-to-income limits of 45, 15, 10, 5, and 2.5 percent in place of the LTV grid.

D.6 Countercyclical Release in Partial Equilibrium

This subsection reports the partial-equilibrium version of the countercyclical experiment discussed in the main text: we hold factor prices at their fixed-regime paths, and only the house price clears its market. The short-run housing payoff of the release is several times larger than in general equilibrium, which identifies the lending-rate feedback—the higher lending rate crowding out firm credit and depressing wages—as the margin that absorbs most of the policy’s potency.

The answer to the section’s question is therefore mixed. Countercyclical LTV policy is more effective at reopening the entry margin during the bust. It raises the buyer rate and homeownership relative to a fixed 60 percent cap. But in general equilibrium it is not more effective at stabilizing aggregate outcomes. The house-price gain is small and temporary, while household debt rises and consumption is weaker for much of the transition. The policy changes the financing of housing demand rather than producing a stronger macroeconomic recovery.

FIGURE D.7 – Countercyclical LTV Policy: Buffer Release during the Bust (Partial Equilibrium)



Notes: Transition dynamics following the benchmark 5 percent negative productivity shock under two LTV regimes. The blue dashed line is the fixed 60 percent LTV economy; the black solid line is the countercyclical regime in which the cap rises immediately from 60 to 100 percent, remains at 100 percent for 20 model periods, and then returns to 60 percent. Panels show the house price, consumption, the homeownership rate, and the buyer rate. The house price and consumption are percentage deviations from steady state; the homeownership and buyer rates are in levels. Panels display the first 10 model periods; the scheduled release and reversion continue beyond the displayed window. In this diagnostic partial-equilibrium experiment, we hold all prices except the house price at their fixed-regime paths.

E Empirical Evidence on Asset Reliance Around Home Purchase

This appendix provides empirical discipline for the asset-reliance channel in the quantitative model. Each dataset has a distinct role: the PSID documents a purchase-timed break in financial-asset accumulation, and the SCF documents differences in asset participation and conditional balances among recent mortgaged buyers in a broader sample. The evidence does not use exogenous LTV-policy variation and therefore does not estimate the regulatory comparative static.

The appendix documents two complementary patterns. First, in the preferred PSID match, buyers who remain stockholders through the purchase window fall roughly 54 percent behind matched renter-stayers in stock growth; the log estimate excludes zero holdings and is scale invariant. Separately, four in ten stockholder buyers exit direct stockholding altogether. Second, SCF recent mortgaged buyers have lower stock and retirement participation and lower balances conditional on participation, while entering with higher leverage. These facts support the existence and sign of an internal-funds margin without identifying its elasticity to regulation; American Housing Survey descriptives on down payments in the national buyer population are in the online appendix.

E.1 Scope and Identification Limits

The empirical evidence is consistent with the asset-reliance margin and with broader balance-sheet patterns around mortgage-financed purchase; it does not identify the LTV-regulation comparative static. The PSID follows households over time and measures purchase-window asset changes relative to matched renter-stayers, but its clean within-household liquidation margin is direct stockholding. That limitation is about measurement, not about the mechanism. Non-stockholder buyers may finance purchase by drawing down deposits, redirecting income flows, using family transfers, selling unmeasured assets, or accepting lower post-purchase buffers. The SCF has larger samples and separates recent cash buyers from recent mortgaged buyers, but it is a repeated cross-section rather than a panel. Its estimates are therefore descriptive financing-margin facts, not within-household liquidation estimates or causal effects of LTV regulation.

This distinction guides the appendix. The PSID provides timing and the SCF provides breadth. The online appendix reports construction, matching, and sensitivity diagnostics, including a small-sample Dutch replication (Appendix F.4).

E.2 PSID Panel Evidence

The PSID sample uses wealth waves from 1999 to 2021, restricted to SRC core, reference-couple families with reference persons aged 25 to 70. A buyer rents in both pre-period waves, becomes an owner in the

event wave, and originates a new purchase mortgage in that window; renter-stayers rent throughout and have never owned. Both members of a candidate pair hold direct stocks before purchase. Within this symmetric pool we match buyers to renter-stayers on a propensity score estimated from age, education, marital status, race, income, wealth, family size, and wave, without replacement within a 0.05 caliper, which yields 119 pairs. This match is separate from the 681-pair match that underlies the broad buyer-sample results and its balance table; re-estimating the score within the symmetric pool leaves a largest absolute standardized mean difference of 0.14 (Table OA.3).

Our preferred estimate excludes zero holdings and uses logs, so it is scale invariant. Among the 49 pairs in which both members remain stockholders through the purchase window, buyers' stock growth is 0.77 log points, roughly 54 percent, below matched renters (buyer-lineage clustered standard error 0.28; pair sign-flip $p = 0.007$). On the extensive margin, 38.7 percent of buyers (46 of 119) and 30.3 percent of matched renter-stayers (36 of 119) exit direct stockholding in the purchase window; the 8.4-percentage-point buyer-renter difference is positive but imprecisely estimated (buyer-lineage clustered $p = 0.18$). Because the log specification conditions on continued positive stockholding, it measures the intensive margin and excludes complete exits from stockholding. The separate exit estimate describes that extensive margin. The pre-period placebo is negative but statistically indistinguishable from zero (Table E.1).

TABLE E.1 – PSID: Stock Adjustment around Home Purchase (Selected Estimates)

	Estimate	SE	p	N
Purchase-window log gap, continuing stockholders	−0.771	0.282	0.007	49 pairs
Pre-period placebo (log)	−0.516	0.317	0.115	54 pairs
Exit from stockholding, buyers minus renters (pp)	8.4	6.3	0.183	119 pairs
Pooled OLS of log change on buyer indicator	−0.279	0.196	0.156	411

Notes: Symmetric stockholder propensity-score match, without replacement (119 pairs). Log estimates exclude zero holdings. We cluster standard errors by buyer lineage; the p -values for the paired estimates are pair sign-flip randomization values. Buyers' and renters' exit rates are 38.7 and 30.3 percent. The pooled row regresses individual purchase-window log changes on a buyer indicator with the matching covariates and wave fixed effects. The pooled sample is the symmetric candidate pool restricted to positive holdings at both dates (411 person-observations), not only the 119 matched pairs.

These are event-time facts about mortgage-financed purchase, not causal effects of LTV regulation. The online appendix reports data construction, match-quality diagnostics, and descriptive statistics.

E.3 SCF Financing-Margin Evidence

The SCF evidence pools the 2004 to 2022 waves. The sample comprises homeowners who purchased their primary residence, excluding gift and inheritance acquisitions; a recent buyer purchased within two years of the survey, and recent buyers split into cash and mortgaged purchasers. Regressions include log income, age, age squared, education, marital status, number of children, race, and wave fixed effects; each implicate carries one-fifth of the household weight, and inference uses the 999 official replicate weights with between-implicate variance correction.

Recent mortgaged buyers are 2.6 percentage points less likely to hold stocks and 2.6 points less likely to hold retirement assets than comparable long-time purchasers, and, conditional on holding, their stock balances are about 24 percent lower and retirement balances about 26 percent lower, log estimates that exclude zeros and are therefore scale invariant. They also enter with higher leverage: LTV ratios about 15 percentage points higher, debt-to-income 0.77 higher, and payment-to-income about 18 percentage points higher. Recent cash buyers do not display the same broad pattern: they have lower retirement participation but higher retirement balances conditional on holding, and are positively selected on liquid wealth (Table E.2).

TABLE E.2 – SCF: Recent Mortgaged Buyers versus Long-Time Buyers (Selected Estimates)

	Estimate	SE	<i>p</i>
Holds stocks (pp)	−2.6	0.9	0.003
Holds retirement assets (pp)	−2.6	0.8	0.001
Log stocks, holders only	−0.278	0.092	0.003
Log retirement, holders only	−0.303	0.036	< 0.001
LTV ratio	0.149	0.006	< 0.001
Debt-to-income	0.771	0.065	< 0.001
Payment-to-income	0.184	0.040	< 0.001

Notes: SCF waves 2004–2022; coefficients on the recent-mortgaged-buyer indicator (leverage rows: recent versus long-time mortgaged purchases). All specifications include the controls listed in the text. Standard errors are design corrected, combining the 999 official replicate weights with between-implicate variance. Log outcomes exclude zero holdings and are scale invariant.

The SCF is cross-sectional and does not prove within-household liquidation; its contribution is breadth. Deposits show no comparable shortfall, consistent with their role as the transaction vehicle, drawn down at closing and rebuilt from income flows between waves, so the persistent footprint of the internal-funds margin sits in stocks and retirement assets. Administrative-data studies with high-frequency within-household data do observe reduced liquid balances when down-payment requirements tighten ([van Bakkum et al., 2024](#); [Aastveit et al., 2024](#)); we read this as a drawdown that is real but rebuilt too quickly for survey waves to record. The online appendix reports sample construction, weighted group summaries, and the full estimation tables, including Table [OA.8](#).

E.4 Interpretation for the Quantitative Model

The empirical appendix provides descriptive discipline for a mechanism, not an empirical test of LTV regulation. The PSID documents a purchase-timed drawdown of stocks among prior stockholders, and the SCF documents lower participation and conditional balances among recent mortgaged buyers. Together, the data show that internal funds matter at purchase without reducing the mechanism to direct-stock liquidation.

The structural model supplies the LTV comparative static. Tighter caps make housing adjustment more dependent on internal funds, so crisis-induced buyers and movers must finance more of the adjustment from assets and current resources. The household evidence documents this margin; it does not estimate its policy elasticity or the crisis-specific response.

The main appendix keeps the compact PSID and SCF evidence needed for the asset-reliance interpretation. The online appendix reports construction, match-quality, and event-pattern materials, together with a small-sample Dutch replication. We do not interpret any survey exercise as an estimate of the effect of LTV regulation.

ONLINE APPENDIX

F Additional Empirical Materials

This online appendix provides additional detail for Appendix E. It reports how we build the PSID and SCF samples, documents covariate balance and summary statistics, shows raw event-pattern figures that motivate the two core tables in Appendix E, and reports the full SCF regression table. These materials document the empirical design and robustness of the descriptive evidence; the main appendix focuses on the compact tables needed for the asset-reliance interpretation.

F.1 PSID Data Construction, Balance, and Descriptive Patterns

We construct the PSID panel from the 1999–2021 wealth waves. We limit the sample to the SRC core sample, reference-couple households, reference persons aged 25–70, and active respondents. Table OA.1 reports the filter path from the Cross-Year Individual File to the family-year analysis panel.

TABLE OA.1 – PSID Sample Construction

Filter step	Person-year observations
Cross-Year Individual File, all individuals 1968–2021	3,533,082
Restrict to wealth waves (1999, 2001, ..., 2021)	1,009,452
SRC core sample	428,076
Reference couple	100,947
Age 25–70	86,083
Active panel response	86,006
Family-year deduplication, one observation per family-year	53,450

Notes: The final analysis panel is at the family-year level. The reference-couple restriction tracks the family-level wealth, consumption, and tenure outcomes reported in the PSID wealth supplement.

Table OA.2 reports match-quality diagnostics for the main propensity-score match. The match improves balance on every covariate: the largest post-match absolute standardized difference is 0.11 (marital status), and age and liquid wealth, the least balanced covariates before matching, fall to 0.05 and 0.03. The main appendix emphasizes the stockholder specification and pre-period placebo as diagnostics for the clean PSID margin. Table OA.3 reports the same diagnostics for the preferred 119-pair symmetric stockholder match, where the largest absolute standardized difference is 0.14.

Table OA.4 summarizes the PSID tenure-transition groups before the purchase window. Became-owner households are wealthier and more likely to be married than always-renters, but much younger and less wealthy than always-owners. The stockholder share among became-owners is 16.8 percent, which explains why the clean direct-stock evidence is a subgroup result; it does not imply that the asset-reliance channel operates only among stockholders.

Figure OA.1 plots raw indexed wealth-component holdings around purchase. The left panel shows all became-owner buyers; the right panel restricts to buyers with direct stocks before purchase. The stockholder panel visually motivates the matched-pair result: direct stocks rise before purchase, fall after the purchase wave, and partially rebuild later.

F.2 SCF Data Construction and Descriptive Patterns

The SCF analysis uses the 2004, 2007, 2010, 2013, 2016, 2019, and 2022 public-use waves. Table OA.5 reports the construction of the homeowner sample and the split between recent cash buyers, recent mortgaged buyers, and long-time buyers.

TABLE OA.2 – PSID Match-Quality Diagnostics at $t - 4$

Covariate at $t - 4$	Post-match mean		Standardized mean difference	
	Buyer	Renter	Pre-match	Post-match
Age	34.2	34.7	−0.287	−0.052
Family income	91,539	90,594	+0.454	+0.043
Net worth excl. home	97,007	69,231	+0.164	+0.017
Liquid Tier 1	32,061	36,257	+0.639	+0.026
Family size	2.46	2.49	+0.046	−0.019
Bachelor’s degree or higher	58.7%	58.1%	+0.422	+0.012
Married	43.9%	49.5%	+0.389	−0.112

Notes: We report buyer and renter means on the matched sample. We compute standardized mean differences before matching on the full candidate pool and after matching on the 681 matched pairs. Post-match differences use the 681 matched pairs with renters entering as matched (with replacement); the pre-match pool contains 684 buyer and 5,176 renter-stayer triplets. For income, net worth, and liquid wealth, standardized differences are computed on the log or inverse-hyperbolic-sine transforms that enter the propensity score; means are in dollars at $t - 4$. The preferred stockholder estimates use a separate 119-pair match with the propensity score re-estimated within the symmetric stockholder pool.

TABLE OA.3 – Match-Quality Diagnostics for the Symmetric Stockholder Match

Covariate at $t - 4$	Buyer mean	Renter mean	SMD pre-match	SMD post-match
Age	35.1	35.7	−0.25	−0.06
Log family income	11.5	11.6	0.23	−0.12
IHS net worth excl. home	7.8	9.0	0.20	−0.14
IHS liquid Tier 1	10.7	11.0	0.21	−0.12
Family size	2.0	2.1	0.06	−0.10
College (education 3 or higher)	0.79	0.76	0.19	0.06
Married	0.40	0.41	0.26	−0.02
Majority racial-ethnic group	0.96	0.94	0.05	0.08

Notes: Balance for the preferred symmetric stockholder match: both members of a candidate pair hold direct stocks before purchase, the propensity score is re-estimated within this pool with the same specification as the main match, and buyers are matched to renter-stayers exactly on the calendar wave within a 0.05 caliper, without replacement (119 pairs). Means are post-match. Standardized mean differences use the pooled standard deviation and are computed on the symmetric candidate pool before matching and on the matched pairs after matching. College denotes education level 3 or higher; the majority racial-ethnic row is the share in the sample’s modal category. Education, marital status, and race enter the propensity score.

Table OA.6 reports weighted summary statistics by buyer group. The descriptive pattern previews the regression evidence. Recent cash buyers are older and extremely wealthy, consistent with positive selection into outright purchase. Recent mortgaged buyers are younger than long-time buyers and have lower non-housing wealth, while still being much wealthier than renters.

F.3 Down Payments in the Broad Buyer Population (AHS)

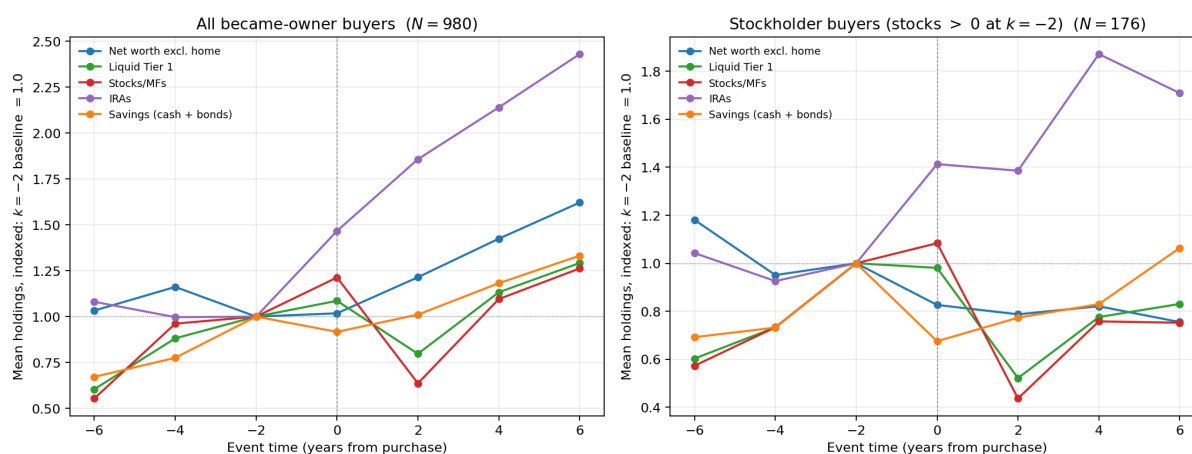
The PSID and SCF measure financial-asset adjustment around purchase in selected survey margins. The 2023 American Housing Survey National Public Use File shows that down payments are widespread in the national owner-buyer population. Among recent owner-buyers who moved between 2019 and 2023, 89.5 percent report a down payment and 43.2 percent report at least 16 percent down; even among first-home buyers, 31 (Table OA.7).3 percent report at least 16 percent down. The AHS does not identify

TABLE OA.4 – PSID Summary Statistics by Tenure-Transition Group

Variable at $t - 2$ unless noted	Always renter	Became owner	Always owner	Lost owner
Family income	52,436	85,335	119,283	82,809
Net worth excl. home	3,171	15,685	80,856	16,042
Liquid Tier 1 wealth	903	4,707	13,888	2,778
Stocks/MFs, conditional on holding	15,231	20,832	61,746	23,907
Stocks > 0	9.1%	16.8%	29.2%	16.7%
Reference-person age	39.5	37.8	47.5	41.2
Married	30.1%	57.6%	78.8%	56.7%
Observations	8,501	1,848	25,959	944

Notes: Dollar variables are in real dollars. Became-owner households rent at the pre-period wave and own at the purchase wave. The stockholder share refers to positive direct stock or mutual fund holdings before purchase.

FIGURE OA.1 – PSID Mean Wealth-Component Holdings around Home Purchase



Notes: Each series plots the mean dollar holding of the indicated wealth component across became-owner buyers at each event time, divided by the same series' value at $k = -2$. The vertical dashed line marks the purchase wave. The left panel uses all became-owner buyers; the right panel restricts to buyers with positive direct stock holdings at $k = -2$.

the source of those funds, but it indicates that the internal-funds requirement is broad rather than a feature of the PSID stockholder sample.

F.4 DNB Household Survey (DHS) Replication

As an external replication, we use the DNB Household Survey (DHS) to compare financial-asset adjustment around renter-to-owner transitions with that of households that remain renters or owners. We require explicit renter-renter-owner responses in the three event-window observations and treat years in which all risky-asset items are missing as nonresponse rather than zero holdings. These screens reduce the usable pre-purchase risky-asset sample to 30 buyer events.

In the resulting sample, the buyer-minus-continuing-owner decline in risky assets is large and statistically significant (household-clustered $p=0.004$) after including year fixed effects; outcomes are inverse-hyperbolic-sine changes in holdings. Adding demographic controls gives -2.660 (SE 1.120, $p=0.018$)

TABLE OA.5 – SCF Sample Construction

Filter step	Household-implicate observations
Pooled SCF waves 2004–2022	190,265
Homeowners	130,585
Owners who purchased, excluding gift/inheritance	120,910
Recent buyer, acquired in last two years	15,708
with current mortgage	12,601
outright / cash buyer	3,107
Long-time buyer, acquired more than two years ago	105,197

Notes: The SCF stores five multiple imputations of each household. Counts are household-implicate observations; unique household counts are approximately one fifth of the reported row counts.

TABLE OA.6 – SCF Summary Statistics by Buyer Group, 2004–2022

Variable	Renter	Recent cash	Recent mortgaged	Long-time buyer
Family income	57,316	217,442	156,220	163,738
Net worth excl. home	105,663	1,776,316	495,831	1,050,598
Liquid Tier 1 wealth	36,413	628,350	144,245	318,712
Stocks/MFs, conditional on holding	19,419	391,168	93,125	213,234
Stocks > 0	10.5%	32.1%	28.2%	30.8%
Age	43.3	57.5	40.3	57.0
Married	39.0%	58.8%	73.9%	68.2%
Households	11,936	624	2,529	21,044

Notes: Weighted means use the per-implicate SCF population weight. Liquid Tier 1 wealth equals deposits, certificates of deposit, stocks, mutual funds, bonds, and savings bonds. Dollar variables are in real dollars.

TABLE OA.7 – Down Payments in the 2023 American Housing Survey

	All owner-buyers	Recent owner-buyers (moved 2019-2023)	First-home owner-buyers	Prior-renter owner-buyers
Weighted households, millions	77.2	20.4	35.1	51.9
Made any down payment	88.3	89.5	84.2	88.0
Reported positive down-payment amount	75.5	80.1	71.9	75.9
No down payment	11.7	10.5	15.8	12.0
Down payment at least 16 percent	43.3	43.2	31.3	41.9
Down payment at least 21 percent	28.8	28.3	18.9	26.1
100 percent down payment / outright	14.3	13.4	10.0	11.4
First home ever owned	45.4	42.6	100.0	46.9
Prior renter purchase	67.2	65.1	69.4	100.0

Notes: 2023 AHS National PUF, household weights. Sample: owner-occupied households that bought or built the home; we exclude gift and inheritance acquisitions. Entries are weighted percentages of each column sample with valid down-payment information. Shares with down payments of at least 16 or 21 percent are lower bounds because a separate category absorbs down payments of unreported size.

for 19 buyer events, the buyers with complete covariates; a separate symmetric 28-pair buyer-renter match gives -2.717 with a sign-flip p-value of 0.015. The result corroborates purchase-linked risky-asset adjustment in a second country.

TABLE OA.8 – SCF Asset Margins and Leverage among Recent Purchasers

Panel A: Asset participation and positive-holder balances				
	Stockholder	Retirement holder	Log stocks	Log retirement
Recent cash buyer	0.029 (0.020)	−0.049** (0.023)	0.245 (0.195)	0.362*** (0.105)
Recent mortgaged buyer	−0.026*** (0.009)	−0.026*** (0.008)	−0.278*** (0.092)	−0.303*** (0.036)
Households	24,181	24,181	10,932	17,668
Panel B: Leverage among purchase mortgages				
	LTV	Debt/income	Payment/income	
Recent mortgaged buyer	0.149*** (0.006)	0.771*** (0.065)	0.184*** (0.040)	
Households	6,303	6,258	6,303	
Panel C: Full-distribution checks for recent mortgaged buyers				
	Liquid fin. (IHS)	Deposits (IHS)		
Recent mortgaged buyer	−0.109*** (0.038)	−0.027 (0.035)		
Households	24,181	24,181		

Notes: Pooled 2004–2022 SCF. The omitted group in Panels A and C is long-time purchasers; Panel B compares recent and long-time purchasers with current purchase mortgages. All specifications include the demographic controls and wave fixed effects described in the text. The first two Panel A outcomes are linear probability models; the next two are log balances among positive holders. Point estimates average five implicate-specific weighted regressions. Sampling variances use 999 official first-implicate replicate weights multiplied by multiplicity factors; total variance adds (6/5) times between-implicate variance. Panel C reports distributional checks in inverse-hyperbolic-sine (IHS) units; we do not translate them into percentages. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.5 SCF Regression Table

This subsection reports the full SCF regression table underlying the compact design-corrected estimates. Appendix E.3 reports the compact design-corrected SCF regression evidence; the online appendix reports sample construction and weighted descriptive statistics.

G Computational Appendix

A household's state is $\theta = (a, h, \tilde{d}, z, j, o)$ where o denotes housing tenure, j denotes the household's age, z denotes the income efficiency shock, $\tilde{d}$ denotes mortgage debt as a ratio to the initial house value, h denotes the owner-occupied house size, and a denotes financial wealth.²² For renters, active or inactive renters ($o \in \{r, e\}$) and $h = d = 0$. The grid for a has 120 exponentially spaced points, and the grid for $\tilde{d}$ has 36. Age j takes values 1 to 30, and h lies on a linear grid with 7 points. The income shock z takes 15 values, with nodes and transition probabilities from the Tauchen method. Because the model has a life-cycle structure, the income nodes vary with age, which sharpens the Markov approximation to the AR(1) process. The retirement system follows the U.S. social-security benefit formula as in Guvenen and Smith (2014). We scale retirement income so that working-age households face a 12 percent payroll tax.

²²We replace the level of debt, d , with loan-to-value ratio, $\tilde{d}$, in the computation as it allows us to use a compact space independent of house grid and house price.

G.1 Steady-State Computation

We compute the steady state in the following steps:

1. The bank's problem pins down the steady-state lending rate, $r_\ell = r + \frac{1-\beta_L(1+r)}{\lambda\beta_L}$.
2. Guess K , H and N_s .
3. Free labor mobility across sectors gives N_h :

$$N_h = \left(\frac{\delta_h H}{\bar{L} \theta_h} \right)^{\frac{1}{1-\theta_h}}$$

4. Labor-market clearing then delivers N_g :

$$N_g = 1 - N_h - N_s$$

5. The goods producer's optimality conditions yield w , u , and r :

$$\begin{aligned} u &= \left(\frac{(1-\alpha) \frac{K}{N_g}}{(1+\mu r_\ell) \vartheta} \right)^{\frac{1}{\alpha+\psi}} \\ w &= \vartheta^{\frac{\alpha-1}{\alpha+\psi}} \left(\frac{(1-\alpha) \left(\frac{K}{N_g} \right)^\alpha}{(1+\mu r_\ell)} \right)^{\frac{1+\psi}{\alpha+\psi}} \\ r &= \alpha \left(\frac{K}{N_g u} \right)^{\alpha-1} - \delta_k \end{aligned}$$

6. The construction firm's problem gives p_h :

$$p_h = \frac{w}{(1-\theta_h) \left(\frac{\bar{L}}{N_h} \right)^{\theta_h}}$$

7. The rental company's problem prices rental services:

$$p_r = \kappa + \frac{r + \delta_h}{1+r} p_h,$$

the steady state of (A.9).

8. With prices in hand, we solve the household problem by backward induction:
 - (a) In the terminal period all dynamic choices equal zero: $a' = \tilde{d}' = 0$. This yields the terminal value, $V_j(\theta)$, and the mortgage contract's continuation value, $p_\ell(\theta; j)$.
 - (b) Given $V_j(\theta)$ and $p_\ell(\theta; j)$, solve $V_{j-1}(\theta)$ and $p_\ell(\theta; j-1)$:
 - i. Given $V_j(\theta)$ and $p_\ell(\theta; j)$, first form the expectations $EV_j(\theta)$ and $Ep_\ell(\theta; j)$.
 - ii. Then compute origination mortgage prices, $q^m(\theta)$.
 - iii. The inactive renter, and the active renter who stays a renter, solve a standard problem over housing services, consumption, and saving. We evaluate expected continuation values by linear interpolation. For saving, a global search on a fine discrete grid for a' brackets the maximum, and Brent's method then locates the optimum. Given saving,

a closed form delivers housing services.²³ Consumption then follows from the budget constraint.

- iv. The renter who buys a house solves the most demanding problem, choosing consumption, saving, house size, and mortgage debt. Down payments and house sizes lie on finite sets: for the down payment, the grid for $\tilde{d}$ defines the choice set, and for house size the grid for h does.²⁴ For each down-payment and house-size pair, we maximize the objective, $V_{j-1}^{d,h}$, over saving, as in step 8(b)iii. Household choices then deliver $q^m(\theta)$. Points off the grid use linear interpolation. A choice of $\tilde{d}$ and h implies mortgage debt $\tilde{d}p_h^{ss}h$ where p_h^{ss} is the initial steady-state equilibrium price. With the objective solved for each pair, the buyer's value is $V_{j-1}(\theta) = \max_{d,h} \{V_{j-1}^{d,h}\}$.
- v. The homeowner's problem splits by adjustment type:
 - A. Stayer: a stayer chooses only consumption and saving, so the solution mirrors the inactive renter's. The one difference is that the continuation value updates the remaining-balance state, $\tilde{d}$. Given current $\tilde{d}$, $\tilde{d}' = (\tilde{d} - m(\tilde{d}))(1 + r_\ell)$ where $m(\tilde{d}) = \tilde{d} \frac{r_\ell(1+r_\ell)^{J-j}}{(1+r_\ell)^{J-j+1}-1}$. Linear interpolation over $\tilde{d}$ gives the expected continuation value.
 - B. Seller: a seller solves the active renter's problem with the sale proceeds added to the budget constraint: $p_h h (1 - \varphi_s - \delta_h) - \tilde{d}p_h^{ss}h$
 - C. Refinancer: a refinancer solves the buyer's problem restricted to the current house.
 - D. Defaulter: a defaulter solves the inactive renter's problem.
- vi. The homeowner's solution also yields the payment on each contract type and the mortgage contract's continuation value, $p_\ell(\theta; j-1)$:

$$p_\ell(\theta; j-1) = m(\theta) + \frac{1}{1+r_\ell} E_{\theta'|\theta} p_\ell(\theta'; j),$$

where

$$m(\theta) = \begin{cases} \tilde{d}hp_h & \text{if } o \in \{hr, hf\} \\ p_h h (1 - \varphi_e) & \text{if } o = he \\ \frac{r_\ell(1+r_\ell)^{J-j}}{(1+r_\ell)^{J-j+1}-1} \tilde{d}hp_h^{ss} & \text{if } o = hh \end{cases}$$

(c) Repeat step (b) for each $j = \{J-1, \dots, 1\}$.

9. Given the household policy functions, we simulate 20,000 households in each of the $J = 30$ generations, for a total of 600,000 simulated households. The resulting cross-section yields the stationary distribution λ , from which we build aggregate saving, A , aggregate housing demand, H^d , aggregate rental demand, H^r , aggregate loan demand $L = \mu w N_g + \int_\theta p_\ell(\theta) \lambda(\theta)$, and residual costs TC .
10. Given aggregate loan demand, the bank's problem determines bank liabilities B .
11. Given rental demand and prices, the rental company's problem values the rental sector: $V^{rc} = \frac{p_r - \kappa - \delta_h p_h}{r} H^r$
12. Given wages and residual costs, solve for residual-sector employment $N_s^0 = \frac{TC}{w}$
13. If the asset-market, housing-market, and residual-employment conditions hold, i.e. $\max(|A - K - B - V^{rc}|, |H - H^d|, |N_s - N_s^0|) < \epsilon$ where $\epsilon = 10^{-4}$, stop.

²³CES utility over consumption and housing services admits a closed form for the housing choice.

²⁴Adding points to the grids for $\tilde{d}$ and h leaves the results unchanged to the reported precision.

Otherwise update the guesses, $K = (1 - \lambda_k) K + \lambda_k (A - B - V^{rc}(H^r))$, $H = (1 - \lambda_h) H + \lambda_h H^d$, and $N_s = (1 - \lambda_s) N_s + \lambda_s N_s^0$ where $\lambda_k = \lambda_h = \lambda_s = 0.1$, and return to step 3.

14. With the equilibrium in hand, the construction firm's problem delivers q_p .

G.2 Transition Algorithm

Two features distinguish the transition problem. Equilibrium prices and allocations now form a path, and the algorithm must handle movements in the lending rate relative to the fixed reference mortgage rate along that path. The second feature matters because we keep the mortgage rate out of the state vector: borrowers pay points at origination that bring the risk-adjusted rate down to the reference mortgage rate. After the shock, all new mortgages are priced at the steady-state reference mortgage rate, so origination cohorts need not carry separate rates, and banks build this into mortgage pricing. After a negative productivity shock the lending rate falls. Discounting future payments at the steady-state risk-free rate then means that, absent default risk, the bank collects more than the principal at origination, so the mortgage price q^m exceeds 1; the excess compensates the borrower for the above-market payments made in later periods. The arrangement resembles lender credits in practice, and the lending rate's deviation from steady state is short-lived in our experiments.

With these modifications, the algorithm proceeds as follows:

1. Fix the transition horizon at T periods; we set $T = 60$, or 120 years.
2. Solve the initial steady state as above and store its distribution, $\lambda_0(\theta)$.
3. Because the shock is temporary, the final steady state coincides with the initial one; store $V_T(\theta)$ and $p_\ell^T(\theta)$.
4. Guess paths for capital, the housing stock, rental demand, the lending rate, and residual sector employment: $\left\{ K_t, H_t, H_t^{r,0}, r_{\ell,t+1}^0, N_{s,t} \right\}_{t=1}^{T-1}$.
5. From these guesses, compute the price vector $\vec{p}_t \equiv \{w_t, r_{t+1}, p_t^h, p_t^r\}$ from the goods producer's, construction firm's, and rental company's problems, and V_t^{rc} from the rental company's problem.
6. Solve every cohort's problem over its lifetime, from the cohort born in period $-J + 2$ through the cohort born in period $T - 1$ ²⁵:
 - (a) Given prices and next-period value functions, solve each generation's problem and the mortgage contract's continuation value (p_t^ℓ) exactly as in the steady state.
 - (b) Using the generations' policy functions, simulate the economy forward from $\lambda_0(\theta)$ for T periods, holding the idiosyncratic shock draws fixed across iterations. The simulation yields the transition-path distributions $\{\lambda_t(\theta)\}_{t=1}^T$.
 - (c) From the simulated path, compute the aggregates: $A_t, H_t^{r,1}, H_t^d, M_t = \int p_\ell^t(\theta) \lambda_t(\theta) TC_t$.
 - (d) The residual-sector problem gives residual-sector employment: $N_{s,t}^0 = \frac{TC_t}{w_t}$.
 - (e) The bank's problem then gives $\{r_{\ell,t+1}^1\}_{t=1}^{T-1}$:

$$L_{t+1} = \frac{(1 + r_{t+1})}{1 + r_{t+1} - (1 - \eta_{t+1})(1 + r_{\ell,t+1}^1)} \beta_L N_t$$

²⁵An age- j member of cohort $g \in \{-J + 2, \dots, T - 1\}$ faces prices $\vec{p}_{g+j-1}$.

where $L_{t+1} = M_{t+1} + \mu w_{t+1} \left(\bar{w}, u_{t+1} \left(r_{\ell,t+1}^1 \right) \right) N_{g,t+1}$ and

$$N_t = \begin{cases} L_t (1 + r_{\ell,t}) - B_t (1 + r_t) & \text{if } t = 1 \\ \left(1 + r_{\ell,t}^0 \right) \eta_t L_t & \text{if } t > 1. \end{cases}$$

where $L_t = M_t + \mu w_t (\bar{w}, u_t) N_{g,t}$. Given $\{L_{t+1}\}_{t=1}^{T-1}$, the bank's problem determines $\{B_{t+1}\}_{t=1}^{T-1}$:

$$B_{t+1} = \frac{\hat{\lambda}_t - 1}{\hat{\lambda}_t} L_{t+1} \text{ where } \hat{\lambda}_t = \frac{(1+r_{t+1})}{1+r_{t+1}-(1-\eta_{t+1})(1+r_{\ell,t+1})} \text{ and } \eta_t = \zeta^{1-\beta_L} \left(\frac{1+r_{t+1}}{1+r_{\ell,t+1}^0} - (1-\eta_{t+1}) \right)^{\beta_L}.$$

(f) Update the guesses for each $t \in \{1, \dots, T-1\}$:

$$\begin{aligned} K_t &= (1 - \lambda_k) K_t + \lambda_k (A_t - B_t - V_t^{rc} (H_t^r)) \\ H_t &= (1 - \lambda_h) H_t + \lambda_h H_t^d \\ H_t^r &= (1 - \lambda_{rc}) H_t^{r,0} + \lambda_{rc} H_t^{r,1} \\ r_{\ell,t+1}^0 &= (1 - \lambda_r) r_{\ell,t+1}^0 + \lambda_r r_{\ell,t+1}^1 \\ N_{s,t} &= (1 - \lambda_s) N_{s,t} + \lambda_s N_{s,t}^0 \end{aligned}$$

(g) Iterate until the guesses converge, with criteria $\max_t |K_t + B_t + V_t^{rc} (H_t^r) - A_t| < \epsilon_k$, $\max_t |H_t^{r,1} - H_t^{r,0}| < \epsilon_h$, $\max_t |H_t^d - H_t| < \epsilon_h$, $\max_t |r_{\ell,t+1}^1 - r_{\ell,t+1}^0| < \epsilon_r$, and $\max_t |N_{s,t} - N_{s,t}^0| < \epsilon_s$ where $\epsilon_k = \epsilon_h = \epsilon_s = 10^{-3}$ and $\epsilon_r = 10^{-4}$.